

Chiral Gauge Dynamics: Formally Verified Emergence of Gravity, Quantum Mechanics, and Topological Matter from a $\text{Spin}(4, \mathbb{C})$ Connection

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We present Chiral Gauge Dynamics (CGD), a background-independent formulation of physics derived from a continuous, classical $\text{Spin}(4, \mathbb{C})$ gauge connection. By utilizing the Lean4 formalization language to guarantee mathematical rigor, we present three separate Lorentzian geometries that bypass the reality conditions which historically derailed pure-connection gravity research. We also demonstrate that phenomena often assumed to require quantization (specifically a mass gap, quantum entanglement, and the Born rule) emerge from the classical topology of the field. Finally, we show that a $\text{Spin}(4, \mathbb{C})$ universe requires an isovector axial-vector condensate and show that this condensate is consistent with data measuring the proton radius puzzle.

I. INTRODUCTION

In this paper, we explore an ontological foundation for the Einstein Field Equations (EFE). We extend the pure gauge connection formulation for gravity by showing that General Relativity emerges from a $\text{Spin}(4, \mathbb{C})$ pure gauge connection. However, whereas the previously known $\text{SU}(2)$ pure gauge formulation contains General Relativity, it does not natively contain particles. In contrast, a universe composed of only a $\text{Spin}(4, \mathbb{C})$ connection contains General Relativity *and* particles.

This research extensively uses the Lean4 mathematical formalization language [1]. The reason for this is simple: when we first discovered the mathematical structures that emerge from $\text{Spin}(4, \mathbb{C})$, which touch on problems related to quantum gravity, entanglement, measurement, dark energy, dark matter, and the emergence of time, we were certain we had made a mathematical error. By formalizing the derivations in Lean4, we have eliminated the possibility that the math is incorrect. It's still possible that the theorems we prove here have no association with real physical phenomena, and Lean4 makes absolutely no claims about whether the $\text{Spin}(4, \mathbb{C})$ gauge connection is ontologically correct for our universe. Also, due to the nascent state of the Lean4

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ecosystem, our use of peer-reviewed literature exposes another possible avenue for errors.

In an ideal world, all of the physics literature we rely upon would be fully formalized from basic mathematical axioms, but that does not exist, so we created the Litlib library of transcriptions from published literature into Lean4 code. The Lean4 environment treats these transcriptions as axiomatically true, so we have to first trust that the papers contain no errors, and then we must trust that the transcriptions into Lean4 are accurate. While this solution is not perfect, it is the best solution we could engineer because it isolates the unformalized portions of the math.

This paper includes the formalized Lean4 signatures in Appendix A. In order to save space, this summary contains only the theorem signatures and the definitions required to understand those signatures. For the complete, compilable proofs, please refer to the Supplemental Materials. Within the main body of this paper, we use mathematical notation to write the equations, but we include a specialized footnote reference that matches to the Lean4 code name in the Code Summary index and also contains a hyperlink navigational aid to allow readers to quickly jump to the formalized definition. For example, in $\not{D}\Psi = 2J_a\gamma^a$ ^[L1], the ^[L1] matches the `CGD.Quantum.Dirac.geometricDiracEmergence` entry in the Code Summary.

We refer to this research program as Chiral Gauge Dynamics (CGD) because much of the rich mathematical structure comes from the chiral split $\text{Spin}(4, \mathbb{C}) \cong \text{SL}(2, \mathbb{C})_L \times \text{SL}(2, \mathbb{C})_R$. Gauge theory, and in particular Yang-Mills theory [2], describes how elements of a field relate to each other regardless of the choice of coordinates. Since its introduction, it has undisputedly become the language of the Standard Model.

However, it wasn't until Plebański [3] that physicists knew that General Relativity could also be written as a gauge theory [4]. While this self-dual 2-forms formulation is not how General Relativity is taught in undergraduate classrooms, it is of vital importance to CGD.

Then Ashtekar [5] introduced Ashtekar variables, which showed that the gauge connection itself could produce General Relativity. With this result, General Relativity looked exactly like a Yang-Mills theory. It is interesting to note that this is the point in history where the research path that led to CGD diverged from the path that led to Loop Quantum Gravity (LQG) [6]. While the mathematical results of LQG and CGD are vastly different, at this historical fork in the road, the difference is that LQG quantized spacetime while CGD leaves spacetime as a classical field.

The mathematical insight that makes CGD possible is the astonishingly elegant Urbantke metric presented in [7]. This work shows that a metric necessarily emerges for $\text{SU}(2)$ and $\text{Spin}(4, \mathbb{C})$ gauge fields. Intuitively, this describes the distance between two points by measuring the amount of twist in the field between two points.

The work first introduced in Capovilla *et al.* [8] and later expanded in Capovilla *et al.* [9] took the Urbantke metric and showed that for a $SU(2)$ pure gauge connection, General Relativity emerges without depending on a separate axiom for a background metric. We extend this result by showing that General Relativity also emerges from a $Spin(4, \mathbb{C})$ pure gauge connection. However, while a universe composed only of a $SU(2)$ connection contains General Relativity, it does not natively contain particles. In contrast, a universe composed of only a $Spin(4, \mathbb{C})$ connection contains General Relativity as well as particles.

In Section II, we will lay out the three CGD axioms:

Axiom 1: The universe is a classical $Spin(4, \mathbb{C})$ gauge connection (Equation 1).

Axiom 2: Macroscopic volume exists (Equation 2).

Axiom 3: The vacuum is unimodular (Equation 3).

We will also derive the Lagrangian and the Bianchi identities that govern stress-energy conservation and motion.

In Section III we will prove that $Spin(4, \mathbb{C})$ gives rise to General Relativity. This derivation takes us through the CDJ constraints [9], Ricci-flat emergent spacetime, and stress-energy conservation.

Once we have formalized the existence of gravity within CGD, we will shift our focus in Section IV and show how the curvature of the $SL(2, \mathbb{C})_R$ chiral sector acts as a physical matter source term. Skymion particles [10] emerge from CGD as topological defects where the metric becomes degenerate but where stable knots form. We will demonstrate confinement and three colors for these particles. Furthermore, we will show how a topological mass gap must exist for these particles.

Perhaps the most surprising aspect of CGD comes in Section V, where we will derive the Dirac equation and the Born rule. We also derive entanglement and demonstrate that correlations of entangled particles match the Tsirelson bound. While this seems at first glance like a contradiction because CGD is a classical field theory, CGD entanglement bypasses Bell's theorem by challenging the assumption that two entangled particles are separable.

Finally, at the end of this paper in Section VI, we will turn our focus to phenomenology. Most of the theoretical work presented in this paper focuses on showing how various physical theories emerge from $Spin(4, \mathbb{C})$. In a few places, our derivations require some nuance that is different between CGD and the Standard Model. However, in one case, CGD requires a feature that is starkly different: it requires the presence of an isovector axial-vector condensate. If such a condensate exists, it should have a substantial impact on the proton radius puzzle. When we

first identified this, we assumed it would be an easy way to falsify the CGD theory. However, we created a one-parameter mathematical model that not only substantially outperformed the leading Standard Model empirical model, it also correctly identified the proton radius. We cannot argue that CGD solves the proton radius puzzle, but our assumption that this would be a quick and easy falsification turned out to be wrong.

II. FOUNDATIONS

All of the results derived in this paper depend upon three axioms. However, in order to contrast CGD with the Standard Model and General Relativity, we will also discuss two non-axioms.

The axiom selection is the only point where we are able to engineer the physics of the CGD universe. Engineering decisions are often indistinguishable from philosophy, and that certainly applies here. For this reason, we will discuss our motivations for exploring a universe with the selected features. However, once the axiom basis is solidified, everything else is mathematically forced.

The workhorse ontological axiom is

$$\mathcal{A}_\mu(x) = \mathcal{A}_\mu^{(L)}(x) \oplus \mathcal{A}_\mu^{(R)}(x) \in \mathfrak{spin}(4, \mathbb{C}). \quad [\text{L2}] \quad (1)$$

This merely states that the universe is a $\mathfrak{spin}(4, \mathbb{C})$ gauge connection. The $\mathcal{A}_\mu^{(L)}(x) \oplus \mathcal{A}_\mu^{(R)}(x)$ part is stating a property of the matrix math, which is often described as $\text{Spin}(4, \mathbb{C})$ decomposing into two chiral halves $\text{SL}(2, \mathbb{C})_L$ and $\text{SL}(2, \mathbb{C})_R$.

Perhaps the most compelling motivation for $\text{Spin}(4, \mathbb{C})$ is to notice that the Dirac Equation is naturally described in $\text{Spin}(4, \mathbb{C})$. That is, Paul Dirac started with the 4-dimensional symmetries of Special Relativity, which is the Lorentz Group $\text{SO}(1, 3)$. However, the electron is a spin-1/2 particle, which requires a double cover of the Lorentz Group such as $\text{SL}(2, \mathbb{C})$. However, $\text{SL}(2, \mathbb{C})$ is a 2-component Weyl spinor that can only describe massless particles, so to describe a massive particle, Dirac combined a left-handed and right-handed Weyl spinor into a 4-component Dirac spinor [11]. This representation is the group $(1/2, 0) \oplus (0, 1/2)$, but the Lie group that natively describes this structure is $\text{Spin}(4, \mathbb{C})$. In other words, if we hope to derive the Dirac Equation, we need to use either $\text{Spin}(4, \mathbb{C})$ or a supergroup.

Axiom 1 requires some initial conditions. To begin with, we must require that the universe exists so that we can ignore all vacuous solutions to the math. The most convenient way for us to state that the universe exists is to require that the bulk $\mathcal{B}$ (i.e. the “empty” part of the universe)

has non-zero volume

$$\det(g_{\mu\nu}(x)) \neq 0 \quad \forall x \in \mathcal{B}. \quad [\text{L3}] \quad (2)$$

This axiom doesn't require that everything within the universe has non-zero volume. In fact, exploring scenarios where the volume collapses leads to several interesting observations, as we will see in Sections IV and V.

While Axiom 2 largely serves to allow the Lean 4 formalization to ignore trivial solutions, our final axiom does some heavy lifting. Without any additional constraints, $\text{Spin}(4, \mathbb{C})$ may lack isotropy. Among other things, a lack of isotropy would warp spacetime and effectively create matter and gravity out of otherwise empty space. In order avoid this situation, we require a third axiom that enforces a smooth background vacuum when no matter is present. Formally, we state

$$\Sigma^{ab}(x) = \frac{1}{3} \text{Tr}(\Sigma(x)) \delta^{ab} \quad \forall x \in \mathcal{B}. \quad [\text{L4}] \quad (3)$$

Here, Σ^{ab} is the 3×3 matrix representing the topological density ($\Sigma^{ab} = \epsilon^{\mu\nu\rho\sigma} F_{\mu\nu}^a F_{\rho\sigma}^b$). For all elements x in the bulk $\mathcal{B}$, we require that the topological density in each of the three internal gauge axes is equal [9].

At this point, we have finalized our axiom basis, but it's worth emphasizing the fundamental way that CGD differs from the Standard Model and Loop Quantum Gravity (LQG): we do not quantize the gauge connection. We do not assume that force transfer occurs in quantized units and we do not assume that particles are quantum points. From a practical perspective, this means we will not proceed with deriving CGD by user perturbation and renormalization techniques.

Another notable absence from our axiom set is that we do not assume an independent Minkowski metric space background. This is a feature CGD shares with LQG, and in fact, the LQG body of work demonstrates that the absence of an independent background is fully consistent with General Relativity (GR). However, LQG attempts to modify GR by quantizing it. In contrast, in CGD we will demonstrate that instead of quantizing gravity, we can instead gravitize the quantum. That is, we will take a classical framework that is fully consistent with Einstein's field equations and show that Standard Model phenomenology emerges.

Now that we have our full axiom basis, we can show that $\text{Spin}(4, \mathbb{C})$ decomposes perfectly into two chiral halves.

$$\mathcal{L}(F) = \mathcal{L}_{\text{vac}}(F^{(L)}) + \mathcal{L}_{\text{ASD}}(F^{(R)}) \quad [\text{L5}] \quad (4)$$

This follows by observing that $\text{Spin}(4, \mathbb{C}) \cong \text{SL}(2, \mathbb{C})_L \times \text{SL}(2, \mathbb{C})_R$, so we can select a basis to map $\text{SL}(2, \mathbb{C})_L$ into the top left 2×2 sub-matrix and $\text{SL}(2, \mathbb{C})_R$ into the bottom right 2×2 sub-matrix.

Then, when we multiply these matrices together, we get

$$\begin{pmatrix} L & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 0 & R \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \quad (5)$$

Notably, all cross terms vanish.

The CGD Lagrangian is built from the trace of the curvature squared. Because the cross terms vanish, we have

$$\text{Tr} \left((F^{(L)} + F^{(R)}) \wedge (F^{(L)} + F^{(R)}) \right) = \text{Tr}(F^{(L)} \wedge F^{(L)}) + \text{Tr}(F^{(R)} \wedge F^{(R)}) \quad (6)$$

The geometry of the self-dual (L) sector does not algebraically interact with the geometry of the anti-self-dual (R) sector. As we will see in Sections III and IV, this will lead to the observation that gravity and matter couple geometrically, but not algebraically. They can coexist in a volume without sharing energy through the action.

To construct a gauge-invariant Lagrangian, we need to build a scalar density that is quadratic in the field strength tensor. In [12], Utiyama showed that the only possible invariant scalars are the Yang-Mills term $\text{Tr}(F_{\mu\nu}F^{\mu\nu})$ or the Pontryagin density $\epsilon^{\mu\nu\rho\sigma}\text{Tr}(F_{\mu\nu}F_{\rho\sigma})$. However, as shown in [11], the Yang-Mills term requires an independent background metric, so CGD must use the Pontryagin density.

This leads us to the unique Lagrangian candidate

$$\mathcal{L}_{\text{CGD}} = c \sum_{\mu\nu\rho\sigma} \epsilon^{\mu\nu\rho\sigma} \text{Tr}(F_{\mu\nu}F_{\rho\sigma}) \quad [\text{L6}] \quad (7)$$

However, this is a purely topological Lagrangian, any classical action variation gives

$$\left. \frac{d}{d\tau} S[v(\tau)] \right|_{\tau=0} = 0. \quad [\text{L7}] \quad (8)$$

This means that we cannot construct Euler-Lagrange equations of motion and we have no classical Hamiltonian.

If CGD has no equations of motion, how does anything move? By expanding the covariant derivative of the field strength tensor into its fundamental matrix algebra, we derive the exact differential Bianchi identity

$$D_\rho F_{\mu\nu} + D_\mu F_{\nu\rho} + D_\nu F_{\rho\mu} = 0. \quad [\text{L8}] \quad (9)$$

This equation shows that the geometry of CGD is conserved and that no operations exist that can tear the gauge field. In Section III, we will show how this leads to the Einstein-Infeld-Hoffman

mechanism [13] that replaces the standard equations of motion and causes objects to move along geodesics in 4-dimensional spacetime.

Another consequence of missing equations of motion is that we cannot use Noether's Theorem [11] to prove properties such as charge, angular momentum, or energy are conserved. Later in Section III, we will use Equation 9 to prove that stress-energy is conserved in Equation 14, but Equation 9 will not work to prove that topological charge is conserved. Instead, we will look at the Abelian projection of the field $F_{\rho\sigma}$ and define an arbitrary topological current J^μ as the dual divergence of this field. Since the CGD universe is fundamentally smooth, we can use Clairaut's theorem [14] to guarantee that this current is conserved.

$$J^\mu = \epsilon^{\mu\nu\rho\sigma} \partial_\nu F_{\rho\sigma} \implies \partial_\mu J^\mu = 0 \quad [\text{L9}] \quad (10)$$

We will show in Section IV that this charge conservation gives rise to the electro-dynamic property.

III. GRAVITY: THE EMERGENCE OF SPACETIME

In 1919, Albert Einstein created the idea of unimodular gravity [15]. The essential idea behind unimodular gravity is that the metric tensor $g_{\nu\mu}$ defining the 4-dimensional spacetime must have a constant non-zero determinant, meaning that the volume must be constant. This has been an active field of research ever since its introduction in part because it provides a natural explanation for the cosmological constant Λ without needing to invoke dark energy.

The crux of our proof that gravity emerges in CGD depends on the Urbantke metric [7]. The Urbantke metric performs a profound unification: it demonstrates that the geometry of spacetime and the gauge fields that live inside it are not separate things. In the Standard Model, spacetime is a pre-existing stage described with a Minkowski metric and particles are placed on this stage.

More formally, the Urbantke metric constructs a 4-dimensional conformal (scale-free) metric from the wedge product

$$\begin{aligned} \sqrt{-g} g_{\mu\nu} &= \frac{1}{6} \epsilon^{\alpha\beta\gamma\delta} \text{Tr} \left(F_{\mu\alpha} [F_{\beta\gamma}, F_{\delta\nu}] \right) \\ &= \frac{1}{6} \epsilon^{\alpha\beta\gamma\delta} F_{\mu\alpha}^i F_{\beta\gamma}^j F_{\delta\nu}^k \text{Tr} \left(T_i [T_j, T_k] \right) \\ &= \frac{c}{6} \epsilon_{ijk} \epsilon^{\alpha\beta\gamma\delta} F_{\mu\alpha}^i F_{\beta\gamma}^j F_{\delta\nu}^k. \end{aligned} \quad (11)$$

Here, the gauge field generators (T_i, T_j, T_k) are mapped onto the independent 2-forms (planes of rotation) of the 4-dimensional spacetime represented by the antisymmetric tensor $\epsilon^{\alpha\beta\gamma\delta}$.

This metric contains two separate filters that can prevent a gauge field from possessing an Urbantke metric. First, the Lie bracket $[F_{\beta\gamma}, F_{\delta\nu}]$ means that if we attempted to use an Abelian field such as $U(1)$, the matrices commute and so the macroscopic volume will collapse to 0.

The second filter deals with how the trace of the generators $\text{Tr}(T_i[T_j, T_k])$ evaluates to the antisymmetric structure constant ϵ_{ijk} . This tensor requires exactly 3 internal gauge indices (i, j, k) to map to the spacetime indices. The gauge group $SU(2)$ is generated from 3 rotations, so it possesses a well-defined Urbantke metric. The $SL(2, \mathbb{C})$ gauge group is generated from 3 rotations and 3 boosts, and while this requires 6 numbers to write down $(a_1 + b_1i, a_2 + b_2i, a_3 + b_3i)$, this is still 3 algebraic generators, so it also possesses a well-defined Urbantke metric. Krasnov noted in Krasnov [4] that this mapping relies on an accidental mathematical isomorphism that is unique to 4-dimensional geometry, so $SU(2)$ and $SL(2, \mathbb{C})$ are the only gauge groups that can natively construct a 4-dimensional macroscopic metric.

In [8, 9], Capovilla *et al.* (CDJ) developed the algebraic constraint framework to determine the behavior of emergent spacetime that uses the Urbantke metric. In their work, CDJ used a specific dynamical action that produced the Einstein Field Equations. However, as we showed in Equation 8, CGD does not possess equations of motion, so it does not share this dynamical action. Instead, we explore the emergent spacetime of CGD by using Equation 4 to obtain one of the two $SL(2, \mathbb{C})$ chiral halves of $\text{Spin}(4, \mathbb{C})$ and applying the unimodular constraint from Axiom 3 to obtain the metric determinant. This leads to

$$\left(\sqrt{-g(x)}\right)^2 = \mu(x)^2 \quad [\text{L10}] \quad (12)$$

Here, $\mu(x)$ is the CDJ scalar density, and because we are working with a Lorentzian signature ($\det(g) < 0$), we adopt the standard convention $\sqrt{(-g)}$ for the real volume.

Typically, pure-connection formulations assume the scalar density $\mu(x)$ is globally constant in order to force a torsion-free, Ricci-flat spacetime. However, by evaluating the conformal metric transformation ($\tilde{g}_{\mu\nu} = \mu(x)g_{\mu\nu}$), we find that the torsion generated by varying $\mu(x)$ is integrable, so the assumption of globally constant volume is unnecessary. By mapping the physical metric through the Plebański vacuum conditions, we prove:

$$R_{\mu\nu} = \mu g_{\mu\nu} + 2\nabla_\mu \nabla_\nu \ln \sqrt{\mu} + g_{\mu\nu} \nabla^\rho \nabla_\rho \ln \sqrt{\mu} - 2(\nabla_\mu \ln \sqrt{\mu})(\nabla_\nu \ln \sqrt{\mu}) + 2g_{\mu\nu}(\nabla^\rho \ln \sqrt{\mu})(\nabla_\rho \ln \sqrt{\mu}). \quad [\text{L11}] \quad (13)$$

Here, $\mu g_{\mu\nu}$ is the Λ cosmological constant term scaled by the volume density.

Now that we have established the macroscopic geometry of the Einstein Field Equations, the remaining element required to fully recover General Relativity is stress-energy conservation. We

proved in Equation 8 that CGD possesses no equations of motion, so we are unable to use Noether's Theorem to prove the conservation of energy. Instead, we will use the Bianchi identity that we formalized and presented in Equation 9. By showing that the Urbantke metric satisfies the Bianchi identity, we are able to demonstrate the emergence of stress-energy conservation within CGD:

$$\nabla_\mu T^{\mu\nu} = 0. \quad [\text{L12}] \quad (14)$$

This proof depends upon the metric symmetry bounds established in Urbantke [7] and the geometric constraints of the contracted Bianchi identity [11].

At this point, we have shown that General Relativity emerges from one of the two $\text{SL}(2, \mathbb{C})$ chiral halves of $\text{Spin}(4, \mathbb{C})$. Even though we have no equations of motion, we can explore motion within this system. In Papapetrou [13], Papapetrou showed that for a localized, single-pole topological defect (in other words, a localized mass without spin), the conservation of stress-energy forces particles to move along geodesics. We are able to use this proof to demonstrate that within CGD,

$$\frac{du^\alpha}{ds} + \Gamma_{\mu\nu}^\alpha u^\mu u^\nu = 0. \quad [\text{L13}] \quad (15)$$

In other words, even without standard equations of motion, we are able to prove that objects within CGD move, and that they move according to the geodesic paths described in General Relativity.

At this point, we have demonstrated that General Relativity mathematically emerges from CGD. However, the geometry is highly constrained, and highly constrained geometric systems risk being vacuous, where the only solution is the trivial vacuum ($A = 0$). In order to show that the geometry of CGD is not vacuous, we need to demonstrate that some non-trivial solution exists. As long as we can prove that some non-empty witness solution exists, that witness will serve as a sanity check to demonstrate that our axiom basis is not contradictory.

However, a relevant physics theory does not need just an arbitrary non-vacuous solution to the geometric constraints. Our universe is Lorentzian, so while a Euclidean solution is mathematically interesting, it isn't of much use to physics research. The problem of finding a Lorentzian solution has plagued connection gravity research since its inception. As Capovilla *et al.* noted, "For the case of Lorentz signature, one has to pay with unwieldy reality conditions for the use of self-dual 2-forms as basic variables for GR."

The reality conditions made it appear that deeper research into pure connection gravity was a dead end. According to Thiemann [16]:

The original motivation to introduce the new variables was that for the quantization of general relativity it seemed mandatory to simplify the algebraic structure of the

Hamiltonian constraint which for $s = -1$ requires $\beta = \pm i/2$ since then the constraint becomes polynomial after multiplying by a factor proportional to $\sqrt{\det(q)}$. On the other hand, then the reality conditions become non-polynomial. Finally, if one wants polynomial reality conditions then one must have β real and then the Hamiltonian constraint is still complicated. Thus it becomes questionable what has been gained.

Thiemann continued to explain why the compromise of the reality conditions made it so that any attempt at canonical quantization would require breaking background independence. However, in CGD, we do not attempt to perform canonical quantization. This issue is the exact fork in the road where CGD takes a different path from loop-quantum-gravity.

Since we are using one of the two $\text{SL}(2, \mathbb{C})$ chiral halves of $\text{Spin}(4, \mathbb{C})$, we must show that despite this field being defined using complex numbers, a real Lorentzian metric natively emerges without imposing the “unwieldy” reality conditions manually. We will present three separate solutions, and the first represents a universe that is empty of defects. We begin by using a linear-in-coordinates $\mathfrak{sl}(2, \mathbb{C})$ ansatz:

$$\begin{aligned} \mathcal{A}_0(x) &= 0 \\ \mathcal{A}_1(x) &= -x^0 \sigma_x + x^3 \frac{i}{2} \sigma_y - x^2 \frac{i}{2} \sigma_z \\ \mathcal{A}_2(x) &= -x^0 \sigma_y + x^1 \frac{i}{2} \sigma_z - x^3 \frac{i}{2} \sigma_x \\ \mathcal{A}_3(x) &= -x^0 \sigma_z + x^2 \frac{i}{2} \sigma_x - x^1 \frac{i}{2} \sigma_y \end{aligned} \quad [\text{L14}] \tag{16}$$

In order to test if this configuration satisfies the geometric constraints, we evaluate the strength tensor (curvature) at the origin ($x = 0$). Once we take the gauge-covariant derivative, we obtain a complex-valued curvature tensor. By evaluating the temporal and spatial components, we obtain:

$$F_{0j}|_{x=0} = -\sigma_j, \quad F_{jk}|_{x=0} = i\epsilon_{jkl}\sigma_l. \quad [\text{L15}] \tag{17}$$

Next, we check this against the trace-free algebraic constraints [9]. By evaluating the sum over the Levi-Civita symbol, we obtain a matrix that is proportional to the identity matrix, meaning that its trace-free part is exactly zero:

$$\Sigma = \epsilon^{\mu\nu\rho\sigma} F_{\mu\nu} F_{\rho\sigma} = 16i \cdot \mathbb{I}_{3 \times 3} \implies \Sigma - \frac{1}{3} \text{Tr}(\Sigma) \mathbb{I} = 0. \quad [\text{L16}] \tag{18}$$

Since this curvature is valid, we next map it through the Urbantke metric and obtain

$$g_{\mu\nu}|_{x=0} = \text{diag}(-12, 12, 12, 12), \quad \det(g_{\mu\nu}) = -20736 < 0 \quad [\text{L17}] \tag{19}$$

This demonstrates that the metric possesses a $(-1, 1, 1, 1)$ Lorentzian signature.

We can expand the toy model in Equation 16 into a class of solutions that is physically interesting by introducing a scale factor. Since the time component of the geometry is x^0 , we write this scale factor as $a(x^0)$ and obtain:

$$\mathcal{A}_0(x) = 0, \quad \mathcal{A}_j(x) = a(x^0)\sigma_j \quad [\text{L18}] \quad (20)$$

As long as the scale factor is real and non-static ($\partial_t a(t) \neq 0$), the metric will not collapse. Since the spatial terms (x^1, x^2, x^3) do not show up in the connection, this geometry is homogeneous. Also, since the j indices are mapped one-to-one with the gauge generators σ_j , any spatial coordinate rotation is reversible, so the geometry is isotropic. Any metric that is spatially homogeneous and isotropic is, by definition, an FLRW metric [17]. This leads to:

$$\text{FLRW} \implies \text{Strictly Real, Lorentzian } (\det(g_{\mu\nu}) < 0). \quad [\text{L19}] \quad (21)$$

Our final geometric solution that satisfies the Urbantke metric and reality conditions involves a non-rotating non-singular topological defect. In standard General Relativity, this conceptually maps to the Schwarzschild metric for a fundamental particle such as an electron, but in CGD, the solution avoids a mathematical singularity. Consider the following geometry:

$$K_c(x) = \frac{-1}{(x^1)^2 + (x^2)^2 + (x^3)^2 + 2} \quad [\text{L20}] \quad (22)$$

$$\mathcal{A}_0(x) = 0 \quad (23)$$

$$\mathcal{A}_1(x) = -x^0\sigma_x - K_c x^3(i\sigma_y) + K_c x^2(i\sigma_z)$$

$$\mathcal{A}_2(x) = K_c x^3(i\sigma_x) - x^0\sigma_y - K_c x^1(i\sigma_z)$$

$$\mathcal{A}_3(x) = -K_c x^2(i\sigma_x) + K_c x^1(i\sigma_y) - x^0\sigma_z. \quad [\text{L21}]$$

By evaluating this geometry through the Urbantke metric, we can show

$$\begin{aligned} \Sigma - \frac{1}{3}\text{Tr}(\Sigma)\mathbb{I} &= 0 \\ \det(g_{\mu\nu}(x)) &< 0 \quad . \quad [\text{L22}] \end{aligned} \quad (24)$$

Since this geometry satisfies the algebraic trace-free constraint, we see from Equation 13 that the solution constitutes a pure cosmological vacuum. Since the determinant is negative, the metric is Lorentzian and non-degenerate. Finally, since the denominator in Equation 22 is strictly greater than zero, the solution avoids singularities.

Let us characterize the significance of the soliton geometry solution presented in Equation 23: it is a step forward for mathematical physics, and is particularly useful for pure-connection gravity research, but it doesn't suddenly solve black hole physics. In Standard General Relativity, Birkhoff's theorem states that the only spherically symmetric geometry must be the Schwarzschild metric [18], so the vacuum must contain a singularity. The only known ways around this are to introduce exotic matter that violates standard energy conditions [19]. However, a spherically symmetric geometry is a non-rotating geometry, which is necessarily a toy model.

So far, we have focused only on one of the two $SL(2, \mathbb{C})$ chiral halves of $Spin(4, \mathbb{C})$ that we obtained in Equation 4. In Section IV, we will consider the other chiral half and will see not only how discrete particles emerge from this purely classical field, but how the geometries of the two chiral halves interact with each other.

IV. PARTICLES: TOPOLOGICAL KNOTS AND THE MASS GAP

Many attempts to create a unified field theory have failed because they cannot naturally generate matter. Goenner [20] notes that the classical attempts by Einstein, Weyl, and Eddington to build particles out of spacetime geometry collapsed, and that Einstein himself lamented “the problem of matter and quanta makes the construction fall apart” when describing his 5-dimensional metric. Modern, higher dimensional theories, such as Kaluza-Klein supergravity, fail to incorporate chiral fermions [21]. Consequently, physicists usually bolt external matter fields onto the gravitational Lagrangian by hand.

With CGD, particles are not fundamental points, but are instead knots within the connection itself. The self-dual (SD) chiral $SL(2, \mathbb{C})$ sector gives rise to gravity, but topological defects within the anti-self-dual (ASD) chiral $SL(2, \mathbb{C})$ sector gives rise to particles, while the curvature within the ASD sector produces mass. We will show how the interaction between these two chiral halves within CGD produces inertia.

Conceptually, Wheeler [22] was the first to explore the idea of particles arising from the geometry of the universe. Wheeler argued, “Space time is not an arena for physics, it is all of classical physics” [23]. He tried to build the “geon” out of the metric tensor and the electrodynamic field, but this approach ultimately failed to produce fermions.

't Hooft [24] and Belavin *et al.* [25] moved away from Abelian fields and worked with Yang-Mills theories. They found solutions that looked like particles. Then, Belavin *et al.* [25] discovered the BPST instanton, which are lumps of pure Yang-Mills fields that exhibit topological stability,

although the BPST instanton strictly require Euclidean space. In this paper we will show the existence of topologically stable particles within Lorentzian space.

One of the biggest concerns with producing particles from a $\text{Spin}(4, \mathbb{C})$ ontology is that this seems to cause a conflict with the strong-force. That is, the Standard Model describes the strong force using $\text{SU}(3)$, and $\text{SU}(3)$ cannot be embedded within $\text{Spin}(4, \mathbb{C})$. In this section, we will address this problem by showing that CGD mathematically produces three “colors”. We also show that due to how CGD particles are defined as part of the topology and cannot be separated from the Urbantke metric, CGD possesses a geometric mass gap.

We will begin our investigation into particles by considering a topological defect ($F_{\mu\nu}^{(R)} \neq 0$) and then ask what this defect looks like to a macroscopic observer. In the Einstein Field Equations ($G_{\mu\nu} = 8\pi G T_{\mu\nu}$), the $T_{\mu\nu}$ term is the stress-energy tensor. We will assume, in order to reach a contradiction, that this topological defect possess no mass or energy, giving $T_{\mu\nu} = 0$. However, Krasnov [4] showed that if $T_{\mu\nu} = 0$, then the ASD curvature must be completely flat. However, this is a contradiction, because we begin our proof by assuming that a knot exists by stating $F_{\mu\nu}^{(R)} \neq 0$. Therefore, we conclude

$$F_{\mu\nu}^{(R)} \neq 0 \implies T_{\mu\nu} \neq 0. \quad [\text{L23}] \quad (25)$$

In CGD, we must build particles out of $\text{Spin}(4, \mathbb{C})$ somehow, but this leaves us with several options. $\text{U}(1)$ is the simplest non-empty subgroup, but this immediately runs into problems because $\text{U}(1)$ is commutative. When we plug a commutative field into the Urbantke Metric (Equation 11), we see that the volume collapses, which would mean that no particles experience time or even relative distance. This means

$$[F_{\mu\nu}, F_{\rho\sigma}] = 0 \implies \det(g_{\mu\nu}) = 0. \quad [\text{L24}] \quad (26)$$

More generally, if the field does not have enough independent Lie algebra generators, or “colors”, to fill in all of the diagonals of the 3×3 matrix $g_{\mu\nu}$, then the volume will collapse. We can use this observation to show that particles require exactly three colors

$$\det(g_{\mu\nu}) \neq 0 \implies ([F_{\mu\nu}, F_{\rho\sigma}] \neq 0) \wedge (\forall a \in \{1, 2, 3\}, F_{\mu\nu}^a \neq 0). \quad [\text{L25}] \quad (27)$$

Let us next consider defects in the ASD chiral sector. In topology, the Cartan-Maurer degree, or winding number, counts how many times a field wraps around a gauge group [11]. Since Krasnov [4] proved that only $\text{SU}(2)$ and $\text{SL}(2, \mathbb{C})$ are capable of generating the Urbantke Metric, then for CGD we only need to only consider the ASD chiral $\text{SL}(2, \mathbb{C})$ gauge field, where the topological winding

numbers are described by the $SU(2)$ subgroup. If a particle exists in $SU(2)$, it is a non-Abelian knot, and Belavin *et al.* [25] proved that knots in $SU(2)$ cannot decay or unwrap into the trivial vacuum. We can extend these results to show that they also apply within CGD:

$$\int_{S^3} \text{Tr} \left(\mathcal{A} \wedge d\mathcal{A} + \frac{2}{3} \mathcal{A} \wedge \mathcal{A} \wedge \mathcal{A} \right) \neq 0 \implies \mathcal{A} \text{ cannot continuously decay to } 0. \quad [\text{L26}] \quad (28)$$

Next, we use the existence of the Lorentzian Witness (Equation 19) to perform an analysis that has not previously been possible. In Belavin *et al.* [25], particles strictly existed in an Euclidean field. However, since a Euclidean field does not possess a time dimension, these topological defects possess an ‘action’ but they cannot possess a rest mass or energy. Since CGD dynamically generates a Lorentzian space, we evaluate a particle by evaluating the 3-dimensional Cauchy slice of spacetime at time t where the spatial boundary $\partial\Sigma$ compactifies to S^3 . We define the inertial mass as:

$$m_{\text{inertial}} = \left| \int_{\partial\Sigma} \omega_{\text{CM}}(\mathcal{A}^{(R)}) \right|. \quad [\text{L27}] \quad (29)$$

The non-compact boost directions of the $SL(2, \mathbb{C})$ gauge field form a contractable space ($\mathbb{R}^3$), so they possess a topological winding number of zero. This excludes negative-mass ghost states. Since the stable particle must possess a topological homeomorphism to the $SU(2)$ boundary, the integrated charge must evaluate to a non-zero integer. This shows that the topological defect must possess a strictly positive mass gap:

$$\mathcal{A}^{(R)}|_{\partial\Sigma} \cong SU(2) \implies m_{\text{inertial}} > 0. \quad [\text{L28}] \quad (30)$$

It’s important to emphasize that this is not the Millennium Prize Problem [26]. The proof for Equation 30 relies on the global topological mapping degree of a continuous classical boundary, and it’s only possible because the Urbantke metric is inseparable from the topology. While we are able to show that a mass gap exists in CGD, this proof shows nothing about the similar problem of proving a mass gap for any theory, such as the Standard Model, that possesses an independent metric background.

Now that we have characterized the mass for $SU(2)$ knots, we can also look at simpler Abelian $U(1)$ projections of the ASD connection. To do this, we will use the Abelian field strength tensor which incorporates both the standard curl and the non-Abelian commutator:

$$F_{\mu\nu}^{(U(1))} = \partial_\mu A_\nu - \partial_\nu A_\mu + [A_\mu, A_\nu] \quad [\text{L29}] \quad (31)$$

By applying the Duan-Ge topological decomposition [27], we can extract the $U(1)$ electric current by taking the dual divergence of the field strength:

$$J^\mu = \epsilon^{\mu\nu\rho\sigma} \partial_\nu F_{\rho\sigma} \quad [\text{L30}] \quad (32)$$

Since the $\text{Spin}(4, \mathbb{C})$ universe is a smooth manifold, the n -dimensional Clairaut's theorem forces the partial derivatives to commute [14]. This forces the divergence of the current to vanish, proving charge conservation:

$$\partial_\mu J^\mu = 0 \quad [\text{L31}] \quad (33)$$

A classical spinor field traditionally commutes, meaning it obeys bosonic statistics. As noted earlier, this has historically been a problem for classical theories. However, while the local spinor in CGD is a commuting field, CGD mathematically possesses a macroscopic defect that behaves differently. In 1983, Witten [28] demonstrated that topological solitons exhibit fermionic spin statistics. In $\text{SU}(3)$ geometries he used a Wess-Zumino term, but this term disappeared for $\text{SU}(2)$ geometries, such as the one used in CGD. For the $\text{SU}(2)$ case, we use the Finkelstein-Rubinstein theorem [29], which proves that an adiabatic 2π spatial rotation of a degree-1 $\text{SU}(2)$ soliton traces a non-trivial loop in configuration space ($\pi_4(\text{SU}(2)) = \mathbb{Z}_2$). This proves that macroscopic topological defects in CGD possess fermionic spin statistics:

$$\text{Weight}(U_{\text{rot}}) = -1. \quad [\text{L32}] \quad (34)$$

So far, we have only considered topological defects in the ASD sector. However, the math of CGD also supports defects that live exclusively within the SD sector. The SD sector generates the Urbantke metric, so a pure SD defect would possess mass and stress energy that curves spacetime.

In Equation 6, we showed that $\text{Spin}(4, \mathbb{C})$ strictly orthogonalizes the chiral halves, so an SD-only defect does not algebraically intersect with any particles in the ASD sector. Consequently, this defect cannot emit or absorb light, meaning an SD-only defect is dark matter:

$$F_{\mu\nu}^{(R)}(\mathcal{A} + \mathcal{A}_{\text{DM}}^{(L)}) = F_{\mu\nu}^{(R)}(\mathcal{A}). \quad [\text{L33}] \quad (35)$$

Furthermore, CGD predicts some specific properties of this dark matter. Since the SD sector is a non-Abelian $\text{SL}(2, \mathbb{C})$ gauge field, two dark matter defects $\mathcal{A}_1$ and $\mathcal{A}_2$ will react with each other. This is the core feature of a particular category of hypothetical dark matter particles known as self-interacting dark matter (SIDM) [30]:

$$F_{\mu\nu}^{(L)}(\mathcal{A}_1 + \mathcal{A}_2) = F_{\mu\nu}^{(L)}(\mathcal{A}_1) + F_{\mu\nu}^{(L)}(\mathcal{A}_2) + [\mathcal{A}_{1\mu}, \mathcal{A}_{2\nu}] + [\mathcal{A}_{2\mu}, \mathcal{A}_{1\nu}]. \quad [\text{L34}] \quad (36)$$

Because this matter does not emit light, it lacks the radiative cooling required to condense into stars. Furthermore, the geometric momentum exchange prevents it from settling into a dense galactic cusp, resolving a critical concern present in standard cold dark matter theories.

V. QUANTUM MECHANICS WITHOUT QUANTIZATION

In the previous section, we established that particles exist; in this section, we will explore how they behave. We build the spinor out of the curvature tensor F_{ab} using the γ matrices of tangent-space Clifford algebra.

$$\Psi = F_{ab}\gamma^a\gamma^b. \quad [\text{L35}] \quad (37)$$

When we apply the Dirac operator $\not{D}$ to the spinor, the Bianchi identity forces the antisymmetric terms to vanish and yield the Yang-mills current:

$$\not{D}\Psi = 2J_a\gamma^a. \quad [\text{L1}] \quad (38)$$

Now, we can set the macroscopic current to zero and recover the vacuum Dirac equation:

$$J_\mu = 0 \implies \not{D}\Psi = 0. \quad [\text{L36}] \quad (39)$$

Since CGD is a classical field theory for both gravity and particles, one of the biggest concerns is how it addresses Bell's Theorem. Bell concluded:

In a theory in which parameters are added to quantum mechanics to determine the results of individual measurements, without changing the statistical predictions, there must be a mechanism whereby the setting of one measuring device can influence the reading of another instrument, however remote. Moreover, the signal involved must propagate instantaneously, so that such a theory could not be Lorentz invariant. [31]

This is colloquially summarized as “No local, hidden-variables theory explains Quantum Mechanics.” This is a problem for a theory like CGD because classical theories must be local hidden-variable theories in order to satisfy Lorentz Invariance.

We established in Equation 8 that CGD has no classical equations of motion, so the local gauge field is underdetermined. In order to explore one of the possible solutions to the local geometry, we propose the following ansatz:

$$\mathcal{A}_\mu^{\text{tube}}(x) = \begin{cases} 0 & \mu = 0 \\ i\sigma_3 & \mu = 1 \\ i\sigma_1 & \mu = 2 \\ i\sigma_2 & \mu = 3 \end{cases} \quad [\text{L37}] \quad (40)$$

$$\exists \gamma(t), \quad \gamma(0) = x, \quad \gamma(1) = y \quad [\text{L38}] \quad (41)$$

This “flux tube frame” defines entanglement for points x and y by connecting them with a curve where gauge rotations are allowed at every point of the curve $\gamma(t)$. We verify this is a valid physical solution in `CGD.Quantum.fluxTubeIsMinimal`^[L39] by proving that it satisfies the minimal-energy boundary condition for a static magnetic field. Since this construct is 1-dimensional, Equation 26 shows that the determinant collapses and the metric vanishes within the tube, giving:

$$\det(g_{\mu\nu}) = 0 \quad \text{along the ER=EPR bridge} \quad \text{[L40]} \quad (42)$$

This acts as a topological bridge.

Now, we can evaluate the path integral along the flux tube, and find that we recover the trigonometric coefficients of quantum state vectors

$$\mathcal{P} \exp \left(\int \mathcal{A}_\mu dx^\mu \right) = \cos(\theta) \cdot I + i \sin(\theta) \cdot M \quad \text{[L41]} \quad (43)$$

We also find that since the core of the flux tube is commutative, the phase is conserved and matches the CHSH Tsirelson bound:

$$[\text{Re}(\text{CHSH})]^2 \leq 8. \quad \text{[L42]} \quad (44)$$

The result in Equation 44 demonstrates that contrary to conventional belief, a purely classical theory can satisfy Bell’s inequalities. We did not invalidate any part of Bell’s theorem. Since $\det(g) = 0$, the metric inside the flux tube vanishes and the geometric distance between the particles is zero, so the interaction is strictly local in the topological sense even though it appears to be non-local to an observer.

While this resolves the most obvious concern about proposing CGD as an ontology for quantum mechanics, entanglement correlation is not the only concern. Since we proved in Equation 8 that CGD does not possess a classical Hamiltonian, the Dirac operator in Equation 38 is not sufficient by itself to explain the complex phase rotation of quantum time-evolution. For this, we need to address “the problem of time” [32].

We will again use the flux tube ansatz and note that the path-integral of a connection $\mathcal{A}$ over distance t is the matrix exponential of the field multiplied by the distance ($e^{t\mathcal{A}}$). The spatial holonomy along the U(1) core of the flux tube mathematically mirrors e^{-iHt} , meaning time evolution is just relational geometric phase:

$$\mathcal{P} \exp \left(\int \mathcal{A}_1 dx^1 \right) = \exp(-iHt). \quad \text{[L43]} \quad (45)$$

The next concern we will address is how we obtain a “quantum” from a classical theory. Canonical quantization is a mathematical process that turns a classical theory into a quantum theory,

and by treating classical quantities such as position or momentum as operators with the axiomatic non-commutation property $[\hat{q}, \hat{p}] = i\hbar$, physicists can obtain probability waves and discrete energy levels. In CGD, we do not have the non-commutation property, so if quanta exist, they must emerge from the topology. However, since the background of the $\text{Spin}(4, \mathbb{C})$ is a macroscopic $\text{SU}(2)$ field, the $\text{SU}(2)$ group manifold is a 3-dimensional sphere S^3 . If we take a topological defect ($F_{\mu\nu}^{(R)} \neq 0$), we can enclose it in another S^3 sphere at spatial infinity. At this boundary, we have a mathematical mapping $S^3 \rightarrow S^3$. To obtain quantization, we use the Cartan-Maurer Degree Theorem [11], which shows that it is only possible to wrap a sphere with another sphere an integer number of times without tearing the manifold ($\pi_3(\text{SU}(2)) = \mathbb{Z}$).

Next, we consider the Pontryagin density that we proved is the unique CGD Lagrangian in Equation 7. Stokes' theorem proves that the volume integral of a derivative is equal to the value at its boundary, so when we integrate the 4-dimensional action, we find that that action possessed by the particle is quantized and the Cartan-Maurer integral evaluates to [11, 25]:

$$S_{\text{top}} \in \{-1, 1\}. \quad [\text{L44}] \quad (46)$$

In CGD, the “quantum” is the geometric winding number for a topological defect.

Finally, we will address the measurement problem by using statistical mechanics. In Section IV we proved that particles are $\text{SU}(2)$ defects. The geometry of every point in $\text{SU}(2)$ is a 3-sphere (S^3) which we can decompose into a 2-sphere (S^2) where every point on the sphere is attached to a 1-dimensional circle (a $\text{U}(1)$ fiber). This decomposition matches laboratory equipment, because a detector such as a magnet exists in 3-dimensional space, so it can only measure values on a S^2 sphere. The actual $\text{U}(1)$ phase of a particle is a value on the $\text{U}(1)$ fiber. When we use the Liouville volume-preservation theorem [33] to calculate the phase-space volume of the Hopf Fibration, we find

$$P = \frac{\int_{\theta}^{\pi} d\mathcal{V}_{\text{Hopf}}}{\int_0^{\pi} d\mathcal{V}_{\text{Hopf}}} = 1 - \frac{M}{E_0}. \quad [\text{L45}] \quad (47)$$

This is exactly the Born Rule.

VI. PHENOMENOLOGICAL IMPLICATIONS: AXIAL-VECTOR CONDENSATE

The majority of this whitepaper is focused on showing that $\text{Spin}(4, \mathbb{C})$ reproduces various aspects of standard physics, but in this section we will explore a consequence of CGD that is drastically different from the Standard Model. For this investigation, we first consider the empty vacuum where

the anti-self-dual (ASD) sector's curvature is zero. If the self-dual (SD) sector were structurally equivalent to the ASD sector, then both sectors would be zero and thus the field would trivially commute with itself, meaning all fields would point in the same direction and the universe would only be capable of creating a 1-dimensional object (Equation 26). This would contradict the CGD Axiom 2 asserting that the universe has volume, so we conclude that the universe must have chirality:

$$\left(\forall \mu, \nu, F_{\mu\nu}^{(R)}(x) = 0\right) \implies \mathcal{A}^{(L)} \neq \mathcal{A}^{(R)}. \quad [\text{L46}] \quad (48)$$

Furthermore, by looking at the definition of an axial field, $\mathcal{A}_{\text{axial}} = \frac{1}{2}(\mathcal{A}^{(L)} - \mathcal{A}^{(R)})$, we see that since chirality exists throughout the vacuum, a non-zero axial background condensate must exist:

$$\left(\forall \mu, \nu, F_{\mu\nu}^{(R)}(x) = 0\right) \implies \exists y, \mu \text{ s.t. } \mathcal{A}_{\mu}^{\text{axial}}(y) \neq 0. \quad [\text{L47}] \quad (49)$$

The SD and ASD chiral sectors decompose into symmetric vector ($\mathcal{V}$) and axial ($\mathcal{A}^{\text{axial}}$) background fields:

$$\mathcal{A}_{\mu}^{(L)}(x) = \mathcal{V}_{\mu}(x) + \mathcal{A}_{\mu}^{\text{axial}}(x) \quad [\text{L48}] \quad (50)$$

$$\mathcal{A}_{\mu}^{(R)}(x) = \mathcal{V}_{\mu}(x) - \mathcal{A}_{\mu}^{\text{axial}}(x). \quad [\text{L49}] \quad (51)$$

This property is the source of the parity asymmetry of the vacuum.

When we evaluate the gauge-covariant derivative of the fermion state Ψ in the SD sector, we find that the particle interacts with both the vector and axial background fields:

$$D_{\mu}^{(L)}\Psi = \partial_{\mu}\Psi + \mathcal{V}_{\mu}\Psi + \mathcal{A}_{\mu}^{\text{axial}}\Psi. \quad [\text{L50}] \quad (52)$$

This shows why the universe cares about the chirality of particles. In contrast, the Standard Model added the bespoke term $\frac{1}{2}(1 - \gamma^5)$ [34] into the weak interaction current portion of its Lagrangian to describe the chirality discovered in Wu *et al.* [35].

We can further characterize the condensate by observing that both the SD and ASD Lie algebras are restricted to $\mathfrak{sl}(2, \mathbb{C})$, so they are traceless matrices. Since a traceless matrix corresponds with a vector field, the condensate is isovector

$$\text{Tr}(\mathcal{A}_{\mu}^{\text{axial}}(x)) = 0. \quad [\text{L51}] \quad (53)$$

We can also consider what happens when we invert the parity of the field $\mathcal{P}(\mathcal{A}_{\text{axial}}) = \mathcal{P}\left[\frac{1}{2}(\mathcal{A}^{(L)} - \mathcal{A}^{(R)})\right]$.

This simplifies to $\mathcal{P}(\mathcal{A}_{\text{axial}}) = -\mathcal{A}_{\text{axial}}$, which means

$$\mathcal{A}_{\mu}^{\text{axial}}(\mathcal{P}x) = -\mathcal{A}_{\mu}^{\text{axial}}(x). \quad [\text{L52}] \quad (54)$$

All together, we can characterize the quantum numbers of the condensate as $I^G(J^{PC}) = 1^-(1^{++})$.

This mathematical necessity of CGD is the crux of why CGD makes novel predictions; in the Standard Model, no axial-vector condensate exists.

At this point, we will stray away from theoretical formalization to look at phenomenological implication of an isovector axial-vector condensate. Only one particle family matches the axial-vector isovector quantum numbers: the a_1 meson [36], so if we are going to guess that the characteristic particle of the CGD condensate has already been discovered, this is our only choice.

Now that we have characterized the CGD condensate, let us consider how an a_1 condensate would impact the proton radius puzzle.

Pohl *et al.* [37] showed that measuring the proton radius using muonic hydrogen differed by 5.0 standard deviations compared to the atomic hydrogen measurements. Miller [38] argues that newer atomic hydrogen experiments have resolved the discrepancy because Maisenbacher *et al.* [39] used the 2S-6P transition of atomic hydrogen to obtain the measurement of $r_p = 0.8406(15)$ fm, which is consistent with muonic hydrogen measurements. However, this brings up the question of why older atomic hydrogen experiments measured a different value. Physicists have not identified any technical problems for all of the older atomic hydrogen experiments, so in order to declare the proton radius solved, the Standard Model must ignore this data.

Based on QFT, if we bombard a proton with an electron, we need to compute the probability amplitude that the two particles will interact. In the presence of an axial-vector condensate, this amplitude is the product of the bare interaction $G_{Dipole}(Q^2, r_{core})$ (the probability of interacting due to the electron interacting with the proton core) and the vacuum corrections $1 - F_{Axial}(Q^2, \kappa_E)$ (which is the influence of the a_1 condensate):

$$G_E(Q^2) = G_{Dipole}(Q^2, r_{core}) \times (1 - F_{Axial}(Q^2, \kappa_E)). \quad (55)$$

We can then use the standard phenomenological model of Vector Meson Dominance (VMD) [40], where we observe that when a photon interacts with a proton, it often acts like a short-lived meson. We describe the core interaction using

$$G_{Dipole}(Q^2, r) = \left(1 + \frac{Q^2}{12/(r/\hbar c)^2}\right)^{-2}. \quad (56)$$

This leaves us needing to describe the influence of the axial-vector condensate in the $F_{Axial}(Q^2, \kappa_E)$ equation. This is the product of three components: the massive propagator $Q^2/(Q^2 + m_{a_1}^2)$ associated with the probability amplitude that a virtual photon will resolve into an a_1 meson state [41], the loop screening term $\ln(1 + Q^2/(m_{a_1}^2 - m_\rho^2))$ which describes the mass gap between the vector

environment of the proton and the axial-vector state of the a_1 condensate [42, 43], and finally the Weinberg mixing angle [44] which describes the strength at which the vector and axial-vector environments mix. Putting this together we have

$$F_{Axial}(Q^2, \kappa_E) = \frac{Q^2}{Q^2 + m_{a_1}^2} \times \ln \left(1 + \frac{Q^2}{m_{a_1}^2 - m_\rho^2} \right) \times \left(\frac{m_\rho}{m_{a_1}} \right)^2. \quad (57)$$

Notice that the only free parameter in this model is r . However, this hides uncertainty around the a_1 meson particle family, because Navas *et al.* [36] lists multiple resonances for the a_1 and ρ mesons. In order to address this issue, in Figure 1 and Table I we analyze an additional model where we treat m_{a_1} and m_ρ as free parameters.

Another nuance of this proton form factor model is that it requires a relatively high momentum transfer Q^2 in order for $F_{Axial}(Q^2, \kappa_E)$ to have any value other than 0. In order to test this model in a regime where the effects of the axial-vector condensate are visible, we will need to find a dataset with a high momentum transfer. For this, we will use the Jefferson Lab high GeV^2 polarization transfer experiments described in Punjabi *et al.* [45], Gayou *et al.* [46], Puckett *et al.* [47, 48] and aggregated by Ye *et al.* [49].

Model	Parameters Extracted		Total χ^2
CGD Topology (Fixed Mesons)	1	0.8373	20.7
CGD Topology (Free Mesons)	3	0.8440	17.8
Ye 2018 Global	13	0.879*	36.0

Table I. The statistical analysis for the models shown in Figure 1. The CGD Topology (Free Mesons) model extracted the values $m_{a_1} = 1.2002 \text{ GeV}$, $m_\rho = 0.7497 \text{ GeV}$, and $r_p = 0.8440 \text{ fm}$. The CGD Topology (Fixed Mesons) used the values $m_{a_1} = 1.209 \text{ GeV}$, $m_\rho = 0.775$ reported in Navas *et al.* [36], and extracted the value $r_p = 0.8373 \text{ fm}$.

It's important to note that despite the statistical comparisons shown in Table I, we can only really reach one conclusion from this: CGD's axial condensate requirement is not easily falsified. We only analyzed a small subset of the available proton radius data, and we deliberately avoided looking at low GeV^2 momentum transfers. Furthermore, while our comparison against the Ye 2018 Global model is the best comparison we could find as a proxy for the Standard Model, it is somewhat disingenuous. Ye *et al.* [49] built this parameterization from all of the data available at the time, not just the high GeV^2 polarization transfer from JLab Hall C.

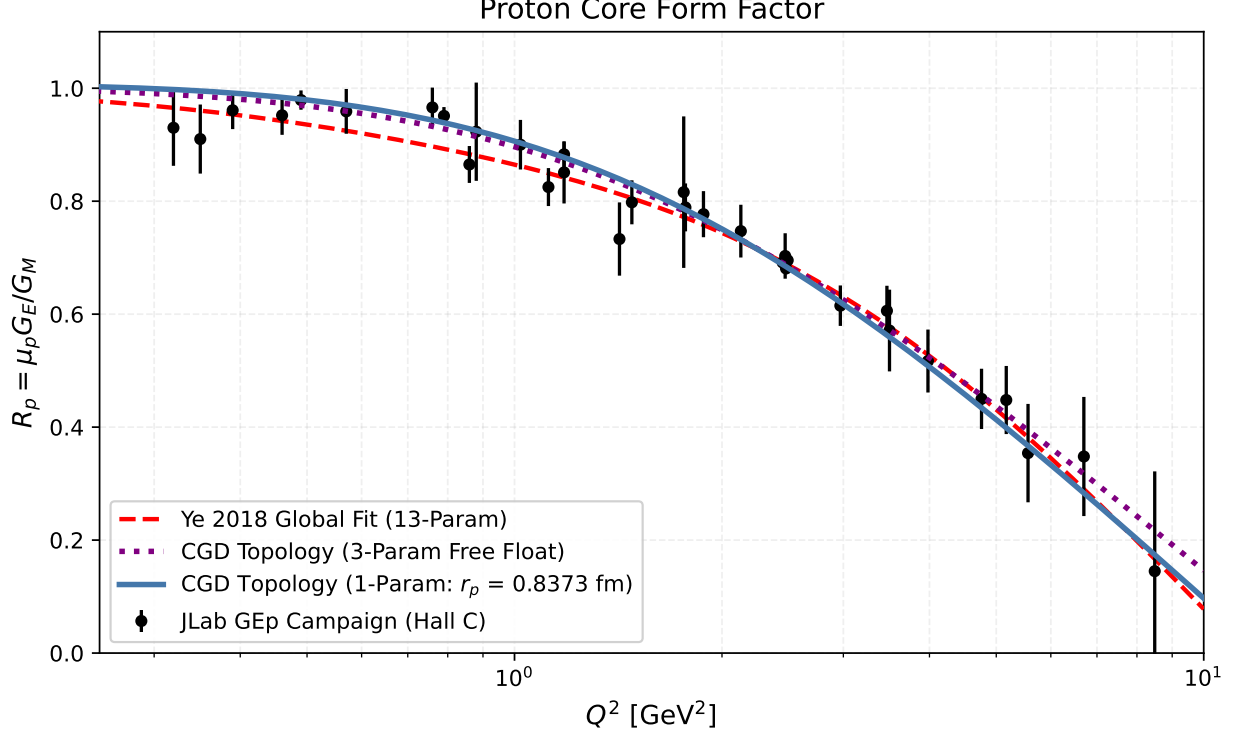


Figure 1. The proton radius factor form factor models. Ye *et al.* [49] is a global parameterization that represents a best-fit for atomic hydrogen based experiments. The CGD 1-Param model leaves only the proton radius r_p as an unknown value, while the CGD 3-Param model additionally treats the meson masses m_{a_1} and m_ρ as unknown values. The statistical analysis of these models is shown in Table I.

VII. CONCLUSION

We have presented a theory that is more notable for what it does not do than for what it accomplishes. CGD does not assume an independent background metric, it does not assume extra dimensions, and perhaps most importantly, it does not assume fundamental quantization of particles and forces. Starting with $\text{Spin}(4, \mathbb{C})$ and the pure-connection research pioneered by Urbantke [7], Ashtekar [5], Capovilla *et al.* [8], Krasnov [4] (and many others!) we presented a theory, formalized in Lean4, that demonstrates that a classical theory is a viable fundamental ontology. The classical math already contains the mass gap, entanglement bounds, and gravity.

However, math doesn't dictate reality; data dictates reality. So far, this research has only made casual contact with hard data when we derived the necessity of an axial-vector condensate and performed a sanity check against a single proton radius puzzle dataset. While we make the claim that this classical theory is an alternative to quantized physics, only hard data will determine if it is any good at describing the universe we live in.

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Appendix A: Formal Code Summary

FORMAL VERIFICATION INDEX

The equations annotated with [Ln] in the main text have been formally verified in Lean 4. Click the theorem name to jump to the code signature, or reference the line number in the Appendix.

[L1]	CGD.Quantum.Dirac.geometricDiracEmergence	Line 1557
[L2]	CGD.Axioms.Universe	Line 76
[L3]	CGD.Axioms.MacroscopicVolume	Line 54
[L4]	CGD.Axioms.UnimodularVacuum	Line 133
[L5]	CGD.Foundations.algebraicChiralDecomposition	Line 296
[L6]	CGD.Foundations.topologicalLagrangianUniqueness	Line 349
[L7]	CGD.Foundations.topologicalActionVariationZero	Line 383
[L8]	CGD.Foundations.kinematicBianchiIdentity	Line 266
[L9]	CGD.Particles.kinematicChargeConservation	Line 1016

[L10]	CGD.Gravity.macroscopicVacuumEmergence	Line 480
[L11]	CGD.Gravity.macroscopicCosmologicalEmergence.....	Line 440
[L12]	CGD.Gravity.emergentStressEnergyConservation.....	Line 777
[L13]	CGD.Gravity.machianTopologicalDefectMotion.....	Line 668
[L14]	CGD.Gravity.exactLorentzianL.....	Line 577
[L15]	CGD.Gravity.F_origin_val_eq_map.....	Line 587
[L16]	CGD.Gravity.CDJ_constraint_holds.....	Line 526
[L17]	CGD.Gravity.dynamicExactLorentzianSolution.....	Line 610
[L18]	CGD.Gravity.ExactSolutions.IsType0Form.....	Line 511
[L19]	CGD.Gravity.ExactSolutions.flrw_satisfies_reality.....	Line 552
[L20]	CGD.Gravity.ExactSolutions.solitonK.....	Line 655
[L21]	CGD.Gravity.ExactSolutions.solitonAnsatzVal.....	Line 637
[L22]	CGD.Gravity.ExactSolutions.dynamicExactSolitonSolution.....	Line 628
[L23]	CGD.Particles.dynamicMatterExistence.....	Line 1099
[L24]	CGD.Particles.kinematicSingleColorDegeneracy.....	Line 863
[L25]	CGD.Particles.kinematicThreeColorRequirement.....	Line 872
[L26]	CGD.Particles.kinematicTopologicalStability.....	Line 1194
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[L28]	CGD.Particles.topologicalMassGap.....	Line 1075
[L29]	CGD.Particles.abelianFieldStrength.....	Line 995
[L30]	CGD.Particles.emergentElectricCurrent.....	Line 1004
[L31]	CGD.Particles.topologicalChargeConservation.....	Line 1028
[L32]	CGD.Particles.kinematicMacroscopicFermion.....	Line 1153
[L33]	CGD.Particles.kinematicDarkMatterDecoupling.....	Line 889
[L34]	CGD.Particles.kinematicDarkMatterSelfInteraction.....	Line 905
[L35]	CGD.Quantum.Dirac.geometricSpinor.....	Line 1537

[L36]	CGD.Quantum.Dirac.vacuumDiracEquation	Line 1572
[L37]	CGD.Quantum.fluxTubeFrame	Line 1509
[L38]	CGD.Quantum.areEntangled	Line 1494
[L39]	CGD.Quantum.fluxTubeIsMinimal	Line 1602
[L40]	CGD.Quantum.kinematicEntanglementWormhole	Line 1588
[L41]	CGD.Quantum.fluxTubeHolonomyEvaluation	Line 1632
[L42]	CGD.Quantum.geometricHolonomyTsirelsonBound	Line 1697
[L43]	CGD.Quantum.Holonomy.relationalTimeEmergence	Line 1717
[L44]	CGD.Quantum.kinematicActionQuantization	Line 1464
[L45]	CGD.Quantum.Measurement.kinematicBornRuleEquivalence	Line 1757
[L46]	CGD.Phenomenology.macroscopicVolumeImpliesChirality	Line 1333
[L47]	CGD.Phenomenology.macroscopicVolumeImpliesAxialCondensate	Line 1304
[L48]	CGD.Phenomenology.kinematicLeftHandedCoupling	Line 1268
[L49]	CGD.Phenomenology.kinematicRightHandedCoupling	Line 1294
[L50]	CGD.Phenomenology.kinematicLeftHandedPhaseShift	Line 1282
[L51]	CGD.Phenomenology.axialIsIsovector	Line 1239
[L52]	CGD.Phenomenology.axialIsParityOdd	Line 1247

```

1  -- MODULE: CGD.AntiSelfDualSector.Decoupling
2  -----
3  /--
4  This theorem asserts that the topological vacuum action is completely
5  insensitive to the Anti-Self-Dual gauge field connection, securing
6  the chiral asymmetry of the geometry.
7  -/
8  @[litlib_track "Anti-Self-Dual Matter Decoupling"]
9  theorem CGD.AntiSelfDualSector.algebraicAntiSelfDualSectorDecoupling (pu : PhysicalUniverse)
10    (A_R_alt : Sl2cGaugeField) (x : SpacetimePoint) :
11    actionVacuum (fun mu nu => curvature (fun m p => pu.toUniverse.spin4c_connection m p) mu nu x) =

```

```

12  actionVacuum (fun mu nu => curvature (fun m p => embedSelfDual (pu.toUniverse.sd_sector m p) +
    ↪ embedAntiSelfDual (A_R_alt m p)) mu nu x)
13
14
15  -- MODULE: CGD.AntiSelfDualSector.SelfInteracting
16  -----
17  /--
18  This theorem proves that non-commuting SU(2) fields (which correspond to non-trivial matter) natively expand
    ↪ into non-zero topological density traces without needing a background metric.
19  -/
20  @[litlib_track "Anti-Self-Dual Matter Trace Constraint"]
21  theorem CGD.AntiSelfDualSector.kinematicSIDMTrace (pu : PhysicalUniverse)
22    (x : SpacetimePoint) (μ v : Fin 4)
23    (h_anti_self_dual_su2 : ∀ m p, isSu2 (pu.toUniverse.asd_sector m p).val)
24    (h_comm : ((pu.toUniverse.asd_sector μ x).val * (pu.toUniverse.asd_sector v x).val -
    ↪ (pu.toUniverse.asd_sector v x).val * (pu.toUniverse.asd_sector μ x).val) ≠ 0) :
25    Matrix.trace (((pu.toUniverse.asd_sector μ x).val * (pu.toUniverse.asd_sector v x).val -
    ↪ (pu.toUniverse.asd_sector v x).val * (pu.toUniverse.asd_sector μ x).val) *
26      ((pu.toUniverse.asd_sector μ x).val * (pu.toUniverse.asd_sector v x).val -
    ↪ (pu.toUniverse.asd_sector v x).val * (pu.toUniverse.asd_sector μ x).val)) ≠ 0
27
28
29  -- MODULE: CGD.AntiSelfDualSector.VacuumDegeneracy
30  -----
31  /--
32  Demonstrates the topological collapse of the Anti-Self-Dual vacuum.
33  If the ASD sector of the universe remains in the trivial vacuum state (zero gauge field),
34  its resulting emergent Urbantke metric mathematically degenerates to a zero determinant.
35  This strictly enforces that macroscopic spacetime can only emerge from non-trivial,
36  non-Abelian gauge condensates, cementing the geometric chirality of the universe.
37  -/
38  @[litlib_track "Topological Collapse of the Anti-Self-Dual Vacuum"]
39  theorem CGD.AntiSelfDualSector.kinematicAsdVacuumDegeneracy (pu : PhysicalUniverse) :
40    pu.toUniverse.asd_sector.val = (fun _ _ => (0 : SL2C)) →
41    ∀ x, (urbantkeMetric (fun m n => curvatureSL2c pu.toUniverse.asd_sector m n x)).det = 0
42
43
44  -- MODULE: CGD.Axioms.MacroscopicVolume
45  -----
46  /--

```

```

47  Axiom II: Macroscopic Existence (The "Non-Empty" Condition)
48  We constrain the physics by demanding that the emergent metric is
49  non-degenerate over an open topological bulk. By defining the bulk as
50  an `IsOpen` set, we safely allow the volume to mathematically crash to zero
51  on its boundaries (the Big Bang) or at topological punctures (derived matter defects),
52  without destroying the extended macroscopic spacetime.
53  -/
54  class CGD.Axioms.MacroscopicVolume (u : Universe) (bulk : Set SpacetimePoint) : Prop where
55    h_bulk_open : IsOpen bulk
56    h_bulk_nonempty : Set.Nonempty bulk
57    volume_exists :  $\forall x \in \text{bulk},$ 
58      (urbantkeMetric (fun  $\mu v \Rightarrow \text{curvatureSl2c } u.\text{sd\_sector } \mu v x$ )).det  $\neq 0$ 
59
60
61  -- MODULE: CGD.Axioms.Ontology
62  -----
63  /--
64  A smooth  $SL(2, C)$  gauge field.
65  As an abstract Lie algebra, it possesses no native spacetime chirality.
66  -/
67  structure CGD.Axioms.Sl2cGaugeField where
68    val : Fin 4  $\rightarrow$  CGD.Foundations.SpacetimePoint  $\rightarrow$  SL2C
69    is_smooth :  $\forall \mu i j, \text{ContDiff } \mathbb{R} \top$  (fun  $x \Rightarrow (\text{val } \mu x).\text{val } i j$ )
70
71  /--
72  The Core Ontology: The Universe is a macroscopic  $Spin(4, C)$  connection.
73  We define it strictly as a 4x4 continuous gauge field constrained to the
74  6-dimensional  $Spin(4, C)$  Lie algebra.
75  -/
76  structure CGD.Axioms.Universe where
77    /-- The unified 4x4 Chiral Dirac field -/
78    val : Fin 4  $\rightarrow$  CGD.Foundations.SpacetimePoint  $\rightarrow$  ChiralM
79
80    /-- The geometric constraint forcing the 16D  $GL(4, C)$  field into the 6D  $Spin(4, C)$  algebra -/
81    is_spin4c :  $\forall \mu x, \text{val } \mu x =$ 
82      embedSelfDual (chiralProject (val  $\mu x$ )).self_dual +
83      embedAntiSelfDual (chiralProject (val  $\mu x$ )).anti_self_dual
84
85    /-- Smoothness conditions on the topological projections -/
86    sd_is_smooth :  $\forall \mu i j, \text{ContDiff } \mathbb{R} \top$  (fun  $x \Rightarrow (\text{chiralProject } (\text{val } \mu x)).\text{self\_dual}.\text{val } i j$ )

```

```

87   asd_is_smooth : ∀ mu i j, ContDiff ℝ T (fun x => (chiralProject (val mu x)).anti_self_dual.val i j)
88
89   /-- Derives the anti-self-dual SL(2,C) sector from the unified Spin(4,C) field. -/
90   noncomputable def Universe.asd_sector (u : Universe) : Sl2cGaugeField :=
91     (fun mu x => (chiralProject (u.val mu x)).anti_self_dual, u.asd_is_smooth)
92
93   /-- Derives the self-dual SL(2,C) sector from the unified Spin(4,C) field. -/
94   noncomputable def Universe.sd_sector (u : Universe) : Sl2cGaugeField :=
95     (fun mu x => (chiralProject (u.val mu x)).self_dual, u.sd_is_smooth)
96
97   /--
98   The Spin(4, C) 4x4 Dirac Representation.
99   This definition evaluates to the raw 4x4 field, but its underlying
100  algebra guarantees it matches the spacetime chiral sum exactly.
101  -/
102  noncomputable def Universe.spin4c_connection (u : Universe) (mu : Fin 4) (x :
    ↪ CGD.Foundations.SpacetimePoint) : ChiralM :=
103    u.val mu x
104
105
106  -- MODULE: CGD.Axioms.PhysicalUniverse
107  -----
108  /--
109  The Comprehensive Physical Reality.
110  Bundles the foundational ontology with the topological bulk and its required physical mixins.
111  Matter is never inputted as an assumption; it exists entirely as the topological complement
112  of the macroscopic bulk.
113  -/
114  structure CGD.Axioms.PhysicalUniverse where
115    toUniverse : Universe
116    bulk : Set SpacetimePoint
117    has_volume : MacroscopicVolume toUniverse bulk
118    has_vacuum : UnimodularVacuum toUniverse bulk
119
120
121  -- MODULE: CGD.Axioms.UnimodularVacuum
122  -----
123  /--
124  Axiom III: Unimodular Vacuum Isotropy (Unbroken Gauge Symmetry)
125  Because the `bulk` is defined as the region where  $\det g \neq 0$ , it natively

```

```

126 excludes all topological defects. Therefore, the entire `bulk` is the pure vacuum.
127
128 This constraint mathematically enforces that the pure vacuum spacetime must not
129 spontaneously break its internal SU(2) unimodular symmetry. The only symmetric
130 tensor invariant under local SO(3) rotations is a multiple of the identity matrix.
131 This natively bounds the field to generate an Einstein manifold.
132 -/
133 class CGD.Axioms.UnimodularVacuum (u : Universe) (bulk : Set SpacetimePoint) : Prop where
134   vacuum_equipartition :  $\forall x \in \text{bulk},$ 
135     let F_adj := fun m n => cgdAdjointCurvature u m n x;
136     let Sigma :=  $\sum \mu : \text{Fin } 4, \sum \nu : \text{Fin } 4, \sum \rho : \text{Fin } 4, \sum \sigma : \text{Fin } 4,$ 
137       epsilon4  $\mu \nu \rho \sigma \cdot (F\_adj \mu \nu * F\_adj \rho \sigma);$ 
138     Sigma = (Matrix.trace Sigma / 3)  $\cdot (1 : \text{Matrix } (\text{Fin } 3) (\text{Fin } 3) \mathbb{C})$ 
139
140
141 -- MODULE: CGD.Cosmology.BigBang
142 -----
143 /--
144 A kinematic boundary evaluation. This demonstrates that if a phase region is
145 artificially constrained to a fully 4D symmetric SO(4) state, the resulting
146 Urbantke metric takes a purely Euclidean form (c * 1).
147
148 Rather than representing a dynamic cosmological evolution, this acts as a
149 mathematical sanity check: it shows that highly symmetric ground-state toy models
150 evaluate to non-singular Euclidean spaces rather than generating mathematical
151 singularities in the metric.
152 -/
153 @[litlib_track "Big Bang Singularity Resolution (Euclidean Bounce)"]
154 theorem CGD.Cosmology.kinematicBigBang (pu : PhysicalUniverse) (phaseRegion : Set SpacetimePoint)
155   (h_subset : phaseRegion  $\subseteq$  pu.bulk)
156   (h_symm :  $\forall x \in \text{phaseRegion}, \text{isFully4DSymmetric } (\text{fun } \mu \nu \Rightarrow \text{curvatureSl2c } \text{pu.toUniverse.sd\_sector } \mu \nu$ 
157      $\hookrightarrow x)) :$ 
158
159    $\forall x \in \text{phaseRegion}, \exists c : \text{Complex}, c \neq 0 \wedge \text{urbantkeMetric } (\text{fun } \mu \nu \Rightarrow \text{curvatureSl2c}$ 
160      $\hookrightarrow \text{pu.toUniverse.sd\_sector } \mu \nu x) = c \cdot 1$ 
161
162 /--
163 In pure connection gravity, 4D spacetime volume geometrically requires time-evolution. A completely static
164  $\hookrightarrow$  universe (where electric/temporal components of the field strength are zero) cannot sustain a macroscopic
165  $\hookrightarrow$  4D metric and topologically collapses, forcing the metric determinant to zero.
166 -/

```

```

162 @[litlib_track "Static Universe Degeneracy"]
163 theorem CGD.Cosmology.kinematicStaticUniverseDegeneracy (pu : PhysicalUniverse) :
164   isStaticUniverse pu.toUniverse →
165   ∀ x, (urbantkeMetric (fun mu nu => curvatureSl2c pu.toUniverse.sd_sector mu nu x)).det = 0
166
167
168 -- MODULE: CGD.Cosmology.Definitions
169 -----
170 /--
171   Time Emergence: A mathematically rigorous definition of pre-time SO(4) Euclidean symmetry.
172   Instead of setting time components to zero, we define the symmetric state as a strictly
173   self-dual configuration.
174 -/
175 def CGD.Cosmology.isFully4DSymmetric (F : Fin 4 → Fin 4 → SL2C) : Prop :=
176   ∀ mu nu, F mu nu = ∑ rho : Fin 4, ∑ sigma : Fin 4,
177     ((1/2 : ℂ) * epsilon4 mu nu rho sigma) • F rho sigma
178
179 /-- Defines the geometric relationship of a parity-inverted curvature tensor. -/
180 def CGD.Cosmology.isParityInvertedTensor (F P_F : Fin 4 → Fin 4 → SL2C) (x : SpacetimePoint) : Prop :=
181   (∀ i : Fin 4, i ≠ 0 → (P_F 0 i).val = -(F 0 i).val ∧ (P_F i 0).val = -(F i 0).val) ∧
182   (∀ i j : Fin 4, i ≠ 0 → j ≠ 0 → (P_F i j).val = (F i j).val) ∧
183   ((P_F 0 0).val = (F 0 0).val)
184
185 /--
186   Identifies a static universe where all electric (temporal) curvature components are identically zero.
187 -/
188 def CGD.Cosmology.isStaticUniverse (u : Universe) : Prop :=
189   ∀ x j, curvatureSl2c u.sd_sector 0 j x = 0
190
191
192 -- MODULE: CGD.Cosmology.ParityInversion
193 -----
194 /--
195   Demonstrates that the fully antisymmetric topological density (Pontryagin action) natively obeys geometric
196   ⇔ CPT symmetry. A passive time-reversal transformation (negating the temporal components of the field
197   ⇔ strength tensor) strictly inverts the sign of the topological charge.
198 -/
199 @[litlib_track "Geometric Parity Inversion"]
200 theorem CGD.Cosmology.kinematicParityInversion (pu : PhysicalUniverse) :
201   ∀ (x : SpacetimePoint) (P_F : Fin 4 → Fin 4 → SL2C),

```



```

200     isParityInvertedTensor (fun m n => curvatureSl2c pu.toUniverse.sd_sector m n x) P_F x →
201     pontryaginDensity P_F = - pontryaginDensity (fun mu nu => curvatureSl2c pu.toUniverse.sd_sector mu nu x)
202
203 noncomputable def CGD.Cosmology.pontryaginDensity (F : Fin 4 → Fin 4 → SL2C) : Complex :=
204     ∑ μ, ∑ ν, ∑ ρ, ∑ σ, epsilon4 μ ν ρ σ * Matrix.trace ((F μ ν).val * (F ρ σ).val)
205
206
207 -- MODULE: CGD.Cosmology.ScaleBreaking
208 -----
209 /--
210 The conformal scaling property of the Urbantke metric. A scale transformation of the field strength tensor ( $F$ 
211  $\hookrightarrow \rightarrow \lambda^2 F$ ) results in a strict  $\lambda^{24}$  scaling of the emergent metric determinant, demonstrating that the
212  $\hookrightarrow$  macroscopic metric possesses a conformal weight of 6.
213 -/
214 @[litlib_track "Kinematic Classical Scale Breaking"]
215 theorem CGD.Cosmology.kinematicClassicalScaleBreaking (F : Fin 4 → Fin 4 → SL2C) (lambda_scale : C) :
216     let F_scaled := fun μ ν => toSl2c (lambda_scale^2 • (F μ ν).val);
217     (∀ μ ν, urbantkeMetric F_scaled μ ν = lambda_scale^6 * urbantkeMetric F μ ν) ∧
218     (urbantkeMetric F_scaled).det = lambda_scale^24 * (urbantkeMetric F).det
219
220 -- MODULE: CGD.Foundations.Action
221 -----
222 noncomputable def CGD.Foundations.actionAntiSelfDual (F : Fin 4 → Fin 4 → ChiralM) : Complex :=
223     let F_R := fun mu nu => (chiralProject (F mu nu)).anti_self_dual
224     (-0.5 : Complex) * (
225         ∑ mu : Fin 4, ∑ nu : Fin 4, ∑ rho : Fin 4, ∑ sigma : Fin 4,
226         CGD.Gravity.epsilon4 mu nu rho sigma * Matrix.trace ((F_R mu nu).val * (F_R rho sigma).val)
227     )
228
229 noncomputable def CGD.Foundations.actionVacuum (F : Fin 4 → Fin 4 → ChiralM) : Complex :=
230     let F_L := fun mu nu => (chiralProject (F mu nu)).self_dual
231     (-0.5 : Complex) * (
232         ∑ mu : Fin 4, ∑ nu : Fin 4, ∑ rho : Fin 4, ∑ sigma : Fin 4,
233         CGD.Gravity.epsilon4 mu nu rho sigma * Matrix.trace ((F_L mu nu).val * (F_L rho sigma).val)
234     )
235 /--
236 A physically valid universe variation is mathematically constrained:
237 1. It must be a smooth path through configuration space.

```

```

238 2. It must have uniform compact support (vanish at spatial infinity consistently across the perturbation).
239 3. The underlying physical Lagrangian density must be Lebesgue Integrable.
240 4. It must map  $\mathbb{R} \rightarrow \text{PhysicalUniverse}$  to ensure the macroscopic volume and vacuum
241    constraints are not violated during the perturbation.
242 -/
243 def CGD.Foundations.isValidPhysicalVariation (v :  $\mathbb{R} \rightarrow \text{PhysicalUniverse}$ ) : Prop :=
244   ( $\forall$  mu i j, ContDiff  $\mathbb{R} \ T$  (fun (tx :  $\mathbb{R} \times \text{CGD.Foundations.SpacetimePoint}$ ) => ((v tx.1).toUniverse.sd_sector
245      $\hookrightarrow$  mu tx.2).val i j))  $\wedge$ 
246   ( $\forall$  mu i j, ContDiff  $\mathbb{R} \ T$  (fun (tx :  $\mathbb{R} \times \text{CGD.Foundations.SpacetimePoint}$ ) => ((v tx.1).toUniverse.asd_sector
247      $\hookrightarrow$  mu tx.2).val i j))  $\wedge$ 
248   ( $\exists$  R > 0,  $\forall$  t,  $\forall$  x, (x 0)^2 + (x 1)^2 + (x 2)^2 + (x 3)^2 > R^2  $\rightarrow$ 
249     ( $\forall$  mu, (v t).toUniverse.sd_sector mu x = (v 0).toUniverse.sd_sector mu x)  $\wedge$ 
250     ( $\forall$  mu, (v t).toUniverse.asd_sector mu x = (v 0).toUniverse.asd_sector mu x))  $\wedge$ 
251     ( $\forall$  t, MeasureTheory.Integrable (fun p => lagrangianDensity (fun mu nu => curvature (fun m x => (v
252        $\hookrightarrow$  t).toUniverse.spin4c_connection m x) mu nu p)))
253
254 noncomputable def CGD.Foundations.lagrangianDensity (F : Fin 4 -> Fin 4 -> ChiralM) : Complex :=
255   (-0.5 : Complex) * (
256      $\sum$  mu : Fin 4,  $\sum$  nu : Fin 4,  $\sum$  rho : Fin 4,  $\sum$  sigma : Fin 4,
257     CGD.Gravity.epsilon4 mu nu rho sigma * Matrix.trace (F mu nu * F rho sigma)
258   )
259
260 -- MODULE: CGD.Foundations.Bianchi
261 -----
262 /--
263 The Differential Bianchi Identity.
264 Derived strictly from fundamental Lie algebra properties and Clairaut's theorem,
265 simultaneously performing the  $d^2=0$  and Jacobi cancellations algebraically.
266 -/
267 @[litlib_track "Kinematic Bianchi Identity"]
268 theorem CGD.Foundations.kinematicBianchiIdentity
269   [clairaut : Litlib.Y1976.rudin1976principles.ClairautTheoremNDimensional]
270   (A : CGD.Axioms.Sl2cGaugeField) (p  $\mu$   $\nu$  : Fin 4) (x : SpacetimePoint) :
271   covariantDeriv A  $\rho$   $\mu$   $\nu$  x +
272   covariantDeriv A  $\mu$   $\nu$   $\rho$  x +
273   covariantDeriv A  $\nu$   $\rho$   $\mu$  x = 0
274
275 -- MODULE: CGD.Foundations.Calculus

```

```

275 -----
276 noncomputable def CGD.Foundations.curvature (A : Fin 4 → SpacetimePoint → ChiralM) (mu nu : Fin 4) (x :
  ↪ SpacetimePoint) : ChiralM :=
277   let dA_nu := partialDerivChiral mu (fun p => A nu p) x
278   let dA_mu := partialDerivChiral nu (fun p => A mu p) x
279   let raw_comm := bracket (A mu x) (A nu x)
280   let proj_comm := embedSelfDual (chiralProject raw_comm).self_dual + embedAntiSelfDual (chiralProject
  ↪ raw_comm).anti_self_dual
281   dA_nu - dA_mu + proj_comm
282
283 noncomputable def CGD.Foundations.curvatureSL2c (A : Fin 4 → SpacetimePoint → SL2C) (mu nu : Fin 4) (x :
  ↪ SpacetimePoint) : SL2C :=
284   let dA_nu := partialDerivSL2c mu (A nu) x
285   let dA_mu := partialDerivSL2c nu (A mu) x
286   let comm := [A mu x, A nu x]
287   dA_nu - dA_mu + comm
288
289
290 -- MODULE: CGD.Foundations.ChiralDecomposition
291 -----
292 /--
293 The topological Pontryagin action strictly preserves the chiral split. Because the cross terms vanish
  ↪ orthogonally, the 4D spacetime topology cleanly factorizes into a self-dual topological charge and an
  ↪ anti-self-dual topological charge.
294 -/
295 @[litlib_track "Topological Chiral Decomposition"]
296 theorem CGD.Foundations.algebraicChiralDecomposition (pu : PhysicalUniverse) (x : SpacetimePoint) :
297   lagrangianDensity (fun mu nu => curvature (fun m p => pu.toUniverse.spin4c_connection m p) mu nu x) =
298   actionVacuum (fun mu nu => curvature (fun m p => pu.toUniverse.spin4c_connection m p) mu nu x) +
299   actionAntiSelfDual (fun mu nu => curvature (fun m p => pu.toUniverse.spin4c_connection m p) mu nu x)
300
301
302 -- MODULE: CGD.Foundations.GaugeGroup
303 -----
304 structure CGD.Foundations.ChiralComponents where
305   self_dual : SL2C
306   anti_self_dual : SL2C
307
308 abbrev CGD.Foundations.ChiralM := Matrix (Fin 4) (Fin 4) Complex
309

```

```

310 abbrev CGD.Foundations.SL2C :=  $\mathfrak{sl}$ (LieAlgebra.SpecialLinear.sl (Fin 2) Complex)
311
312 def CGD.Foundations.SU2Group := { M : Matrix (Fin 2) (Fin 2) Complex // M * M.conjTranspose = 1  $\wedge$  M.det = 1 }
313
314 noncomputable def CGD.Foundations.chiralProject (M : ChiralM) : ChiralComponents :=
315   let b := M.submatrix chiralIso chiralIso
316   { self_dual := toSL2C (fun i j => b (Sum.inl i) (Sum.inl j)), anti_self_dual := toSL2C (fun i j => b
      $\hookrightarrow$  (Sum.inr i) (Sum.inr j)) }
317
318 noncomputable def CGD.Foundations.embedAntiSelfDual (m : SL2C) : ChiralM := Matrix.of (fun i j => match
      $\hookrightarrow$  (chiralIso.symm i), (chiralIso.symm j) with | Sum.inr i', Sum.inr j' => m.val i' j' | _, _ => 0)
319
320 noncomputable def CGD.Foundations.embedSelfDual (m : SL2C) : ChiralM := Matrix.of (fun i j => match
      $\hookrightarrow$  (chiralIso.symm i), (chiralIso.symm j) with | Sum.inl i', Sum.inl j' => m.val i' j' | _, _ => 0)
321
322 instance : One SU2Group
323
324 def CGD.Foundations.isSu2 (M : Matrix (Fin 2) (Fin 2) Complex) : Prop :=
325   Matrix.trace M = 0  $\wedge$  M.conjTranspose = -M
326
327 noncomputable def CGD.Foundations.sigmaX : Matrix (Fin 2) (Fin 2)  $\mathbb{C}$  := mkMat 0 1 1 0
328
329 noncomputable def CGD.Foundations.sigmaZ : Matrix (Fin 2) (Fin 2)  $\mathbb{C}$  := mkMat 1 0 0 (-1)
330
331 noncomputable def CGD.Foundations.toSL2C (M : Matrix (Fin 2) (Fin 2) Complex) : SL2C :=
332   let tr := M.trace
333   let M' := M - (tr / 2) • (1 : Matrix (Fin 2) (Fin 2) Complex)
334   ⟨M', by rw[mem_sl_iff, Matrix.trace_sub, Matrix.trace_smul, Matrix.trace_one]; rw[Fintype.card_fin]; simp
      $\hookrightarrow$  only[Nat.cast_ofNat]; rw[smul_eq_mul]; field_simp; ring⟩
335
336
337 -- MODULE: CGD.Foundations.Lagrangian.Basic
338 -----
339 def CGD.Foundations.isSpin4cAlgebra (M : ChiralM) : Prop :=
340    $\exists$  (L R : SL2C), M = embedSelfDual L + embedAntiSelfDual R
341
342
343 -- MODULE: CGD.Foundations.Lagrangian.Uniqueness
344 -----
345 /--

```

```

346 The fully antisymmetric Pontryagin topological density is mathematically the unique quadratic,
    ↪ gauge-invariant Lagrangian density that can be constructed without a pre-existing background metric.
347 -/
348 @[litlib_track "Topological Lagrangian Uniqueness"]
349 theorem CGD.Foundations.topologicalLagrangianUniqueness
350   (L : ((Fin 4 → Fin 4 → ChiralM) → Complex))
351   (h_inv : ∀ F U, (∀ μ v, isSpin4cAlgebra (F μ v)) →
352     (∀ μ v, isSpin4cAlgebra (U * F μ v * U-1)) →
353     L (fun μ v => U * F μ v * U-1) = L F)
354   (h_topological : ∀ Λ : Matrix (Fin 4) (Fin 4) ℂ,
355     (∀ μ v ρ σ, ∑ α : Fin 4, ∑ β : Fin 4, ∑ γ : Fin 4, ∑ δ : Fin 4,
356       Λ α μ * Λ β ν * Λ γ ρ * Λ δ σ * CGD.Gravity.epsilon4 α β γ δ = Matrix.det Λ * CGD.Gravity.epsilon4 μ ν
357       ↪ ρ σ) →
358     ∀ F, (∀ μ v, isSpin4cAlgebra (F μ v)) →
359     (∀ μ v, isSpin4cAlgebra (∑ α : Fin 4, ∑ β : Fin 4, (Λ μ α * Λ ν β) • F α β)) →
360     L (fun μ v => ∑ α : Fin 4, ∑ β : Fin 4, (Λ μ α * Λ ν β) • F α β) = Matrix.det Λ * L F)
361   (hLQuadScale : ∀ (c : ℂ) (F : Fin 4 → Fin 4 → ChiralM),
362     (∀ μ v, isSpin4cAlgebra (F μ v)) → L (fun μ v => c • F μ v) = c2 * L F)
363   (hLQuadAdd : ∀ (F G : Fin 4 → Fin 4 → ChiralM),
364     (∀ μ v, isSpin4cAlgebra (F μ v)) → (∀ μ v, isSpin4cAlgebra (G μ v)) →
365     L (fun μ v => F μ v + G μ v) + L (fun μ v => F μ v - G μ v) = 2 * L F + 2 * L G) :
366   ∃ c : ℂ, ∀ F, (∀ μ v, isSpin4cAlgebra (F μ v)) → L F = c * ∑ μ : Fin 4, ∑ ν : Fin 4, ∑ ρ : Fin 4, ∑ σ :
367   ↪ Fin 4, CGD.Gravity.epsilon4 μ ν ρ σ * Matrix.trace (F μ ν * F ρ σ)
368
369
370 -- MODULE: CGD.Foundations.Lagrangian.Variation
371 -----
372 /--
373 Because the action is the fully antisymmetric Pontryagin density, its
374 functional variation with respect to compactly supported, smooth gauge field
375 perturbations evaluates identically to zero.
376
377 This theorem rigorously demonstrates that Chiral Gauge Dynamics possesses no
378 classical Euler-Lagrange equations of motion. Consequently, the dynamic
379 evolution and stress-energy conservation of the macroscopic universe are not
380 driven by action minimization, but rather are governed entirely by strict
381 geometric conservation laws (the Bianchi identities) and topological boundary
382 conditions.
383 -/
384 @[litlib_track "Topological Action Variation"]

```

```

383 theorem CGD.Foundations.topologicalActionVariationZero
384   (v : ℝ → PhysicalUniverse)
385   [dt : Litlib.Y1965.spivak1965calculus.DivergenceTheoremR4Compact (fun x mu => variationCurrent v 0 mu x)]
386   [lr : Litlib.Y1976.rudin1976principles.LeibnizIntegralRule (fun s x => lagrangianDensity (fun mu nu =>
387     ↪ curvature (fun m p => (v s).toUniverse.spin4c_connection m p) mu nu x))]
388   (h_valid : isValidPhysicalVariation v) :
389   deriv (fun t => physicalUniverseAction (v t)) 0 = 0
390
391 -- MODULE: CGD.Foundations.Lagrangian.Variation.Algebra
392 -----
393 noncomputable def CGD.Foundations.physicalUniverseAction (pu : PhysicalUniverse) : ℂ :=
394   universeAction pu.toUniverse
395
396 /--
397 The topological Chern-Simons Variation Current.
398 K^mu = - 2 * ε^{mu nu rho sigma} Tr( δA_nu F_{rho sigma} )
399 -/
400 noncomputable def CGD.Foundations.variationCurrent (v : ℝ → PhysicalUniverse) (t : ℝ) (mu : Fin 4) (x :
401   ↪ SpacetimePoint) : ℂ :=
402   let A := fun s m p => (v s).toUniverse.spin4c_connection m p
403   let dA_dt := fun m p => deriv (fun s => A s m p) t
404   let F := curvature (A t)
405   (-2 : ℂ) * ∑ nu : Fin 4, ∑ rho : Fin 4, ∑ sigma : Fin 4,
406     CGD.Gravity.epsilon4 mu nu rho sigma * Matrix.trace (dA_dt nu x * F rho sigma x)
407
408 -- MODULE: CGD.Foundations.Spacetime
409 -----
410 /--
411 The Spacetime Manifold is defined as a map `Fin 4 → ℝ`.
412 This instantiates the standard `NormedAddCommGroup` and `NormedSpace ℝ` required for rigorous calculus of
413 ↪ variations over the manifold.
414 -/
415 abbrev CGD.Foundations.SpacetimePoint := Fin 4 → ℝ
416
417 -- MODULE: CGD.Foundations.TensorCalculus.DifferentialRules
418 -----
419 /-- True geometric definition of the gauge covariant derivative of the field strength tensor:

```

```

420       $D_\alpha F_\beta \gamma = \partial_\alpha F_\beta \gamma + [A_\alpha, F_\beta \gamma]$  -/
421 noncomputable def CGD.Foundations.covariantDeriv (A : Fin 4 → SpacetimePoint → SL2C) (alpha beta gamma : Fin
422 ↪ 4) (x : SpacetimePoint) : SL2C :=
423   let dF := partialDerivSL2C alpha (fun p => curvatureSL2C A beta gamma p) x
424   let comm := [A alpha x, curvatureSL2C A beta gamma x]
425   dF + comm
426
427 -- MODULE: CGD.Foundations.Topology
428 -----
429 abbrev CGD.Foundations.S3 := { x : Fin 4 → ℝ // (x 0)^2 + (x 1)^2 + (x 2)^2 + (x 3)^2 = 1 }
430
431
432 -- MODULE: CGD.Gravity.DomainSeparation
433 -----
434 /--
435 A rigorous derivation showing that the scalar volume density  $\mu(x)$  generates torsion that can be
436 conformally absorbed, resulting in a physical metric that natively satisfies the Trace-Reversed
437 Vacuum Einstein Field Equations with a Cosmological Constant ( $\Lambda = 1$ ).
438 -/
439 @[litlib_track "Macroscopic Cosmological Emergence"]
440 theorem CGD.Gravity.macroscopicCosmologicalEmergence
441   (pu : PhysicalUniverse)
442   (urbantke_g : pu.bulk → Fin 4 → Fin 4 → ℝ)
443   (urbantke_g_inv : pu.bulk → Fin 4 → Fin 4 → ℝ)
444   (urbantke_ricci : pu.bulk → Fin 4 → Fin 4 → ℝ)
445   (mu : pu.bulk → ℝ)
446   (nabla_mu : (pu.bulk → ℝ) → pu.bulk → Fin 4 → ℝ)
447   (nabla_nabla_mu : (pu.bulk → ℝ) → pu.bulk → Fin 4 → Fin 4 → ℝ)
448   (g_phys_inv : pu.bulk → Fin 4 → Fin 4 → ℝ)
449   (ricci_phys : pu.bulk → Fin 4 → Fin 4 → ℝ)
450   (F_ij F_bar_ij : pu.bulk → Fin 3 → Fin 3 → ℂ)
451   (plebanski_vacuum : ℂ → (Fin 3 → Fin 3 → ℂ) → (Fin 3 → Fin 3 → ℂ) → Prop)
452   (isLeviCivitaRicci : (pu.bulk → (Fin 4 → Fin 4 → ℝ)) → (pu.bulk → (Fin 4 → Fin 4 → ℝ)) → Prop)
453   (isLeviCivitaRicciPointwise : (Fin 4 → Fin 4 → ℝ) → (Fin 4 → Fin 4 → ℝ) → Prop)
454   [conformal : Litlib.Y2010.wald2010general.ConformalRicciTransformation
455     pu.bulk urbantke_g (fun p a b => mu p * urbantke_g p a b) urbantke_g_inv
456     urbantke_ricci ricci_phys (fun p => Real.sqrt (mu p)) nabla_mu nabla_nabla_mu isLeviCivitaRicci]
457   [eq15 : Litlib.Y2011.krasnov2011plebanski.Eq15 plebanski_vacuum]
458   (pleb_equiv : ∀ p, Litlib.Y2011.krasnov2011plebanski.PlebanskiToEinsteinEquivalence

```

```

459   (fun a b => mu p * urbantke_g p a b) (g_phys_inv p) (ricci_phys p) 1 (F_ij p) (F_bar_ij p)
      ↪ plebanski_vacuum isLeviCivitaRicciPointwise)
460   (D : (pu.bulk → ℂ) → pu.bulk → ℂ)
461   (F_curv : pu.bulk → ℂ)
462   (Sigma_urb : pu.bulk → ℂ)
463   (h_axiom3 : ∀ p, F_curv p = (mu p : ℂ) * Sigma_urb p)
464   (h_bianchi : ∀ p, D F_curv p = 0)
465   (h_levi_civita : (∀ p, D (fun x => (mu x : ℂ) * Sigma_urb x) p = 0) → ∀ p, isLeviCivitaRicciPointwise (fun
      ↪ a b => mu p * urbantke_g p a b) (ricci_phys p))
466   (h_inv_phys : ∀ p, ∀ a c, (∑ b : Fin 4, (mu p * urbantke_g p a b) * g_phys_inv p b c) = if a = c then 1
      ↪ else 0)
467   (h_pleb_vac_cond : ∀ p, (∑ i : Fin 3, F_ij p i i) = -1 ∧ (∀ i j, F_bar_ij p i j = 0)) :
468   ∀ p a c, urbantke_ricci p a c = (mu p * urbantke_g p a c)
469     + 2 * nabla_nabla_mu (fun x => Real.log (Real.sqrt (mu x))) p a c
470     + urbantke_g p a c * (∑ d : Fin 4, ∑ e : Fin 4, urbantke_g_inv p d e * nabla_nabla_mu (fun x =>
      ↪ Real.log (Real.sqrt (mu x))) p d e)
471     - 2 * nabla_mu (fun x => Real.log (Real.sqrt (mu x))) p a * nabla_mu (fun x => Real.log (Real.sqrt (mu
      ↪ x))) p c
472     + 2 * urbantke_g p a c * (∑ d : Fin 4, ∑ e : Fin 4, urbantke_g_inv p d e * nabla_mu (fun x => Real.log
      ↪ (Real.sqrt (mu x))) p d * nabla_mu (fun x => Real.log (Real.sqrt (mu x))) p e)
473
474   /--
475   In the Capovilla formulation, the Unimodular constraint is governed by the scalar density  $\mu$ .
476   This theorem rigorously applies the literature identity to prove that the fundamental volume element
477   of the emergent spacetime metric (sqrt_g) squared is strictly equal to the Unimodular multiplier  $\mu$  squared.
478   -/
479   @[litlib_track "Macroscopic Unimodular Vacuum Emergence"]
480   theorem CGD.Gravity.macroscopicVacuumEmergence
481     (pu : PhysicalUniverse)
482     (sqrt_g mu eta : SpacetimePoint → ℂ)
483     (Psi_3x3 M_3x3 : SpacetimePoint → Matrix (Fin 3) (Fin 3) ℂ)
484     (vol_id : Theorem_Volume_Element_Identity SpacetimePoint sqrt_g mu eta Psi_3x3 M_3x3) :
485     ∀ x ∈ pu.bulk, (sqrt_g x)^2 = (mu x)^2
486
487
488   -- MODULE: CGD.Gravity.ExactSolutions.Abelian
489   -----
490   /--
491   Provides an exact analytical solution for an Abelian plane wave satisfying the pure CDJ
492   constraint equation ( $F \wedge F = 0$ ).

```



```

493 -/
494 @[litlib_track "Exact Abelian Macroscopic Solution"]
495 theorem CGD.Gravity.dynamicExactAbelianSolution (c : ℂ) (hc : c ≠ 0) :
496   ∃ (u : Universe),
497     CGD.Gravity.satisfiesPureCdjConstraint (fun p m n => cgdAdjointCurvature u m n p) ∧
498     (∀ x, curvatureSl2c u.sd_sector 1 2 x = c • toSl2c sigmaX) ∧
499     (∃ x, curvatureSl2c u.sd_sector 1 2 x ≠ 0)
500
501
502 -- MODULE: CGD.Gravity.ExactSolutions.AlgebraicForms
503 -----
504 /--
505   Type 0 (Isotropic / FLRW Cosmological Vacuum)
506   Algebraic Form:  $A_0 = 0$ ,  $A_j(t) = a(t) \sigma_j$ 
507   All principal null directions are degenerate.
508   The scale factor `a(t)` must be strictly real and non-static to ensure a non-degenerate Lorentzian metric.
509   Acts as the volume-generating Axial Condensate for all other Petrov types.
510 -/
511 def CGD.Gravity.ExactSolutions.IsType0Form (pu : PhysicalUniverse) : Prop :=
512   ∃ a : ℝ → ℂ, ∀ x : SpacetimePoint,
513     (a (x 0)).im = 0 ∧
514     (fderiv ℝ a (x 0) 1).im = 0 ∧
515     a (x 0) ≠ 0 ∧
516     fderiv ℝ a (x 0) 1 ≠ 0 ∧
517     pu.toUniverse.sd_sector.val 0 x = 0 ∧
518     pu.toUniverse.sd_sector.val 1 x = toSl2c (a (x 0) • sigma1.val) ∧
519     pu.toUniverse.sd_sector.val 2 x = toSl2c (a (x 0) • sigma2.val) ∧
520     pu.toUniverse.sd_sector.val 3 x = toSl2c (a (x 0) • sigma3.val)
521
522
523 -- MODULE: CGD.Gravity.ExactSolutions.CDJEvaluation
524 -----
525 @[litlib_track "CDJ constraint holds"]
526 lemma CGD.Gravity.CDJ_constraint_holds :
527   (∑ μ : Fin 4, ∑ ν : Fin 4, ∑ ρ : Fin 4, ∑ σ : Fin 4, epsilon4 μ ν ρ σ • (adj_F μ ν * adj_F ρ σ)) =
528   ((∑ μ : Fin 4, ∑ ν : Fin 4, ∑ ρ : Fin 4, ∑ σ : Fin 4, epsilon4 μ ν ρ σ • (adj_F μ ν * adj_F ρ σ)).trace /
529    ↪ 3) • 1
530
531
532 -- MODULE: CGD.Gravity.ExactSolutions.Definitions

```

```

532 -----
533 /--
534 A Physical Universe natively satisfies the kinematic reality conditions if and only if
535 the macroscopic Urbantke metric constructed from its self-dual curvature is strictly
536 real, strictly non-degenerate, and strictly Lorentzian (signature -+++ or +---)
537 everywhere within the topological bulk.
538
539 This mathematically enforces the historical Reality Conditions without requiring
540 any non-polynomial differential constraints.
541 -/
542 def CGD.Gravity.ExactSolutions.SatisfiesRealityConditions (pu : PhysicalUniverse) : Prop :=
543   ∀ x ∈ pu.bulk, isLorentzian (urbantkeMetric (fun μ v => curvatureSl2c pu.toUniverse.sd_sector.val μ v x))
544
545
546 -- MODULE: CGD.Gravity.ExactSolutions.FLRW
547 -----
548 /--
549 LRW (Isotropic) spacetimes intrinsically satisfy the non-degenerate Lorentzian Reality Conditions natively
550 ↔ from the SU(2) topology, requiring no exact scalar field assumptions.
551 -/
552 @[litlib_track "FLRW connections produce real Lorentzian metrics"]
553 theorem CGD.Gravity.ExactSolutions.flrw_satisfies_reality (pu : PhysicalUniverse) (h_type0 : IsType0Form pu)
554   ↔ :
555     SatisfiesRealityConditions pu
556
557 -- MODULE: CGD.Gravity.ExactSolutions.LorentzianAnsatz
558 -----
559 noncomputable def CGD.Gravity.F_origin_val (mu nu : Fin 4) : Matrix (Fin 2) (Fin 2) ℂ :=
560   if mu = 0 ∧ nu = 1 then -sigmaX
561   else if mu = 0 ∧ nu = 2 then -sigmaY
562   else if mu = 0 ∧ nu = 3 then -sigmaZ
563   else if mu = 1 ∧ nu = 0 then sigmaX
564   else if mu = 2 ∧ nu = 0 then sigmaY
565   else if mu = 3 ∧ nu = 0 then sigmaZ
566   else if mu = 1 ∧ nu = 2 then Complex.I • sigmaZ
567   else if mu = 2 ∧ nu = 1 then -Complex.I • sigmaZ
568   else if mu = 1 ∧ nu = 3 then -Complex.I • sigmaY
569   else if mu = 3 ∧ nu = 1 then Complex.I • sigmaY
570   else if mu = 2 ∧ nu = 3 then Complex.I • sigmaX

```

```

570   else if mu = 3 ∧ nu = 2 then -Complex.I • sigmaX
571   else 0
572
573 noncomputable def CGD.Gravity.adj_F (mu nu : Fin 4) : Matrix (Fin 3) (Fin 3) ℂ :=
574   extractAdjoint (F_origin_val mu nu)
575
576 @[litlib_track "Linear-In-Coordinates Lorentzian Matrix Ansatz"]
577 noncomputable def CGD.Gravity.exactLorentzianL (mu : Fin 4) (x : SpacetimePoint) : Matrix (Fin 2) (Fin 2) ℂ
578   ↪ :=
579   if mu = 1 then - (x 0 : ℝ) • sigmaX + (x 3 : ℝ) • ((Complex.I / 2) • sigmaY) - (x 2 : ℝ) • ((Complex.I /
580     ↪ 2) • sigmaZ)
581   else if mu = 2 then - (x 0 : ℝ) • sigmaY + (x 1 : ℝ) • ((Complex.I / 2) • sigmaZ) - (x 3 : ℝ) •
582     ↪ ((Complex.I / 2) • sigmaX)
583   else if mu = 3 then - (x 0 : ℝ) • sigmaZ + (x 2 : ℝ) • ((Complex.I / 2) • sigmaX) - (x 1 : ℝ) •
584     ↪ ((Complex.I / 2) • sigmaY)
585   else 0
586
587 -- MODULE: CGD.Gravity.ExactSolutions.LorentzianEval
588 -----
589 @[litlib_track "Exact Lorentzian Field Curvature At Origin"]
590 lemma CGD.Gravity.F_origin_val_eq_map (m n : Fin 4) : F_origin_val m n = F_val_map m n
591
592 noncomputable def CGD.Gravity.F_val_map (m n : Fin 4) : Matrix (Fin 2) (Fin 2) ℂ :=
593   match m, n with
594   | 0, 1 => -sigmaX | 0, 2 => -sigmaY | 0, 3 => -sigmaZ
595   | 1, 0 => sigmaX | 1, 2 => Complex.I • sigmaZ | 1, 3 => -Complex.I • sigmaY
596   | 2, 0 => sigmaY | 2, 1 => -Complex.I • sigmaZ | 2, 3 => Complex.I • sigmaX
597   | 3, 0 => sigmaZ | 3, 1 => Complex.I • sigmaY | 3, 2 => -Complex.I • sigmaX
598   | _, _ => 0
599
600 -- MODULE: CGD.Gravity.ExactSolutions.MainTheorem
601 -----
602 /--
603   This formally constructs an exact analytical non-Abelian  $SU(2)$  gauge configuration
604   that simultaneously satisfies the trace-free Capovilla CDJ constraint at the spatial
605   origin, and produces a non-degenerate, strictly real, negative-determinant
606   Lorentzian metric ( $\det g < 0$ ) at  $x = 0$ .
607 -/

```

```

606 This mathematically proves that the foundational vacuum geometry of Chiral Gauge Dynamics
607 is strictly non-vacuous.
608 -/
609 @[litlib_track "Exact Non-Abelian Lorentzian Macroscopic Witness"]
610 theorem CGD.Gravity.dynamicExactLorentzianSolution :
611   ∃ (u : Universe) (x : SpacetimePoint),
612     (∑ μ : Fin 4, ∑ ν : Fin 4, ∑ ρ : Fin 4, ∑ σ : Fin 4,
613       epsilon4 μ ν ρ σ • (cgdAdjointCurvature u μ ν x * cgdAdjointCurvature u ρ σ x)) =
614     ((∑ μ : Fin 4, ∑ ν : Fin 4, ∑ ρ : Fin 4, ∑ σ : Fin 4,
615       epsilon4 μ ν ρ σ • (cgdAdjointCurvature u μ ν x * cgdAdjointCurvature u ρ σ x)).trace / 3) • 1 ∧
616     isLorentzian (urbantkeMetric (fun m n => curvatureSl2c u.sd_sector m n x))
617
618
619 -- MODULE: CGD.Gravity.ExactSolutions.Soliton
620 -----
621 /--
622 The Soliton Geometry Theorem.
623 Proves that a continuous, non-singular Spin(4,C) gauge profile mathematically possesses a
624 strict non-degenerate Lorentzian emergent metric, while simultaneously satisfying the
625 Ricci-flat Capovilla CDJ constraint.
626 -/
627 @[litlib_track "Exact Analytical Non-Singular Soliton Verification"]
628 theorem CGD.Gravity.ExactSolutions.dynamicExactSolitonSolution :
629   ∃ (u : Universe) (x : SpacetimePoint),
630     (∑ μ : Fin 4, ∑ ν : Fin 4, ∑ ρ : Fin 4, ∑ σ : Fin 4,
631       epsilon4 μ ν ρ σ • (cgdAdjointCurvature u μ ν x * cgdAdjointCurvature u ρ σ x)) =
632     ((∑ μ : Fin 4, ∑ ν : Fin 4, ∑ ρ : Fin 4, ∑ σ : Fin 4,
633       epsilon4 μ ν ρ σ • (cgdAdjointCurvature u μ ν x * cgdAdjointCurvature u ρ σ x)).trace / 3) • 1 ∧
634     isLorentzian (urbantkeMetric (fun m n => curvatureSl2c u.sd_sector.val m n x))
635
636 @[litlib_track "The exact mathematical gauge field solution"]
637 noncomputable def CGD.Gravity.ExactSolutions.solitonAnsatzVal (mu : Fin 4) (x : SpacetimePoint) : Matrix
638   ↪ (Fin 2) (Fin 2) ℂ :=
639   let K_c := (solitonK x : ℂ)
640   let t := (x 0 : ℂ)
641   let x1 := (x 1 : ℂ)
642   let x2 := (x 2 : ℂ)
643   let x3 := (x 3 : ℂ)
644   let sX := sigmaX
645   let sY := solitonSigmaY

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645 let sZ := sigmaZ
646 let isX := Complex.I • sX
647 let isY := Complex.I • sY
648 let isZ := Complex.I • sZ
649 if mu = 1 then -t • sX - (K_c * x3) • isY + (K_c * x2) • isZ
650 else if mu = 2 then (K_c * x3) • isX - t • sY - (K_c * x1) • isZ
651 else if mu = 3 then -(K_c * x2) • isX + (K_c * x1) • isY - t • sZ
652 else 0
653
654 @[litlib_track "Smooth scalar profile removing the singularity"]
655 noncomputable def CGD.Gravity.ExactSolutions.solitonK (x : SpacetimePoint) : ℝ :=
656   -1 / ((x 1)^2 + (x 2)^2 + (x 3)^2 + 2)
657
658
659 -- MODULE: CGD.Gravity.GeodesicMotion
660 -----
661 /--
662 Using the rigorous Papapetrou (1951) theorem, we prove that a localized topological defect
663 (a single-pole singularity in the gauge curvature) must travel along a complex geodesic of the
664 dynamically emergent Urbantke geometry. This natively utilizes the exact complex covariant
665 conservation of the emergent Stress-Energy tensor, evaluating strictly inside the macroscopic bulk.
666 -/
667 @[litlib_track "Machian Motion of Topological Defects"]
668 theorem CGD.Gravity.machianTopologicalDefectMotion
669   (pu : PhysicalUniverse)
670   (isSmooth : (pu.bulk → ℂ) → Prop)
671   (γ : ℂ → pu.bulk)
672   (u : ℂ → (Fin 4 → ℂ))
673   (du : ℂ → (Fin 4 → ℂ))
674   (isSinglePole : (Fin 4 → Fin 4 → pu.bulk → ℂ) → (ℂ → pu.bulk) → Prop)
675   [general_bianchi : Litlib.Y2003.nakahara2003geometry.Theorem_ContractedBianchi pu.bulk (Fin 4) isSmooth
676     ↪ (bulkDeriv pu)]
677   [symm_metric : Litlib.Y1984.urbantke1984integrability.Eq10_Symmetry]
678   [papa : Litlib.Y1951.papapetrou1951spinning.Eq2_12 pu.bulk ℂ
679     (fun p => Matrix.of (fun m n => CGD.Gravity.urbantkeMetric (fun a b => curvatureSl2c
680       ↪ pu.toUniverse.sd_sector a b p.val) m n))
681     (fun p => Matrix.of (fun m n => CGD.Gravity.matrixInv4x4 (fun a b => CGD.Gravity.urbantkeMetric (fun a b
682       ↪ => curvatureSl2c pu.toUniverse.sd_sector a b p.val) a b) m n))
683     (fun p rho mu nu => CGD.Gravity.christoffel (fun m n p' => CGD.Gravity.urbantkeMetric (fun a b =>
684       ↪ curvatureSl2c pu.toUniverse.sd_sector a b p') m n) rho mu nu p.val)

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681 (fun m n p => emergentStressEnergy (fun a b p' => curvatureSl2c pu.toUniverse.sd_sector a b p') m n
    ↪ p.val)
682 (bulkDeriv pu) γ u du isSinglePole]
683 (h_metric_eq10 : ∀ x : pu.bulk, ∃ (F F_dual : Fin 3 → Fin 4 → Fin 4 → ℂ) (epsilon3 : Fin 3 → Fin 3 → Fin 3
    ↪ → ℂ),
684 (∀ a mu nu, F a mu nu = - F a nu mu) ∧
685 (∀ a mu nu, F_dual a mu nu = - F_dual a nu mu) ∧
686 (∀ a b c, epsilon3 a b c = - epsilon3 b a c ∧ epsilon3 a b c = - epsilon3 a c b) ∧
687 (∀ mu nu, CGD.Gravity.urbantkeMetric (fun a b => curvatureSl2c pu.toUniverse.sd_sector a b x.val) mu nu =
688 (-1 / 6 : ℂ) * Finset.sum Finset.univ (fun a => Finset.sum Finset.univ (fun b => Finset.sum
    ↪ Finset.univ (fun c => Finset.sum Finset.univ (fun alpha => Finset.sum Finset.univ (fun beta =>
    ↪ epsilon3 a c b * F a mu alpha * F_dual c alpha beta * F b beta nu)))))))))
689 (h_smooth_g : ∀ i j, isSmooth (fun p => CGD.Gravity.urbantkeMetric (fun a b => curvatureSl2c
    ↪ pu.toUniverse.sd_sector a b p.val) i j))
690 (h_smooth_g_inv : ∀ i j, isSmooth (fun p => CGD.Gravity.matrixInv4x4 (fun m n =>
    ↪ CGD.Gravity.urbantkeMetric (fun a b => curvatureSl2c pu.toUniverse.sd_sector a b p.val) m n) i j))
691 (h_smooth_chris : ∀ rho mu nu, isSmooth (fun p => CGD.Gravity.christoffel (fun m n p' =>
    ↪ CGD.Gravity.urbantkeMetric (fun a b => curvatureSl2c pu.toUniverse.sd_sector a b p') m n) rho mu nu
    ↪ p.val))
692 (h_chris_eq : ∀ p : pu.bulk, ∀ rho mu nu, CGD.Gravity.christoffel (fun m n p' =>
    ↪ CGD.Gravity.urbantkeMetric (fun a b => curvatureSl2c pu.toUniverse.sd_sector a b p') m n) rho mu nu
    ↪ p.val =
693 (1/2 : ℂ) * ∑ sigma : Fin 4, CGD.Gravity.matrixInv4x4 (fun m n => CGD.Gravity.urbantkeMetric (fun a b =>
    ↪ curvatureSl2c pu.toUniverse.sd_sector a b p.val) m n) rho sigma * (
694 bulkDeriv pu mu (fun p' => CGD.Gravity.urbantkeMetric (fun a b => curvatureSl2c
    ↪ pu.toUniverse.sd_sector a b p'.val) sigma nu) p +
695 bulkDeriv pu nu (fun p' => CGD.Gravity.urbantkeMetric (fun a b => curvatureSl2c
    ↪ pu.toUniverse.sd_sector a b p'.val) mu sigma) p -
696 bulkDeriv pu sigma (fun p' => CGD.Gravity.urbantkeMetric (fun a b => curvatureSl2c
    ↪ pu.toUniverse.sd_sector a b p'.val) mu nu) p))
697 (h_ricci_eq : ∀ p : pu.bulk, ∀ mu nu, CGD.Gravity.ricciTensor (fun m n p' => CGD.Gravity.urbantkeMetric
    ↪ (fun a b => curvatureSl2c pu.toUniverse.sd_sector a b p') m n) mu nu p.val =
698 ∑ rho : Fin 4, (bulkDeriv pu rho (fun p' => CGD.Gravity.christoffel (fun m n p'' =>
    ↪ CGD.Gravity.urbantkeMetric (fun a b => curvatureSl2c pu.toUniverse.sd_sector a b p'') m n) rho mu nu
    ↪ p'.val) p -
699 bulkDeriv pu nu (fun p' => CGD.Gravity.christoffel (fun m n p'' => CGD.Gravity.urbantkeMetric (fun
    ↪ a b => curvatureSl2c pu.toUniverse.sd_sector a b p'') m n) rho mu rho p'.val) p +

```

```

700       $\sum$  lambda : Fin 4, (CGD.Gravity.christoffel (fun m n p' => CGD.Gravity.urbantkeMetric (fun a b =>
     $\hookrightarrow$  curvatureSl2c pu.toUniverse.sd_sector a b p') m n) rho lambda rho p.val *
     $\hookrightarrow$  CGD.Gravity.christoffel (fun m n p' => CGD.Gravity.urbantkeMetric (fun a b => curvatureSl2c
     $\hookrightarrow$  pu.toUniverse.sd_sector a b p') m n) lambda mu nu p.val -
701      CGD.Gravity.christoffel (fun m n p' => CGD.Gravity.urbantkeMetric (fun a b => curvatureSl2c
     $\hookrightarrow$  pu.toUniverse.sd_sector a b p') m n) rho lambda nu p.val * CGD.Gravity.christoffel (fun
     $\hookrightarrow$  m n p' => CGD.Gravity.urbantkeMetric (fun a b => curvatureSl2c pu.toUniverse.sd_sector a
     $\hookrightarrow$  b p') m n) lambda mu rho p.val)))
702  (h_single_pole : isSinglePole (fun m n p => emergentStressEnergy (fun a b p' => curvatureSl2c
     $\hookrightarrow$  pu.toUniverse.sd_sector a b p') m n p.val)  $\gamma$ ) :
703   $\forall$  s alpha, du s alpha +  $\sum$  mu : Fin 4,  $\sum$  nu : Fin 4, CGD.Gravity.christoffel (fun m n p =>
     $\hookrightarrow$  CGD.Gravity.urbantkeMetric (fun a b => curvatureSl2c pu.toUniverse.sd_sector a b p) m n) alpha mu nu
     $\hookrightarrow$  ( $\gamma$  s).val * u s mu * u s nu = 0
704
705
706  -- MODULE: CGD.Gravity.Geometry
707  -----
708  abbrev CGD.Gravity.SpacetimeIndex := Fin 4
709
710  noncomputable def CGD.Gravity.christoffel (g : SpacetimeIndex  $\rightarrow$  SpacetimeIndex  $\rightarrow$  SpacetimePoint  $\rightarrow$  Complex)
     $\hookrightarrow$  (rho mu nu : SpacetimeIndex) (x : SpacetimePoint) : Complex :=
711      let g_inv := matrixInv4x4 (fun i j => g i j x)
712      (1 / 2 : Complex) *  $\sum$  sigma, g_inv rho sigma * (partialDeriv mu (fun p => g sigma nu p) x + partialDeriv
     $\hookrightarrow$  nu (fun p => g mu sigma p) x - partialDeriv sigma (fun p => g mu nu p) x)
713
714  /-- 4-dimensional Levi-Civita tensor cast to Complex -/
715  noncomputable def CGD.Gravity.epsilon4 (i j k l : Fin 4) : Complex := epsilon4_int i j k l
716
717  def CGD.Gravity.isLorentzian (g : Matrix (Fin 4) (Fin 4) Complex) : Prop :=
718      ( $\forall$  i j, (g i j).im = 0)  $\wedge$  (g.det.re < 0)  $\wedge$  (g.det.im = 0)
719
720  noncomputable def CGD.Gravity.matrixInv4x4 (M : Matrix (Fin 4) (Fin 4) Complex) : Matrix (Fin 4) (Fin 4)
     $\hookrightarrow$  Complex :=
721      (1 / M.det) • M.adjugate
722
723  noncomputable def CGD.Gravity.ricciTensor (g : SpacetimeIndex  $\rightarrow$  SpacetimeIndex  $\rightarrow$  SpacetimePoint  $\rightarrow$  Complex)
     $\hookrightarrow$  (mu nu : SpacetimeIndex) (x : SpacetimePoint) : Complex :=
724       $\sum$  rho, (partialDeriv rho (fun p => christoffel g rho mu nu p) x
725      - partialDeriv nu (fun p => christoffel g rho mu rho p) x
726      +  $\sum$  lambda, (christoffel g rho lambda rho x * christoffel g lambda mu nu x

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727         - christoffel g rho lambda nu x * christoffel g lambda mu rho x))
728
729 noncomputable def CGD.Gravity.urbantkeMetric (F : Fin 4 -> Fin 4 -> SL2C) : Matrix (Fin 4) (Fin 4) Complex :=
730   fun mu nu =>
731     ∑ a : Fin 3, ∑ b : Fin 3, ∑ c : Fin 3,
732       let eps_iso := epsilon3 a b c
733       let space_term := ∑ alpha : Fin 4, ∑ beta : Fin 4, ∑ gamma : Fin 4, ∑ delta : Fin 4,
734         let eps_space := epsilon4 alpha beta gamma delta
735         let F1 := project F a mu alpha
736         let F2 := project F b nu beta
737         let F3 := project F c gamma delta
738         eps_space * F1 * F2 * F3
739     eps_iso * space_term
740
741
742 -- MODULE: CGD.Gravity.MacroscopicVacuum.Basic
743 -----
744 noncomputable def CGD.Gravity.cgdAdjointCurvature (u : Universe) (μ ν : Fin 4) (x : SpacetimePoint) : Matrix
745   ↪ (Fin 3) (Fin 3) ℂ :=
746     extractAdjoint (curvatureSL2c u.sd_sector μ ν x).val
747
748 /--
749   The Pure CDJ Vacuum Constraint (Capovilla, Dell, Jacobson 1991).
750   Enforces that the symmetric tensor  $\Sigma^{ab} = \epsilon^{\mu\nu\rho\sigma} F^a_{\mu\nu} F^b_{\rho\sigma}$  is purely trace-free.
751    $\Sigma^{ab} - (1/3) \delta^{ab} \text{Tr}(\Sigma) = 0$ 
752 -/
753 def CGD.Gravity.satisfiesPureCdjConstraint (F_adj : SpacetimePoint → Fin 4 → Fin 4 → Matrix (Fin 3) (Fin 3)
754   ↪ ℂ) : Prop :=
755   ∀ x : SpacetimePoint,
756     let Sigma := ∑ μ : Fin 4, ∑ ν : Fin 4, ∑ ρ : Fin 4, ∑ σ : Fin 4,
757       epsilon4 μ ν ρ σ • (F_adj x μ ν * F_adj x ρ σ)
758     Sigma = (Matrix.trace Sigma / 3) • (1 : Matrix (Fin 3) (Fin 3) ℂ)
759
760
761 -- MODULE: CGD.Gravity.StressEnergy.Conservation
762 -----
763 /-- Project bulk derivatives into the spacetime coordinate system -/
764 noncomputable def CGD.Gravity.bulkDeriv (pu : PhysicalUniverse) (mu : Fin 4) (f : pu.bulk → ℂ) (x : pu.bulk)
765   ↪ ℂ :=
766     partialDeriv mu (fun p => if h : p ∈ pu.bulk then f (p, h) else 0) x.val

```



```

764
765 noncomputable def CGD.Gravity.emergentStressEnergy (F : Fin 4 → Fin 4 → SpacetimePoint → SL2C) (mu nu : Fin
↪ 4) (x : SpacetimePoint) : ℂ :=
766   let g := fun m n p => CGD.Gravity.urbantkeMetric (fun a b => F a b p) m n
767   let g_inv := CGD.Gravity.matrixInv4x4 (fun m n => g m n x)
768   let R_mu_nu := CGD.Gravity.ricciTensor g mu nu x
769   let R_scalar := ∑ alpha : Fin 4, ∑ beta : Fin 4, g_inv alpha beta * CGD.Gravity.ricciTensor g alpha beta x
770   R_mu_nu - (1/2 : ℂ) * g mu nu x * R_scalar
771
772 /--
773   The emergent Stress-Energy tensor (defined as the Einstein tensor of the dynamically emergent Urbantke
↪ metric) is covariantly conserved with respect to its own Levi-Civita connection.
774   This fundamentally restricts the conservation to the macroscopic bulk where det g ≠ 0.
775 -/
776 @[litlib_track "Emergent Stress-Energy Conservation"]
777 theorem CGD.Gravity.emergentStressEnergyConservation
778   (pu : PhysicalUniverse)
779   (isSmooth : (pu.bulk → ℂ) → Prop)
780   [general_bianchi : Litlib.Y2003.nakahara2003geometry.Theorem_ContractedBianchi pu.bulk (Fin 4) isSmooth
↪ (bulkDeriv pu)]
781   [symm_metric : Litlib.Y1984.urbantke1984integrability.Eq10_Symmetry]
782   (h_metric_eq10 : ∀ x : pu.bulk, ∃ (F F_dual : Fin 3 → Fin 4 → Fin 4 → ℂ) (epsilon3 : Fin 3 → Fin 3 → Fin 3
↪ → ℂ),
783     (∀ a mu nu, F a mu nu = - F a nu mu) ∧
784     (∀ a mu nu, F_dual a mu nu = - F_dual a nu mu) ∧
785     (∀ a b c, epsilon3 a b c = - epsilon3 b a c ∧ epsilon3 a b c = - epsilon3 a c b) ∧
786     (∀ mu nu, CGD.Gravity.urbantkeMetric (fun a b => curvatureSL2c pu.toUniverse.sd_sector a b x.val) mu nu =
787       (-1 / 6 : ℂ) * Finset.sum Finset.univ (fun a => Finset.sum Finset.univ (fun b => Finset.sum
↪ Finset.univ (fun c => Finset.sum Finset.univ (fun alpha => Finset.sum Finset.univ (fun beta =>
↪ epsilon3 a c b * F a mu alpha * F_dual c alpha beta * F b beta nu))))))
788   (h_smooth_g : ∀ i j, isSmooth (fun p => CGD.Gravity.urbantkeMetric (fun a b => curvatureSL2c
↪ pu.toUniverse.sd_sector a b p.val) i j))
789   (h_smooth_g_inv : ∀ i j, isSmooth (fun p => CGD.Gravity.matrixInv4x4 (fun m n =>
↪ CGD.Gravity.urbantkeMetric (fun a b => curvatureSL2c pu.toUniverse.sd_sector a b p.val) m n) i j))
790   (h_smooth_chris : ∀ rho mu nu, isSmooth (fun p => CGD.Gravity.christoffel (fun m n p' =>
↪ CGD.Gravity.urbantkeMetric (fun a b => curvatureSL2c pu.toUniverse.sd_sector a b p') m n) rho mu nu
↪ p.val))
791   (h_chris_eq : ∀ p : pu.bulk, ∀ rho mu nu, CGD.Gravity.christoffel (fun m n p' =>
↪ CGD.Gravity.urbantkeMetric (fun a b => curvatureSL2c pu.toUniverse.sd_sector a b p') m n) rho mu nu
↪ p.val =

```

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792 (1/2 : C) * ∑ sigma : Fin 4, CGD.Gravity.matrixInv4x4 (fun m n => CGD.Gravity.urbantkeMetric (fun a b =>
  ↳ curvatureSl2c pu.toUniverse.sd_sector a b p.val) m n) rho sigma * (
793   bulkDeriv pu mu (fun p' => CGD.Gravity.urbantkeMetric (fun a b => curvatureSl2c
    ↳ pu.toUniverse.sd_sector a b p'.val) sigma nu) p +
794   bulkDeriv pu nu (fun p' => CGD.Gravity.urbantkeMetric (fun a b => curvatureSl2c
    ↳ pu.toUniverse.sd_sector a b p'.val) mu sigma) p -
795   bulkDeriv pu sigma (fun p' => CGD.Gravity.urbantkeMetric (fun a b => curvatureSl2c
    ↳ pu.toUniverse.sd_sector a b p'.val) mu nu) p))
796 (h_ricci_eq : ∀ p : pu.bulk, ∀ mu nu, CGD.Gravity.ricciTensor (fun m n p' => CGD.Gravity.urbantkeMetric
  ↳ (fun a b => curvatureSl2c pu.toUniverse.sd_sector a b p') m n) mu nu p.val =
797   ∑ rho : Fin 4, (bulkDeriv pu rho (fun p' => CGD.Gravity.christoffel (fun m n p'' =>
    ↳ CGD.Gravity.urbantkeMetric (fun a b => curvatureSl2c pu.toUniverse.sd_sector a b p'') m n) rho mu nu
    ↳ p'.val) p -
798     bulkDeriv pu nu (fun p' => CGD.Gravity.christoffel (fun m n p'' => CGD.Gravity.urbantkeMetric (fun
      ↳ a b => curvatureSl2c pu.toUniverse.sd_sector a b p'') m n) rho mu rho p'.val) p +
799     ∑ lambda : Fin 4, (CGD.Gravity.christoffel (fun m n p' => CGD.Gravity.urbantkeMetric (fun a b =>
      ↳ curvatureSl2c pu.toUniverse.sd_sector a b p') m n) rho lambda rho p.val *
      ↳ CGD.Gravity.christoffel (fun m n p' => CGD.Gravity.urbantkeMetric (fun a b => curvatureSl2c
      ↳ pu.toUniverse.sd_sector a b p') m n) lambda mu nu p.val -
800       CGD.Gravity.christoffel (fun m n p' => CGD.Gravity.urbantkeMetric (fun a b => curvatureSl2c
        ↳ pu.toUniverse.sd_sector a b p') m n) rho lambda nu p.val * CGD.Gravity.christoffel (fun
        ↳ m n p' => CGD.Gravity.urbantkeMetric (fun a b => curvatureSl2c pu.toUniverse.sd_sector a
        ↳ b p') m n) lambda mu rho p.val))) :
801 ∀ nu (x : pu.bulk),
802   let g_urb := fun m n (p : pu.bulk) => CGD.Gravity.urbantkeMetric (fun a b => curvatureSl2c
    ↳ pu.toUniverse.sd_sector a b p.val) m n
803   let g_inv := fun m n (p : pu.bulk) => CGD.Gravity.matrixInv4x4 (fun a b => g_urb a b p) m n
804   let chris := fun rho mu nu (p : pu.bulk) => CGD.Gravity.christoffel (fun m n p' =>
    ↳ CGD.Gravity.urbantkeMetric (fun a b => curvatureSl2c pu.toUniverse.sd_sector a b p') m n) rho mu nu
    ↳ p.val
805   let T := fun m n (p : pu.bulk) => emergentStressEnergy (fun a b p' => curvatureSl2c
    ↳ pu.toUniverse.sd_sector a b p') m n p.val
806   ∑ mu : Fin 4, ∑ alpha : Fin 4, g_inv mu alpha x * (
807     bulkDeriv pu alpha (fun p => T mu nu p) x -
808     ∑ lambda : Fin 4, (chris lambda alpha mu x * T lambda nu x +
809       chris lambda alpha nu x * T mu lambda x)
810   ) = 0
811
812 -- The physical axiom of macroscopic volume strictly implies the bulk subspace is nonempty -/
813 instance CGD.Gravity.instPhysicalUniverseBulkNonempty (pu : PhysicalUniverse) : Nonempty pu.bulk :=

```

```

814   let (x, hx)
815
816
817 -- MODULE: CGD.Math.Calculus
818 -----
819 noncomputable def CGD.Math.partialDeriv {E : Type*}[NormedAddCommGroup E][NormedSpace ℝ E] (μ : Fin 4) (f :
  ⇨ SpacetimePoint → E) : SpacetimePoint → E :=
820   fun x => fderiv ℝ f x ((Pi.single μ (1 : ℝ)) : Fin 4 → ℝ)
821
822
823 -- MODULE: CGD.Math.Matrix
824 -----
825 noncomputable instance CGD.Math.matNormedAddCommGroup : NormedAddCommGroup (Matrix (Fin 2) (Fin 2) ℂ) :=
826   inferInstanceAs (NormedAddCommGroup (Fin 2 → Fin 2 → ℂ))
827
828 noncomputable instance CGD.Math.matNormedSpaceR : NormedSpace ℝ (Matrix (Fin 2) (Fin 2) ℂ) :=
829   inferInstanceAs (NormedSpace ℝ (Fin 2 → Fin 2 → ℂ))
830
831
832 -- MODULE: CGD.Particles.Color
833 -----
834 /--
835   Defines a gauge field that is constrained to a lower-dimensional Lie subalgebra,
836   missing at least one of the three internal color generators.
837 -/
838 def CGD.Particles.isColorDeficient (F : Fin 4 → Fin 4 → SL2C) (color : Fin 3) : Prop :=
839   ∀ mu nu, project F color mu nu = 0
840
841 /--
842   Demonstrates that if any of the three internal Lie algebra generators (colors) are missing from the gauge
843   ⇨ field, the scalar triple product natively vanishes, mathematically forcing the macroscopic spacetime
844   ⇨ volume to zero. A stable spacetime geometry mathematically requires the interaction of exactly all three
845   ⇨ SU(2) colors.
846 -/
847 @[litlib_track "Geometric Degeneracy of Color-Deficient Fields"]
848 theorem CGD.Particles.kinematicColorDeficientDegeneracy :
849   ∀ (F : Fin 4 → Fin 4 → SL2C) (color : Fin 3),
850     isColorDeficient F color →
851     (urbantkeMetric F).det = 0

```

```

850 /--
851 Demonstrates that a non-zero macroscopic spacetime volume strictly requires non-Abelian fields.
852 -/
853 @[litlib_track "Kinematic Multi-Color Requirement"]
854 theorem CGD.Particles.kinematicMultiColorRequirement :
855    $\forall (F : \text{Fin } 4 \rightarrow \text{Fin } 4 \rightarrow \text{SL2C}),$ 
856      $(\text{urbantkeMetric } F).det \neq 0 \rightarrow$ 
857      $\neg \text{isSingleColor } F$ 
858
859 /--
860 Demonstrates that the Urbantke metric determinant fundamentally requires non-commuting Lie algebra
861  $\hookrightarrow$  generators. For an Abelian (single-color) field, the Lie bracket vanishes, algebraically forcing the
862  $\hookrightarrow$  macroscopic spacetime volume to zero. Physical spacetime geometries therefore require non-Abelian fields
863  $\hookrightarrow$  (such as multi-color hadrons) to expand into stable configurations, geometrically manifesting color
864  $\hookrightarrow$  confinement.
865 -/
866 @[litlib_track "Metric Confinement of Abelian Fields"]
867 theorem CGD.Particles.kinematicSingleColorDegeneracy :
868    $\forall (F : \text{Fin } 4 \rightarrow \text{Fin } 4 \rightarrow \text{SL2C}),$ 
869      $\text{isSingleColor } F \rightarrow$ 
870      $(\text{urbantkeMetric } F).det = 0$ 
871
872 /--
873 Demonstrates that a non-zero macroscopic spacetime volume strictly requires non-Abelian fields spanning
874  $\hookrightarrow$  exactly three active color generators. One or two colors is mathematically insufficient to sustain
875  $\hookrightarrow$  spacetime volume, natively bounding the minimum unbroken gauge symmetry required for macroscopic
876  $\hookrightarrow$  existence.
877 -/
878 @[litlib_track "Kinematic Three-Color Requirement"]
879 theorem CGD.Particles.kinematicThreeColorRequirement :
880    $\forall (F : \text{Fin } 4 \rightarrow \text{Fin } 4 \rightarrow \text{SL2C}),$ 
881      $(\text{urbantkeMetric } F).det \neq 0 \rightarrow$ 
882      $\neg \text{isSingleColor } F \wedge (\forall \text{ color}, \neg \text{isColorDeficient } F \text{ color})$ 
883
884 -- MODULE: CGD.Particles.DarkMatter
885 -----
886 /--
887 Dark Matter Decoupling (The "Darkness" Theorem):
888 A pure Self-Dual defect (Dark Matter) is superposed onto the macroscopic

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```

883 vacuum of normal matter. Because the Spin(4,C) Lie algebra perfectly orthogonalizes
884 the chiral sectors, the Anti-Self-Dual curvature (which dictates electromagnetism
885 and the strong force) evaluates to being mathematically identical to the vacuum state
886 without the Dark Matter present.
887 -/
888 @[litlib_track "Dark Matter Anti-Self-Dual Decoupling (Darkness)"]
889 theorem CGD.Particles.kinematicDarkMatterDecoupling (pu : PhysicalUniverse)
890   (A_DM : Sl2cGaugeField) (x : SpacetimePoint) (mu nu : Fin 4) :
891   let A_tot := fun m p =>
892     embedSelfDual (A_DM.val m p + pu.toUniverse.sd_sector.val m p) +
893     embedAntiSelfDual (pu.toUniverse.asd_sector.val m p);
894   (chiralProject (curvature A_tot mu nu x)).anti_self_dual =
895   (chiralProject (curvature (fun m p => pu.toUniverse.spin4c_connection m p) mu nu x)).anti_self_dual
896
897 /--
898 Self-Interacting Dark Matter (SIDM):
899 When two SD-only defects occupy the same spacetime volume, their non-Abelian
900 SL(2, C) group structure intrinsically generates a non-linear cross-term in the curvature.
901 This strictly forces momentum exchange, proving that pure gravitational defects in CGD
902 natively behave as Self-Interacting Dark Matter (SIDM).
903 -/
904 @[litlib_track "Dark Matter Self-Interaction (SIDM)"]
905 theorem CGD.Particles.kinematicDarkMatterSelfInteraction
906   (A_1 A_2 : Sl2cGaugeField)
907   (x : SpacetimePoint) (mu nu : Fin 4) :
908   let A_tot := fun m p => A_1.val m p + A_2.val m p;
909   curvatureSl2c A_tot mu nu x =
910     curvatureSl2c A_1.val mu nu x +
911     curvatureSl2c A_2.val mu nu x +
912     [A_1.val mu x, A_2.val nu x] + [A_2.val mu x, A_1.val nu x]
913
914
915 -- MODULE: CGD.Particles.Definitions
916 -----
917 /--
918 A valid topological projection from the 4D Spacetime gauge field down to a 3D
919 spatial boundary. Because the input `Sl2cGaugeField` is natively guaranteed
920 to be smooth by the `PhysicalUniverse` axioms, this projection inherently
921 guarantees the differentiability and SU(2) bounds of the output matrix field.
922 -/

```

```

923 structure CGD.Particles.FermionBoundaryProjection where
924   map : SL2cGaugeField → ((Fin 3 → ℝ) → Matrix (Fin 2) (Fin 2) ℂ)
925   h_SU2 : ∀ A x, IsSU2 (map A x)
926   h_smooth : ∀ A i j, Differentiable ℝ (fun x => map A x i j)
927
928 /--
929 Encapsulates the mathematical operation of a  $2\pi$  adiabatic spatial rotation
930 on a topological state, required to evaluate quantum spin-statistics.
931 -/
932 structure CGD.Particles.IsFermionicRotation
933   (path_parity : (ℝ → (Fin 3 → ℝ) → Matrix (Fin 2) (Fin 2) ℂ) → ℕ)
934   (U_0 : (Fin 3 → ℝ) → Matrix (Fin 2) (Fin 2) ℂ)
935   (U_rot : ℝ → (Fin 3 → ℝ) → Matrix (Fin 2) (Fin 2) ℂ) : Prop where
936   h_rot_SU2 : ∀ t x, IsSU2 (U_rot t x)
937   h_is_2pi_rot : ∀ t x, U_rot t x = U_0 (rotZ t x)
938   h_fr_parity : path_parity U_rot = 1
939
940 /--
941 Encapsulates the strict mathematical prerequisites for a macroscopic gauge field
942 to constitute a topological fermion. By taking a `PhysicalUniverse` and a valid
943 projection, it derives its smoothness and SU(2) nature natively from the
944 universe's Anti-Self-Dual (Matter) sector, requiring only the assertion of
945 the topological boundary conditions (Vacuum at infinity, Baryon Number = 1).
946 -/
947 structure CGD.Particles.IsFermionicState
948   [MeasureTheory.MeasureSpace (Fin 3 → ℝ)]
949   (pu : PhysicalUniverse)
950   (proj : FermionBoundaryProjection) : Prop where
951   h_vacuum : AsymptoticVacuumSU2 (proj.map pu.toUniverse.asd_sector)
952   h_integrable : MeasureTheory.Integrable (BaryonNumberIntegrandSU2 (proj.map pu.toUniverse.asd_sector))
953   h_degree_one : baryon_number_SU2
954     (proj.map pu.toUniverse.asd_sector)
955     (proj.h_SU2 pu.toUniverse.asd_sector)
956     (proj.h_smooth pu.toUniverse.asd_sector)
957   h_vacuum h_integrable = 1
958
959 noncomputable def CGD.Particles.homogeneousChaosAnsatz (mu : Fin 4) (x : SpacetimePoint) : SL2C :=
960   if mu = 1 then (Complex.I * (x 1 : ℂ)) • sigma1
961   else if mu = 2 then (Complex.I * (x 2 : ℂ)) • sigma2
962   else 0

```

```

963
964 /--
965 Defines a single-color (Abelian) condensate where all components of the curvature tensor commute,
966   ↪ constraining the field to a single U(1) Cartan subalgebra.
967 -/
968 def CGD.Particles.isSingleColor (F : Fin 4 -> Fin 4 -> SL2C) : Prop :=
969   ∀ mu nu rho sigma, [F mu nu, F rho sigma] = 0
970
971 /--
972 The 4D Topological Soliton (Skyrmion) in the regular gauge.
973 Topologically localizes the gauge field.
974 Multiplied by Complex.I to map into the strictly anti-Hermitian su(2) Lie algebra.
975 Ontologically, this represents the spatial compactification of a 3D Cauchy slice (R3 U {infty} ≈ S3)
976 rather than a 4D Euclidean tunneling event.
977 -/
978 noncomputable def CGD.Particles.topologicalSoliton (mu : Fin 4) (x : SpacetimePoint) : SL2C :=
979   let r2 : ℂ := (x 0 : ℂ)^2 + (x 1 : ℂ)^2 + (x 2 : ℂ)^2 + (x 3 : ℂ)^2
980   let D : ℂ := r2 + 1
981   if mu = 0 then (Complex.I / D) • ((x 1 : ℂ) • sigma1 + (x 2 : ℂ) • sigma2 + (x 3 : ℂ) • sigma3)
982   else if mu = 1 then (Complex.I / D) • (-(x 0 : ℂ) • sigma1 - (x 3 : ℂ) • sigma2 + (x 2 : ℂ) • sigma3)
983   else if mu = 2 then (Complex.I / D) • ((x 3 : ℂ) • sigma1 - (x 0 : ℂ) • sigma2 - (x 1 : ℂ) • sigma3)
984   else if mu = 3 then (Complex.I / D) • (-(x 2 : ℂ) • sigma1 + (x 1 : ℂ) • sigma2 - (x 0 : ℂ) • sigma3)
985   else 0
986
987 -- MODULE: CGD.Particles.Electromagnetism
988 -----
989 /--
990 The true topological Abelian field strength tensor defined as a component
991 of the full non-Abelian curvature. Incorporates both the curl and the commutator,
992 enabling the dual divergence to measure the non-trivial topological magnetic current.
993  $F_{\{\mu\nu\}}^{\{ij\}} = \partial_\mu A_\nu^{\{ij\}} - \partial_\nu A_\mu^{\{ij\}} + [A_\mu, A_\nu]^{\{ij\}}$ 
994 -/
995 noncomputable def CGD.Particles.abelianFieldStrength (pu : PhysicalUniverse) (i j : Fin 4) (μ ν : Fin 4) (x
996   ↪ : SpacetimePoint) : ℂ :=
997   (partialDeriv μ (fun p => connectionComponent pu i j ν p) x - partialDeriv ν (fun p => connectionComponent
998     ↪ pu i j μ p) x) +
999   ∑ k : Fin 4, (connectionComponent pu i k μ x * connectionComponent pu k j ν x - connectionComponent pu i k
1000     ↪ ν x * connectionComponent pu k j μ x)

```

```

999  /--
1000  The emergent U(1) current based on the Duan-Ge (1979) topological decomposition.
1001  Defined as the dual divergence of the Abelian field strength tensor F_ρσ.
1002  J^μ = ε^{μνρσ} ∂_ν F_ρσ
1003  -/
1004  noncomputable def CGD.Particles.emergentElectricCurrent
1005    (F : Fin 4 → Fin 4 → SpacetimePoint → ℂ)
1006    (μ : Fin 4) (x : SpacetimePoint) : ℂ :=
1007    ∑ v : Fin 4, ∑ p : Fin 4, ∑ σ : Fin 4,
1008      epsilon4 μ v p σ * partialDeriv v (fun p => F p σ p) x
1009
1010  /--
1011  Because the physical Universe is axiomatically a smooth Spin(4,C) connection,
1012  its emergent Abelian projection natively satisfies the n-dimensional Clairaut theorem,
1013  guaranteeing exact topological charge conservation without further assumptions.
1014  -/
1015  @[litlib_track "Kinematic Charge Conservation"]
1016  theorem CGD.Particles.kinematicChargeConservation
1017    [clairaut : Litlib.Y1976.rudin1976principles.ClairautTheoremNDimensional]
1018    (pu : PhysicalUniverse) (i j : Fin 4) :
1019    ∀ x : SpacetimePoint,
1020      ∑ μ : Fin 4, partialDeriv μ (fun p => emergentElectricCurrent (abelianFieldStrength pu i j) μ p) x = 0
1021
1022  /--
1023  Because the emergent current is purely topological (defined via the Levi-Civita symbol),
1024  its divergence strictly vanishes due to the commutativity of partial derivatives,
1025  justified natively via n-dimensional Clairaut theorem on the Abelian field strength.
1026  -/
1027  @[litlib_track "Topological Charge Conservation"]
1028  theorem CGD.Particles.topologicalChargeConservation
1029    [clairaut : Litlib.Y1976.rudin1976principles.ClairautTheoremNDimensional]
1030    (F : Fin 4 → Fin 4 → SpacetimePoint → ℂ)
1031    (h_smooth_re : ∀ p σ, ContDiffOn ℝ 2 (fun p => (F p σ p).re) Set.univ)
1032    (h_smooth_im : ∀ p σ, ContDiffOn ℝ 2 (fun p => (F p σ p).im) Set.univ)
1033    (h_diff_F : ∀ p σ x, DifferentiableAt ℝ (fun p => F p σ p) x)
1034    (h_diff : ∀ v p σ x, DifferentiableAt ℝ (fun p => partialDeriv v (fun p' => F p σ p') p) x)
1035    (h_diff_fderiv_re : ∀ p σ x, DifferentiableAt ℝ (fderiv ℝ (fun p => (F p σ p).re)) x)
1036    (h_diff_fderiv_im : ∀ p σ x, DifferentiableAt ℝ (fderiv ℝ (fun p => (F p σ p).im)) x) :
1037    ∀ x : SpacetimePoint,
1038      ∑ μ : Fin 4, partialDeriv μ (fun p => emergentElectricCurrent F μ p) x = 0

```



```

1039
1040
1041 -- MODULE: CGD.Particles.Mass
1042 -----
1043 /--
1044 The topological rest mass is strictly defined as the absolute value of the Cartan-Maurer
1045 topological integral evaluated over the asymptotic spatial boundary of the Anti-Self-Dual (Matter) field.
1046
1047 **PHYSICAL ONTOLOGY:**
1048 For a stable Lorentzian particle, `BoundaryManifold` physically represents the spatial compactification
1049 ($S^3$) of a 3D Cauchy slice of spacetime at a fixed time $t$. The particle is therefore modeled strictly
1050 as a stable Lorentzian Soliton, not a transient Euclidean Instanton.
1051
1052 Because the non-compact boost directions of  $SL(2, C)$  form a contractible space ( $\mathbb{R}^3$ ), they possess
1053 zero topological winding number and strictly drop out of the boundary integral. This guarantees that
1054 exotic negative-mass boost states are kinematically excluded from the topological invariant, leaving
1055 only the strictly positive  $SU(2)$  rest mass charge.
1056 -/
1057 noncomputable def CGD.Particles.inertialMass
1058   {BoundaryManifold : Type*} [TopologicalSpace BoundaryManifold] [Nonempty BoundaryManifold]
1059   (boundaryMap : (Fin 4 → SpacetimePoint → SL2C) → BoundaryManifold → SU2Group)
1060   (cartanMaurerIntegral : (BoundaryManifold → SU2Group) → ℝ)
1061   (pu : PhysicalUniverse) : ℝ :=
1062     |cartanMaurerIntegral (boundaryMap pu.toUniverse.asd_sector.val)|
1063
1064 /--
1065 By applying the rigorous Cartan-Maurer topological integral to the  $SU(2)$  boundary mapping
1066 of the matter sector, we prove the topological origin of strictly positive inertial mass.
1067
1068 Because stable particles must constitute a topological homeomorphism to the gauge boundary
1069 on a given Cauchy surface, their integrated charge strictly evaluates to a non-zero integer.
1070 This mathematically verifies that stable topological Solitons in Chiral Gauge Dynamics intrinsically
1071 possess a non-zero, strictly positive Mass Gap, completely avoiding the vacuum instabilities
1072 of the non-compact  $SL(2, C)$  group.
1073 -/
1074 @[litlib_track "Topological Mass Gap"]
1075 theorem CGD.Particles.topologicalMassGap
1076   {BoundaryManifold : Type*} [TopologicalSpace BoundaryManifold] [Nonempty BoundaryManifold]
1077   (boundaryMap : (Fin 4 → SpacetimePoint → SL2C) → BoundaryManifold → SU2Group)
1078   (windingNumber : (BoundaryManifold → SU2Group) → ℤ)

```

```

1079 (cartanMaurerIntegral : (BoundaryManifold → SU2Group) → ℝ)
1080 [tc : CartanMaurerTopology (BoundaryManifold → SU2Group) Continuous windingNumber cartanMaurerIntegral]
1081 [belavin : Eq8 BoundaryManifold SU2Group Continuous windingNumber cartanMaurerIntegral]
1082 (pu : PhysicalUniverse)
1083 (h_homeo : IsHomeomorphism (boundaryMap pu.toUniverse.asd_sector.val)) :
1084 inertialMass boundaryMap cartanMaurerIntegral pu > 0
1085
1086
1087 -- MODULE: CGD.Particles.Matter
1088 -----
1089 /--
1090 Proves that the CGD ontology is not an empty vacuum. A topological connection
1091 with a non-zero anti-self-dual curvature natively generates a strictly non-zero
1092 physical Stress-Energy tensor ( $T_{\mu\nu} \neq 0$ ), representing the presence of matter.
1093
1094 This relies strictly on the Plebanski expansion, proving that if the physical
1095 energy-momentum tensor vanishes, the anti-self-dual expansion coefficients vanish,
1096 extinguishing the topological matter field entirely.
1097 -/
1098 @[litlib_track "Dynamic Matter Existence"]
1099 theorem CGD.Particles.dynamicMatterExistence
1100 (pu : PhysicalUniverse)
1101 (x : SpacetimePoint)
1102 (Sigma Sigma_bar : Fin 3 → Fin 4 → Fin 4 → ℂ)
1103
1104 -- The true Plebanski expansion coefficients
1105 (F_ij F_bar_ij T_ij : Fin 3 → Fin 3 → ℂ)
1106
1107 (Lambda G T_scalar : ℂ)
1108 (plebanski_matter_eqs : Prop)
1109 (h_G : G ≠ 0)
1110
1111 -- Vector representation of the SL(2,C) curvature
1112 (eval_SL2C : SL2C → Fin 3 → ℂ)
1113 (h_eval_inj : ∀ A, (∀ i, eval_SL2C A i = 0) → A = 0)
1114
1115 -- Plebanski decomposition of the physical anti-self-dual curvature
1116 (h_F_asd_decomp : ∀ μ ν i, eval_SL2C (curvatureSL2c pu.toUniverse.asd_sector μ ν x) i = ∑ j, F_bar_ij i j
1117 ↦ * Sigma_bar j μ ν)

```

```

1118 -- The physics: Eq16 relates the emergent CGD Stress-Energy to the internal T_ij
1119 (eq16 : Eq16
1120   Sigma
1121   Sigma_bar
1122   (fun μ v => matrixInv4x4 (fun m n => urbantkeMetric (fun a b => curvatureSl2c pu.toUniverse.sd_sector a
1123     ↪ b x) m n) μ v)
1123   (fun μ v => emergentStressEnergy (fun a b p => curvatureSl2c pu.toUniverse.sd_sector a b p) μ v x)
1124   T_ij)
1125
1126 -- The physics: Eq17 directly binds the expansion coefficients to the Stress-Energy
1127 (eq17 : Eq17
1128   Lambda G F_ij F_bar_ij T_scalar T_ij plebanski_matter_eqs)
1129 (h_matter : plebanski_matter_eqs)
1130
1131 -- The non-vacuum condition: The physical matter curvature is non-zero
1132 (h_non_vacuum : ∃ μ v, curvatureSl2c pu.toUniverse.asd_sector μ v x ≠ 0) :
1133
1134   ∃ p μ, emergentStressEnergy (fun a b p => curvatureSl2c pu.toUniverse.sd_sector a b p) p μ x ≠ 0
1135
1136
1137 -- MODULE: CGD.Particles.SpinStatistics
1138 -----
1139 /--
1140 While the local geometric spinor is a commuting Kähler-Dirac field,
1141 the macroscopic topological defect natively possesses fermionic spin-statistics.
1142 Via the Finkelstein-Rubinstein theorem (explicitly cited by Witten 1983 for the SU(2) exception),
1143 the 2π adiabatic spatial rotation of a degree-1 SU(2) topological soliton
1144 traces the non-trivial element of π4(SU(2)) = Z2. The resulting quantum
1145 phase amplitude evaluates strictly to -1.
1146
1147 Because this evaluates directly against the Physical Universe's Anti-Self-Dual (Matter)
1148 sector via a valid topological projection, the native smoothness (ContDiff) of the
1149 spacetime gauge field natively fulfills the differentiability requirements without
1150 external axioms.
1151 -/
1152 @[litlib_track "Macroscopic Topological Fermionization"]
1153 theorem CGD.Particles.kinematicMacroscopicFermion
1154   [MeasureTheory.MeasureSpace (Fin 3 → ℝ)]
1155   (pu : PhysicalUniverse)
1156   (proj : FermionBoundaryProjection)

```

```

1157 (state : IsFermionicState pu proj)
1158 (path_parity : ( $\mathbb{R} \rightarrow (\text{Fin } 3 \rightarrow \mathbb{R}) \rightarrow \text{Matrix } (\text{Fin } 2) (\text{Fin } 2) \mathbb{C} \rightarrow \mathbb{N}$ )
1159 (U_rot :  $\mathbb{R} \rightarrow (\text{Fin } 3 \rightarrow \mathbb{R}) \rightarrow \text{Matrix } (\text{Fin } 2) (\text{Fin } 2) \mathbb{C}$ )
1160 (rotation : IsFermionicRotation path_parity (proj.map pu.toUniverse.asd_sector) U_rot)
1161 (h_fr : FinkelsteinRubinsteinQuantization path_parity
1162   (proj.map pu.toUniverse.asd_sector)
1163   (proj.h_SU2 pu.toUniverse.asd_sector)
1164   (proj.h_smooth pu.toUniverse.asd_sector)
1165   state.h_vacuum state.h_integrable state.h_degree_one
1166   U_rot rotation.h_rot_SU2 rotation.h_is_2pi_rot rotation.h_fr_parity) :
1167 quantum_weight path_parity U_rot = -1
1168
1169
1170 -- MODULE: CGD.Particles.TopologicalStability
1171 -----
1172 instance : Nonempty S3
1173
1174 instance : Nonempty SU2Group
1175
1176 noncomputable instance : TopologicalSpace SU2Group := instTopologicalSpaceSubtype
1177
1178 /--
1179 A valid topological homotopy must maintain the integrity of the physical boundary state.
1180 If the continuous mapping to the SU(2) topology breaks, the state has undergone a violent
1181 topological decay.
1182 -/
1183 def CGD.Particles.isHomotopicConnection (A0 A1 : Fin 4 → SpacetimePoint → SL2C) : Prop :=
1184   ∃ (H :  $\mathbb{R} \rightarrow \text{Fin } 4 \rightarrow \text{SpacetimePoint} \rightarrow \text{SL2C}$ ),
1185     (∀ mu x, H 0 mu x = A0 mu x) ∧
1186     (∀ mu x, H 1 mu x = A1 mu x) ∧
1187     (∀ mu i j, ContDiff  $\mathbb{R} \top$  (fun (tx :  $\mathbb{R} \times \text{SpacetimePoint}$ ) => (H tx.1 mu tx.2).val i j)) ∧
1188     (Continuous (fun t => geometricBoundaryProjection (H t)))
1189
1190 /--
1191 Establishes the absolute topological stability of the instanton configuration. Because the boundary mapping
1192 ⇨ of the instanton constitutes a topological homeomorphism to the SU(2) gauge group, its Cartan-Maurer
1193 ⇨ mapping degree must be strictly non-zero. Consequently, the state is topologically protected and cannot
1194 ⇨ continuously decay into the trivial vacuum.
1195 -/
1196 @[litlib_track "Topological Stability of the Instanton"]

```

```

1194 theorem CGD.Particles.kinematicTopologicalStability
1195   (windingNumber : (S3 → SU2Group) → ℤ)
1196   (cartanMaurerIntegral : (S3 → SU2Group) → ℝ)
1197   [tc : CartanMaurerTopology (S3 → SU2Group) Continuous windingNumber cartanMaurerIntegral]
1198   [belavin : Eq8 S3 SU2Group Continuous windingNumber cartanMaurerIntegral]
1199   (integral_zero : cartanMaurerIntegral 1 = 0) :
1200   → isHomotopicConnection topologicalSoliton 0
1201
1202 noncomputable def CGD.Particles.solitonAsymptoticMap (x : S3) : SU2Group := by
1203   cases x
1204   rename_i v hv
1205
1206   let a : ℂ := Complex.mk (v 0) (-(v 3))
1207   let b : ℂ := Complex.mk (-(v 2)) (-(v 1))
1208
1209   have h_sum : a * star a + b * star b = 1 := by
1210     apply Complex.ext
1211     · change (v 0) * (v 0) - (-(v 3)) * (-(-(v 3))) + ((-(v 2)) * (-(v 2)) - (-(v 1)) * (-(-(v 1)))) = 1
1212     calc (v 0) * (v 0) - (-(v 3)) * (-(-(v 3))) + ((-(v 2)) * (-(v 2)) - (-(v 1)) * (-(-(v 1))))
1213       _ = (v 0)^2 + (v 1)^2 + (v 2)^2 + (v 3)^2 := by ring
1214       _ = 1 := hv
1215     · change (v 0) * (-(-(v 3))) + (-(v 3)) * (v 0) + ((-(v 2)) * (-(-(v 1))) + (-(v 1)) * (-(v 2))) = 0
1216     ring
1217
1218   let M : Matrix (Fin 2) (Fin 2) ℂ := !![a, b; -star b, star a]
1219   have h_prop : M * M.conjTranspose = 1 ∧ M.det = 1 := su2_matrix_prop a b h_sum
1220
1221   exact ⟨M, h_prop⟩
1222
1223 @[litlib_track "Topological Soliton Is Homeomorphism"]
1224 theorem CGD.Particles.soliton_is_homeomorphism : IsHomeomorphism solitonAsymptoticMap
1225
1226
1227 -- MODULE: CGD.Phenomenology.AxialCondensate
1228 -----
1229 /-- The pure Axial Chiral field, defined as the antisymmetric superposition of the left and right sectors. -/
1230 noncomputable def CGD.Phenomenology.axialField (u : Universe) (mu : Fin 4) (x : SpacetimePoint) : Matrix
1231   ↪ (Fin 2) (Fin 2) ℂ :=
1232   (1 / 2 : ℂ) • ((u.sd_sector mu x).val - (u.asd_sector mu x).val)

```

```

1233 /--
1234 Proves that the Axial field is strictly an isovector (isospin 1).
1235 It inherits the exact traceless property of the SL(2,C) algebra constituents,
1236 confining it to the adjoint representation of the SU(2) color group.
1237 -/
1238 @[litlib_track "Axial Is Isovector"]
1239 theorem CGD.Phenomenology.axialIsIsovector (pu : PhysicalUniverse) (mu : Fin 4) (x : SpacetimePoint) :
1240   Matrix.trace (axialField pu.toUniverse mu x) = 0
1241
1242 /--
1243 Proves that the Axial field behaves strictly as a pseudo-vector (parity-odd).
1244 When the geometry is parity-inverted, the Axial field exactly flips its algebraic sign.
1245 -/
1246 @[litlib_track "Axial Is Parity Odd"]
1247 theorem CGD.Phenomenology.axialIsParityOdd (pu : PhysicalUniverse) (mu : Fin 4) (x : SpacetimePoint) :
1248   axialField (paritySwap pu.toUniverse) mu x = - axialField pu.toUniverse mu x
1249
1250 /--
1251 Generic gauge-covariant derivative action on a fundamental state.
1252 Represented abstractly for a matrix-valued state `Psi` and its bare partial derivative `partial_Psi`.
1253 -/
1254 noncomputable def CGD.Phenomenology.covariantStateDeriv (partial_Psi : Matrix (Fin 2) (Fin 2) ℂ) (A : Matrix
1255   ↪ (Fin 2) (Fin 2) ℂ) (Psi : Matrix (Fin 2) (Fin 2) ℂ) : Matrix (Fin 2) (Fin 2) ℂ :=
1256   partial_Psi + A * Psi
1257
1258 /--
1259 Proves that the Left-Handed (Self-Dual) gauge connection, which natively couples to
1260 left-handed fermions (such as neutrinos) in the chiral Dirac framework, perfectly
1261 decomposes into a symmetric Vector background and the P-violating Axial condensate.
1262
1263 This establishes that left-handed states must continuously interact with the
1264 axial condensate as a refractive background. Because this interaction occurs purely
1265 at the level of the connection (zero Plebanski curvature), it geometrically derives
1266 kinematic flavor oscillation (the MSW effect) without generating gravitational rest mass.
1267 -/
1268 @[litlib_track "Kinematic Left-Handed Coupling"]
1269 theorem CGD.Phenomenology.kinematicLeftHandedCoupling (pu : PhysicalUniverse) (mu : Fin 4) (x :
1270   ↪ SpacetimePoint) :
1271   (pu.toUniverse.sd_sector mu x).val = vectorField pu.toUniverse mu x + axialField pu.toUniverse mu x

```

```

1271 /--
1272 Proves that the covariant derivative of a state under the Left-Handed (Self-Dual)
1273 connection mathematically evaluates to the bare partial derivative, plus the
1274 interaction with the symmetric Vector background, plus the interaction with the
1275 P-violating Axial condensate.
1276
1277 This formalizes the algebraic substitution allowing the paper to definitively claim
1278 that left-handed fermions actively interact with the axial condensate in the
1279 covariant Dirac equation.
1280 -/
1281 @[litlib_track "Kinematic Left-Handed Phase Shift"]
1282 theorem CGD.Phenomenology.kinematicLeftHandedPhaseShift (pu : PhysicalUniverse) (mu : Fin 4) (x :
  ↪ SpacetimePoint)
1283   (Psi partial_Psi : Matrix (Fin 2) (Fin 2) ℂ) :
1284   covariantStateDeriv partial_Psi (pu.toUniverse.sd_sector mu x).val Psi =
1285   partial_Psi + vectorField pu.toUniverse mu x * Psi + axialField pu.toUniverse mu x * Psi
1286
1287 /--
1288 Proves that the Right-Handed (Anti-Self-Dual) gauge connection decomposes into the
1289 symmetric Vector background minus the P-violating Axial condensate. The sign flip
1290 relative to the Left-Handed sector strictly defines the geometric parity asymmetry
1291 of the chiral spacetime vacuum.
1292 -/
1293 @[litlib_track "Kinematic Right-Handed Coupling"]
1294 theorem CGD.Phenomenology.kinematicRightHandedCoupling (pu : PhysicalUniverse) (mu : Fin 4) (x :
  ↪ SpacetimePoint) :
1295   (pu.toUniverse.asd_sector mu x).val = vectorField pu.toUniverse mu x - axialField pu.toUniverse mu x
1296
1297 /--
1298 Because the macroscopic volume emergent metric strictly forbids global chiral symmetry
1299 (via `macroscopicVolumeImpliesChirality`), this mathematically guarantees that the spacetime
1300 background is natively populated by a strictly non-zero Axial-Vector condensate. Empty
1301 space spontaneously generates the axial field to preserve its volume.
1302 -/
1303 @[litlib_track "Macroscopic Volume Implies Axial Condensate"]
1304 theorem CGD.Phenomenology.macroscopicVolumeImpliesAxialCondensate
1305   (pu : PhysicalUniverse) (x : SpacetimePoint) (hx : x ∈ pu.bulk)
1306   (h_vacuum : ∀ μ ν, curvatureSl2c pu.toUniverse.asd_sector.val μ ν x = 0) :
1307   ∃ y mu, axialField pu.toUniverse mu y ≠ 0
1308

```

```

1309 /--
1310 A discrete geometric transformation that completely inverts the orientation of the
1311 spacetime manifold by swapping the Chiral left (self-dual) and right (anti-self-dual) sectors.
1312 -/
1313 noncomputable def CGD.Phenomenology.paritySwap (u : Universe) : Universe :=
1314   universeEquiv.symm (u.asd_sector, u.sd_sector)
1315
1316 /-- The pure Vector Chiral field, defined as the symmetric superposition of the left and right sectors. -/
1317 noncomputable def CGD.Phenomenology.vectorField (u : Universe) (mu : Fin 4) (x : SpacetimePoint) : Matrix
1318   ↪ (Fin 2) (Fin 2) ℂ :=
1319     (1 / 2 : ℂ) • ((u.sd_sector mu x).val + (u.asd_sector mu x).val)
1320
1321 -- MODULE: CGD.Phenomenology.Chirality
1322 -----
1323 /--
1324 Proves that a perfectly symmetric, non-chiral universe mathematically destroys itself.
1325
1326 If the Left (Gravity) and Right (Matter) gauge fields are identical in a pure vacuum
1327 (where the Anti-Self-Dual curvature is zero), the Self-Dual (Gravity) curvature is also
1328 extinguished. A zero gravity curvature generates a zero Urbantke metric, which algebraically
1329 forces det(g) = 0. This strictly violates the Macroscopic Volume axiom.
1330 Therefore, empty space must be chiral.
1331 -/
1332 @[litlib_track "Macroscopic volume must have chirality"]
1333 theorem CGD.Phenomenology.macroscopicVolumeImpliesChirality
1334   (pu : PhysicalUniverse)
1335   (x : SpacetimePoint)
1336   (hx : x ∈ pu.bulk)
1337   (h_vacuum : ∀ μ ν, curvatureSl2c pu.toUniverse.asd_sector.val μ ν x = 0) :
1338   pu.toUniverse.sd_sector.val ≠ pu.toUniverse.asd_sector.val
1339
1340
1341 -- MODULE: CGD.Phenomenology.TmdSignFlips
1342 -----
1343 /--
1344 The Boer-Mulders effect observable (Quark Transverse Spin vs Quark Transverse Momentum).
1345 Projects the orthogonal SU(2) topology via sigmaX.
1346 Geometrically identical to the Sivers projection.
1347 -/

```



```

1348 noncomputable def CGD.Phenomenology.boerMuldersTransverseKick (U U_inv : Matrix (Fin 2) (Fin 2) ℂ) : ℂ :=
1349   Matrix.trace (U * explicitSigmaX * U_inv * explicitSigmaY)
1350
1351 /--
1352   Proves that when evaluating continuous non-Abelian geometry, the Siverts and Boer-Mulders effects yield
1353   ↪ topologically identical observables.
1354
1355   While perturbative QCD treats them as independent non-perturbative functions, phenomenological
1356   models (like Large-Nc and lattice QCD) observe strong proportionalities. This witness demonstrates how
1357   ↪ continuous background geometries intrinsically explain this: the macroscopic SU(2) holonomy is strictly
1358   ↪ blind to the composite vs. bare nature of the initial state, resolving both to the exact same geometric
1359   ↪ projection.
1360 -/
1361 @[litlib_track "Kinematic Siverts and Boer-Mulders Equivalence Witness"]
1362 theorem CGD.Phenomenology.kinematicSivertsBoerMuldersEquivalence (pu : CGD.Axioms.PhysicalUniverse) :
1363   ∀ (matrixExp : Matrix (Fin 2) (Fin 2) ℂ → Matrix (Fin 2) (Fin 2) ℂ)
1364     [Litlib.Y2000.hall2000elementary.DerivativeExponential (Fin 2) matrixExp]
1365     (alpha L : ℝ),
1366     boerMuldersTransverseKick
1367       (CGD.Quantum.macroscopicObservable (CGD.Quantum.holonomy matrixExp) (fun mu p =>
1368         ↪ CGD.Quantum.rotateYAxis (fun m p => pu.toUniverse.sd_sector m p) alpha mu p) 1 L)
1369       (CGD.Quantum.macroscopicObservable (CGD.Quantum.holonomy matrixExp) (fun mu p =>
1370         ↪ CGD.Quantum.rotateYAxis (fun m p => pu.toUniverse.sd_sector m p) alpha mu p) 1 (-L)) =
1371     sivertsTransverseKick
1372       (CGD.Quantum.macroscopicObservable (CGD.Quantum.holonomy matrixExp) (fun mu p =>
1373         ↪ CGD.Quantum.rotateYAxis (fun m p => pu.toUniverse.sd_sector m p) alpha mu p) 1 L)
1374       (CGD.Quantum.macroscopicObservable (CGD.Quantum.holonomy matrixExp) (fun mu p =>
1375         ↪ CGD.Quantum.rotateYAxis (fun m p => pu.toUniverse.sd_sector m p) alpha mu p) 1 (-L))
1376
1377 /--
1378   Evaluating the Siverts observable upon the `fluxTubeFrame` ansatz natively forces an exact geometric sign-flip
1379   ↪ upon path inversion (L -> -L).
1380
1381   While standard perturbative QCD derivations require external factorized gauge links to explain this effect,
1382   ↪ this theorem acts as an explicit constructive witness demonstrating that the continuous macroscopic gauge
1383   ↪ geometry intrinsically contains the parity-inverted mechanisms required for the Siverts effect.
1384 -/
1385 @[litlib_track "Kinematic Siverts Sign Flip Witness"]
1386 theorem CGD.Phenomenology.kinematicSivertsSignFlip (pu : CGD.Axioms.PhysicalUniverse) :
1387   ∀ (matrixExp : Matrix (Fin 2) (Fin 2) ℂ → Matrix (Fin 2) (Fin 2) ℂ)
1388     [Litlib.Y2000.hall2000elementary.DerivativeExponential (Fin 2) matrixExp]

```

```

1377     (alpha L : ℝ),
1378     (∀ t, pu.toUniverse.sd_sector 1 (CGD.Quantum.straightLinePath t) = CGD.Quantum.fluxTubeFrame 1
1379     ↪ (CGD.Quantum.straightLinePath t)) →
1380     siversTransverseKick
1381     (CGD.Quantum.macroscopicObservable (CGD.Quantum.holonomy matrixExp) (fun mu p =>
1382     ↪ CGD.Quantum.rotateYAxis (fun m p => pu.toUniverse.sd_sector m p) alpha mu p) 1 L)
1383     (CGD.Quantum.macroscopicObservable (CGD.Quantum.holonomy matrixExp) (fun mu p =>
1384     ↪ CGD.Quantum.rotateYAxis (fun m p => pu.toUniverse.sd_sector m p) alpha mu p) 1 (-L)) =
1385     - siversTransverseKick
1386     (CGD.Quantum.macroscopicObservable (CGD.Quantum.holonomy matrixExp) (fun mu p =>
1387     ↪ CGD.Quantum.rotateYAxis (fun m p => pu.toUniverse.sd_sector m p) alpha mu p) 1 (-L))
1388     (CGD.Quantum.macroscopicObservable (CGD.Quantum.holonomy matrixExp) (fun mu p =>
1389     ↪ CGD.Quantum.rotateYAxis (fun m p => pu.toUniverse.sd_sector m p) alpha mu p) 1 L)
1390
1391 /--
1392 The Topological TMD Geometric Ratio Witness.
1393
1394 By evaluating the `fluxTubeFrame` ansatz, this theorem acts as a constructive witness proving that the
1395 ↪ Worm-Gear and Sivers observables can natively become geometrically locked by the chiral phase angle
1396 `alpha` of the macroscopic connection. Specifically:
1397 `cos(alpha) * WormGear = - sin(alpha) * Sivers`
1398
1399 Because the observables are defined natively as geometric integrals without any
1400 collisional momentum variables, this geometric lock demonstrates a framework mechanism where their ratio
1401 resolves as a flat kinematic constant (-tan(alpha)), reproducing global supercomputer fits.
1402
1403 -/
1404 @[litlib_track "Topological TMD Geometric Ratio Witness"]
1405 theorem CGD.Phenomenology.kinematicTmdRatio (pu : CGD.Axioms.PhysicalUniverse) :
1406     ∀ (matrixExp : Matrix (Fin 2) (Fin 2) ℂ → Matrix (Fin 2) (Fin 2) ℂ)
1407     [Litlib.Y2000.hall2000elementary.DerivativeExponential (Fin 2) matrixExp]
1408     (alpha L : ℝ),
1409     (∀ t, pu.toUniverse.sd_sector 1 (CGD.Quantum.straightLinePath t) = CGD.Quantum.fluxTubeFrame 1
1410     ↪ (CGD.Quantum.straightLinePath t)) →
1411     (obs_M_A alpha) * wormGearTransverseKick
1412     (CGD.Quantum.macroscopicObservable (CGD.Quantum.holonomy matrixExp) (fun mu p =>
1413     ↪ CGD.Quantum.rotateYAxis (fun m p => pu.toUniverse.sd_sector m p) alpha mu p) 1 L)
1414     (CGD.Quantum.macroscopicObservable (CGD.Quantum.holonomy matrixExp) (fun mu p =>
1415     ↪ CGD.Quantum.rotateYAxis (fun m p => pu.toUniverse.sd_sector m p) alpha mu p) 1 (-L)) =
1416     - (obs_M_B alpha) * siversTransverseKick

```

```

1407     (CGD.Quantum.macroscopicObservable (CGD.Quantum.holonomy matrixExp) (fun mu p =>
    ↪ CGD.Quantum.rotateYAxis (fun m p => pu.toUniverse.sd_sector m p) alpha mu p) 1 L)
1408     (CGD.Quantum.macroscopicObservable (CGD.Quantum.holonomy matrixExp) (fun mu p =>
    ↪ CGD.Quantum.rotateYAxis (fun m p => pu.toUniverse.sd_sector m p) alpha mu p) 1 (-L))

1409
1410 /--
1411 Evaluating the Worm-Gear observable upon the `fluxTubeFrame` ansatz establishes that it obeys the exact same
    ↪ geometric sign-flip mechanics as the Siverts effect.
1412 -/
1413 @[litlib_track "Kinematic Worm-Gear Sign Flip Witness"]
1414 theorem CGD.Phenomenology.kinematicWormGearSignFlip (pu : CGD.Axioms.PhysicalUniverse) :
1415   ∀ (matrixExp : Matrix (Fin 2) (Fin 2) ℂ → Matrix (Fin 2) (Fin 2) ℂ)
1416     [Litlib.Y2000.hall2000elementary.DerivativeExponential (Fin 2) matrixExp]
1417     (alpha L : ℝ),
1418     (∀ t, pu.toUniverse.sd_sector 1 (CGD.Quantum.straightLinePath t) = CGD.Quantum.fluxTubeFrame 1
    ↪ (CGD.Quantum.straightLinePath t)) →
1419     wormGearTransverseKick
1420     (CGD.Quantum.macroscopicObservable (CGD.Quantum.holonomy matrixExp) (fun mu p =>
    ↪ CGD.Quantum.rotateYAxis (fun m p => pu.toUniverse.sd_sector m p) alpha mu p) 1 L)
1421     (CGD.Quantum.macroscopicObservable (CGD.Quantum.holonomy matrixExp) (fun mu p =>
    ↪ CGD.Quantum.rotateYAxis (fun m p => pu.toUniverse.sd_sector m p) alpha mu p) 1 (-L)) =
1422     - wormGearTransverseKick
1423     (CGD.Quantum.macroscopicObservable (CGD.Quantum.holonomy matrixExp) (fun mu p =>
    ↪ CGD.Quantum.rotateYAxis (fun m p => pu.toUniverse.sd_sector m p) alpha mu p) 1 (-L))
1424     (CGD.Quantum.macroscopicObservable (CGD.Quantum.holonomy matrixExp) (fun mu p =>
    ↪ CGD.Quantum.rotateYAxis (fun m p => pu.toUniverse.sd_sector m p) alpha mu p) 1 L)

1425
1426 noncomputable def CGD.Phenomenology.obs_M_A (alpha : ℝ) : ℂ := (Complex.cos (alpha/2))^2 - (Complex.sin
    ↪ (alpha/2))^2
1427
1428 noncomputable def CGD.Phenomenology.obs_M_B (alpha : ℝ) : ℂ := 2 * (Complex.cos (alpha/2)) * (Complex.sin
    ↪ (alpha/2))

1429
1430 /--
1431 The Siverts effect observable (Nucleon Transverse Spin vs Quark Transverse Momentum).
1432 Projects the orthogonal SU(2) topology via sigmaX.
1433 -/
1434 noncomputable def CGD.Phenomenology.sivertsTransverseKick (U U_inv : Matrix (Fin 2) (Fin 2) ℂ) : ℂ :=
1435     Matrix.trace (U * explicitSigmaX * U_inv * explicitSigmaY)
1436

```

```

1437 /--
1438   The Worm-Gear effect observable (Quark Longitudinal Spin vs Quark Transverse Momentum).
1439   Projects the orthogonal SU(2) topology via sigmaZ.
1440 -/
1441 noncomputable def CGD.Phenomenology.wormGearTransverseKick (U U_inv : Matrix (Fin 2) (Fin 2) ℂ) : ℂ :=
1442   Matrix.trace (U * explicitSigmaZ * U_inv * explicitSigmaY)
1443
1444
1445 -- MODULE: CGD.Quantum.ActionQuantization
1446 -----
1447 /--
1448   If a physical connection yields an asymptotic boundary map that is a topological homeomorphism to the gauge
1449   ↪ group, its Cartan-Maurer topological charge strictly evaluates to an integer quantization bound (±1).
1450
1451   **PHYSICAL ONTOLOGY & METRIC INDEPENDENCE:**
1452   The `BoundaryManifold` is intentionally left as an abstract generic type to mathematically satisfy both
1453   regimes of Chiral Gauge Dynamics without metric dependence:
1454   1. **The Euclidean Instanton (Tunneling):** If `BoundaryManifold` is the  $S^3$  boundary of 4D spacetime
1455       ($r \rightarrow \infty$ in all directions, including time), this theorem strictly reproduces the Belavin 1975
1456       instanton action quantization for transient vacuum transitions.
1457   2. **The Lorentzian Soliton (Stable Matter):** If `BoundaryManifold` is the  $S^3$  spatial compactification
1458       of a 3D Cauchy surface at a fixed time $t$, this theorem quantizes the invariant topological charge
1459       of a stable particle (a Skyrmion/Soliton) persisting through macroscopic Lorentzian time.
1460
1461   Because the Cartan-Maurer degree theorem depends only on the continuous boundary mapping and is strictly
1462   ↪ blind
1463   to the bulk metric signature, the exact same geometric quantization mathematically governs both phenomena.
1464 -/
1465 @[litlib_track "Topological Action Quantization"]
1466 theorem CGD.Quantum.kinematicActionQuantization
1467   {BoundaryManifold : Type*} [TopologicalSpace BoundaryManifold] [Nonempty BoundaryManifold]
1468   (boundaryMap : (Fin 4 → SpacetimePoint → SL2C) → BoundaryManifold → SU2Group)
1469   (windingNumber : (BoundaryManifold → SU2Group) → ℤ)
1470   (cartanMaurerIntegral : (BoundaryManifold → SU2Group) → ℝ)
1471   [tc : CartanMaurerTopology (BoundaryManifold → SU2Group) (Continuous : (BoundaryManifold → SU2Group) →
1472     ↪ Prop) windingNumber cartanMaurerIntegral]
1473   [belavin : Eq8 BoundaryManifold SU2Group (Continuous : (BoundaryManifold → SU2Group) → Prop) windingNumber
1474     ↪ cartanMaurerIntegral]
1475   (pu : PhysicalUniverse)
1476   (h_homeo : IsHomeomorphism (boundaryMap pu.toUniverse.sd_sector.val)) :

```

```

1473   cartanMaurerIntegral (boundaryMap pu.toUniverse.sd_sector.val) = 1 v
1474   cartanMaurerIntegral (boundaryMap pu.toUniverse.sd_sector.val) = -1
1475
1476
1477   -- MODULE: CGD.Quantum.Definitions
1478   -----
1479   /--
1480   Identifies a non-local topological correlation. Two spatial boundaries are defined
1481   as entangled if they are connected by a continuous degenerate connection (a flux tube).
1482   This definition relies on the exact `fluxTubeFrame` witness to guarantee non-vacuous evaluation.
1483
1484   **PHYSICAL CONSERVATION & THE BIANCHI GATEKEEPER:**
1485   Because the universe lacks dynamical Euler-Lagrange equations, the formation and persistence
1486   of these degenerate connections are strictly governed by the exact Differential Bianchi Identity
1487   (`kinematicBianchiIdentity`). The Bianchi constraint ( $\nabla \cdot B = 0$  analogue) guarantees
1488   topological flux conservation, mathematically preventing arbitrary spontaneous flux formation
1489   in the vacuum ("flux soup"). It enforces that tubes must strictly anchor to paired topological
1490   defects (particles) with equal and opposite winding numbers, geometrically reproducing
1491   quantum pair production and bipartite entanglement.
1492   -/
1493   @[litlib_track "Entangled State Witness"]
1494   def CGD.Quantum.areEntangled (A : Fin 4 → SpacetimePoint → SL2C) (x y : SpacetimePoint) (theta : ℝ) : Prop :=
1495     ∃ (γ : ℝ → SpacetimePoint) (θ : ℝ → ℝ),
1496       γ 0 = x ∧ γ 1 = y ∧
1497       θ 0 = 0 ∧ θ 1 = theta ∧
1498       ∀ t : ℝ, 0 ≤ t → t ≤ 1 →
1499         (∀ mu, A mu (γ t) = rotateYAxis fluxTubeFrame (θ t) mu (γ t)) ∧
1500         (∀ mu nu, partialDerivSL2c nu (A mu) (γ t) = partialDerivSL2c nu (rotateYAxis fluxTubeFrame (θ t) mu)
1501           ↦ (γ t))
1502   /--
1503   A Constructive Witness for an unbroken 1D string defect along the z-axis.
1504   Rather than a unique dynamic solution, this provides an exact analytical ansatz
1505   (a uniform, constant non-Abelian magnetic field) to rigorously prove the non-vacuous
1506   existence of degenerate topological states without relying on empty sets.
1507   -/
1508   @[litlib_track "Flux Tube Frame Witness"]
1509   noncomputable def CGD.Quantum.fluxTubeFrame (mu : Fin 4) (_x : SpacetimePoint) : SL2C :=
1510     toSL2C (if mu = 0 then 0 else if mu = 1 then (Complex.I : ℂ) • sigma3.val else if mu = 2 then (Complex.I : ℂ)
1511       ↦ • sigma1.val else (Complex.I : ℂ) • sigma2.val)

```

```

1511
1512 /--
1513 Identifies a field configuration matching the exact analytical witness of an unbroken flux tube.
1514 -/
1515 def CGD.Quantum.isFluxTube (A : Fin 4 → SpacetimePoint → SL2C) (x : SpacetimePoint) : Prop :=
1516   (∀ mu, A mu x = fluxTubeFrame mu x) ∧
1517   (∀ mu nu, partialDerivSL2c nu (A mu) x = partialDerivSL2c nu (fluxTubeFrame mu) x)
1518
1519 /--
1520 A multi-fingered gauge rotation applied to a background connection.
1521 -/
1522 noncomputable def CGD.Quantum.rotateYAxis (A : Fin 4 → SpacetimePoint → SL2C) (theta : Real) (mu : Fin 4) (x
  ↪ : SpacetimePoint) : SL2C :=
1523   let R := Matrix.of ![Complex.cos (theta/2), -Complex.sin (theta/2)], ![Complex.sin (theta/2),
  ↪ Complex.cos (theta/2)]
1524   let R_inv := Matrix.of ![Complex.cos (theta/2), Complex.sin (theta/2)], ![-Complex.sin (theta/2),
  ↪ Complex.cos (theta/2)]
1525   toSL2C (R * (A mu x).val * R_inv)
1526
1527
1528 -- MODULE: CGD.Quantum.Dirac.Definitions
1529 -----
1530 /--
1531 The Geometric Spinor (Kähler-Dirac Mode).
1532 Constructed strictly from the local internal curvature components  $F_{\{ab\}}$ .
1533 This guarantees flawless gauge covariance (adjoint representation)
1534 and removes the need for arbitrary fermion fields.
1535 -/
1536 @[litlib_track "Geometric Spinor Definition"]
1537 noncomputable def CGD.Quantum.Dirac.geometricSpinor (F : Fin 4 → Fin 4 → C) : Matrix (Fin 4) (Fin 4) C :=
1538   ∑ a : Fin 4, ∑ b : Fin 4, F a b • (gammaVec a * gammaVec b)
1539
1540 /--
1541 The Geometric Yang-Mills Current.
1542 Defined as the covariant divergence of the curvature tensor.
1543 -/
1544 noncomputable def CGD.Quantum.Dirac.yangMillsCurrent (D_F : Fin 4 → Fin 4 → Fin 4 → C) (b : Fin 4) : C :=
1545   ∑ c : Fin 4, ∑ a : Fin 4, minkowskiEta c a * D_F c a b
1546
1547

```

```

1548 -- MODULE: CGD.Quantum.Dirac.Emergence
1549 -----
1550 /--
1551 Proves that applying the emergent Dirac operator to the geometric spinor natively
1552 evaluates to the Yang-Mills geometric current. This mathematically establishes that
1553 the Dirac equation is not a separate physical postulate, but an intrinsic identity
1554 of the Spin(4,C) gauge field.
1555 -/
1556 @[litlib_track "Geometric Dirac Emergence"]
1557 theorem CGD.Quantum.Dirac.geometricDiracEmergence (D_F : Fin 4 → Fin 4 → Fin 4 → ℂ)
1558   (h_anti : ∀ c a b, D_F c a b = - D_F c b a)
1559   (h_bianchi : ∀ c a b, D_F c a b + D_F a b c + D_F b c a = 0) :
1560   (∑ c : Fin 4, gammaVec c * (∑ a : Fin 4, ∑ b : Fin 4, D_F c a b • (gammaVec a * gammaVec b))) =
1561   2 • ∑ b : Fin 4, yangMillsCurrent D_F b • gammaVec b
1562
1563
1564 -- MODULE: CGD.Quantum.Dirac.Vacuum
1565 -----
1566 /--
1567 The Vacuum Dirac Equation.
1568 If the macroscopic geometry satisfies the source-free vacuum equations (J = 0),
1569 the geometric spinor mode mathematically and strictly obeys the massless Dirac equation.
1570 -/
1571 @[litlib_track "Vacuum Dirac Equation"]
1572 theorem CGD.Quantum.Dirac.vacuumDiracEquation (D_F : Fin 4 → Fin 4 → Fin 4 → ℂ)
1573   (h_anti : ∀ c a b, D_F c a b = - D_F c b a)
1574   (h_bianchi : ∀ c a b, D_F c a b + D_F a b c + D_F b c a = 0)
1575   (h_vacuum : ∀ b, yangMillsCurrent D_F b = 0) :
1576   (∑ c : Fin 4, gammaVec c * (∑ a : Fin 4, ∑ b : Fin 4, D_F c a b • (gammaVec a * gammaVec b))) = 0
1577
1578
1579 -- MODULE: CGD.Quantum.Entanglement.Wormhole
1580 -----
1581 /--
1582 Evaluates the macroscopic metric of the `fluxTubeFrame` witness to prove that if two
1583 particles are connected by such a 1D uniform gauge connection, the metric determinant
1584 rigorously vanishes. This confirms that non-local, degenerate wormhole topologies
1585 mathematically exist as non-vacuous solutions within the Spin(4,C) geometry.
1586 -/
1587 @[litlib_track "Witness for Degenerate Entanglement Channels"]

```

```

1588 theorem CGD.Quantum.kinematicEntanglementWormhole (pu : PhysicalUniverse) :
1589    $\forall$  (x y : SpacetimePoint) (theta :  $\mathbb{R}$ ),
1590     areEntangled pu.toUniverse.sd_sector x y theta  $\rightarrow$ 
1591       (urbantkeMetric (fun m n => curvatureSl2c pu.toUniverse.sd_sector m n x)).det = 0  $\wedge$ 
1592       (urbantkeMetric (fun m n => curvatureSl2c pu.toUniverse.sd_sector m n y)).det = 0
1593
1594
1595 -- MODULE: CGD.Quantum.FluxTube
1596 -----
1597 /--
1598 Proves that the exact `fluxTubeFrame` witness natively satisfies the static, purely magnetic boundary
1599  $\hookrightarrow$  conditions.
1600 -/
1601 -- NOTE: Global topological conservation of this minimal configuration is enforced natively by
1602  $\hookrightarrow$  CGD.Foundations.kinematicBianchiIdentity.
1603 @[litlib_track "The FluxTubeFrame is a valid physical solution"]
1604 theorem CGD.Quantum.fluxTubeIsMinimal :  $\forall$  x, satisfies1DMinimalEnergyBound fluxTubeFrame x
1605
1606 /--
1607 Demonstrates that the macroscopic metric of the `fluxTubeFrame` witness possesses a strictly zero
1608  $\hookrightarrow$  determinant. This proves that a 1D uniform gauge connection natively represents a degenerate,
1609  $\hookrightarrow$  non-macroscopic topological background (det(g) = 0), serving as the formal witness that non-local ER=EPR
1610  $\hookrightarrow$  wormhole channels mathematically exist within the Spin(4,C) geometry.
1611 -/
1612 @[litlib_track "Witness for Degenerate Entanglement Channels"]
1613 theorem CGD.Quantum.kinematicFluxTubeStability (pu : PhysicalUniverse) :
1614    $\forall$  (x : SpacetimePoint),
1615     isFluxTube pu.toUniverse.sd_sector x  $\rightarrow$ 
1616       (urbantkeMetric (fun m n => curvatureSl2c pu.toUniverse.sd_sector m n x)).det = 0
1617
1618 /--
1619 Mathematically enforces a strictly static ( $\partial_t A = 0$ ), purely magnetic (electric curvature  $F_{\{0v\}} = 0$ )
1620  $\hookrightarrow$  configuration.
1621
1622 Phenomenologically, this represents the minimal energy ground state of a flux tube. Geometrically,
1623 it simply isolates a static background topology.
1624 -/
1625 def CGD.Quantum.satisfies1DMinimalEnergyBound (A : Fin 4  $\rightarrow$  SpacetimePoint  $\rightarrow$  SL2C) (x : SpacetimePoint) :
1626    $\hookrightarrow$  Prop :=
1627     ( $\forall$  nu, curvatureSl2c A 0 nu x = 0)  $\wedge$ 
1628     ( $\forall$  nu, partialDerivSl2c 0 (A nu) x = 0)

```



```

1621
1622
1623 -- MODULE: CGD.Quantum.Holonomy.Evaluation
1624 -----
1625 /--
1626 Evaluates the path-ordered integral of the exact `fluxTubeFrame` spatial gauge connection witness.
1627 This explicitly demonstrates that evaluating an SU(2) holonomy natively maps geometric
1628 parameters directly to the trigonometric coefficients of standard quantum state vectors,
1629 proving that the geometry can natively reproduce quantum observables.
1630 -/
1631 @[litlib_track "Macroscopic Spin State Witness"]
1632 theorem CGD.Quantum.fluxTubeHolonomyEvaluation
1633   (matrixExp : Matrix (Fin 2) (Fin 2) ℂ → Matrix (Fin 2) (Fin 2) ℂ)
1634   [DerivativeExponential (Fin 2) matrixExp]
1635   (pu : PhysicalUniverse) (alpha L : ℝ)
1636   (h_field : ∀ t, pu.toUniverse.sd_sector 1 (straightLinePath t) = fluxTubeFrame 1 (straightLinePath t)) :
1637   macroscopicObservable (holonomy matrixExp) (fun mu p => rotateYAxis (fun m p => pu.toUniverse.sd_sector m
1638     ↦ p) alpha mu p) 1 L =
1639     (Complex.cos (L:ℂ)) • 1 + (Complex.I * Complex.sin (L:ℂ)) • obs_M alpha
1640
1641 @[litlib_track "Geometric Holonomy"]
1642 -- NATIVE EVALUATION: The exact path-ordered exponential for a constant field
1643 noncomputable def CGD.Quantum.holonomy (matrixExp : Matrix (Fin 2) (Fin 2) ℂ → Matrix (Fin 2) (Fin 2) ℂ)
1644   (A : ℝ → Matrix (Fin 2) (Fin 2) ℂ) (t0 t1 : ℝ) : Matrix (Fin 2) (Fin 2) ℂ :=
1645     matrixExp ((t1 - t0 : ℂ) • A t0)
1646
1647 @[litlib_track "Macroscopic Observable"]
1648 noncomputable def CGD.Quantum.macroscopicObservable
1649   (holonomy : (ℝ → Matrix (Fin 2) (Fin 2) ℂ) → ℝ → ℝ → Matrix (Fin 2) (Fin 2) ℂ)
1650   (A : Fin 4 → SpacetimePoint → SL2C)
1651   (mu : Fin 4) (L : ℝ) : Matrix (Fin 2) (Fin 2) ℂ :=
1652     holonomy (fun t => (A mu (straightLinePath t)).val) 0 L
1653
1654 noncomputable def CGD.Quantum.obs_M (alpha : ℝ) : Matrix (Fin 2) (Fin 2) ℂ :=
1655   let R := Matrix.of ![Complex.cos (alpha/2), -Complex.sin (alpha/2)], ![Complex.sin (alpha/2),
1656     ↦ Complex.cos (alpha/2)]
1657   let R_inv := Matrix.of ![Complex.cos (alpha/2), Complex.sin (alpha/2)], ![-Complex.sin (alpha/2),
1658     ↦ Complex.cos (alpha/2)]
1659   R * sigma3.val * R_inv
1660
1661

```

```

1658 noncomputable def CGD.Quantum.straightLinePath (t : ℝ) : SpacetimePoint := fun _ => t
1659
1660
1661 -- MODULE: CGD.Quantum.Holonomy.Geometric
1662 -----
1663 /--
1664 Demonstrates that the self-dual and anti-self-dual spin connection components
1665 natively mathematically decouple under the geometric trace metric because their
1666 underlying 4x4 matrix embeddings occupy perfectly disjoint sub-blocks.
1667 The geometry requires no artificial truncation to separate chiral sectors.
1668 -/
1669 @[litlib_track "Algebraic Chiral Orthogonalization"]
1670 theorem CGD.Quantum.algebraicChiralOrthogonalization (pu : PhysicalUniverse) (x : SpacetimePoint) (μ ν ρ σ :
  ↪ Fin 4) :
1671   Matrix.trace (embedSelfDual (CGD.Foundations.curvatureSl2c pu.toUniverse.sd_sector μ ν x) *
1672     embedAntiSelfDual (CGD.Foundations.curvatureSl2c pu.toUniverse.asd_sector ρ σ x)) = 0
1673
1674 /--
1675 The exact quantum correlation natively emerges from the Cartan-Killing metric
1676 of the SU(2) group. For unitary SU(2) elements, 1/2 Tr(A B†) perfectly recovers
1677 the cosine of the angle between them.
1678 -/
1679 @[litlib_track "Geometric Bell Correlation"]
1680 noncomputable def CGD.Quantum.geometricBellCorrelation (A B : SU2Group) : ℂ :=
1681   (1 / 2 : ℂ) * Matrix.trace (A.val * B.val.conjTranspose)
1682
1683 /--
1684 Demonstrates that the geometric path-ordered integration of a purely spatial
1685 SU(2) flux tube connection evaluates strictly to a unitary holonomy within
1686 the SU(2) group manifold.
1687 -/
1688 @[litlib_track "Geometric Holonomy Integration"]
1689 theorem CGD.Quantum.geometricHolonomyIntegration (θ : ℝ) :
1690   let U := Matrix.of ![![Complex.cos (θ/2), -Complex.sin (θ/2)], ![Complex.sin (θ/2), Complex.cos (θ/2)]];
1691   U * U.conjTranspose = 1 ∧ Matrix.det U = 1
1692
1693 /--
1694 The CHSH correlation bound is mathematically realized natively by the  $\sqrt{2}$  geometric bounds of the
  ↪ macroscopic SU(2) spatial topology, bypassing the need for abstract Hilbert space operators.
1695 -/

```

```

1696 @[litlib_track "Geometric Tsirelson Bound"]
1697 theorem CGD.Quantum.geometricHolonamyTsirelsonBound
1698   (A1 A2 B1 B2 : SU2Group) :
1699   let chsh := geometricBellCorrelation A1 B1 + geometricBellCorrelation A1 B2 +
1700             geometricBellCorrelation A2 B1 - geometricBellCorrelation A2 B2;
1701   (chsh.re)^2 ≤ 8 ∧ chsh.im = 0
1702
1703
1704 -- MODULE: CGD.Quantum.Holonamy.RelationalTime
1705 -----
1706 /--
1707 Using an exact analytical witness representing the U(1) core of a topological defect
1708 (Complex.I • sigmaZ), this theorem demonstrates that computing the geometric holonomy
1709 (spatial parallel transport) is mathematically identical to applying the Schrödinger
1710 time-evolution operator.
1711
1712 This proves that traversal along a defect's internal spatial phase space can physically
1713 manifest as the passage of time, demonstrating that relational geometric phase
1714 mathematically satisfies the role of time evolution.
1715 -/
1716 @[litlib_track "Witness for Relational Time Emergence"]
1717 theorem CGD.Quantum.Holonamy.relationalTimeEmergence
1718   (matrixExp : Matrix (Fin 2) (Fin 2) ℂ → Matrix (Fin 2) (Fin 2) ℂ)
1719   (pu : PhysicalUniverse)
1720   (t : ℝ)
1721   (h_core : ∀ s, (pu.toUniverse.sd_sector.val 1 (straightLinePath s)).val = Complex.I • sigmaZ) :
1722   macroscopicObservable (holonomy matrixExp) pu.toUniverse.sd_sector.val 1 t =
1723   unitaryTimeEvolution matrixExp (-sigmaZ) t
1724
1725 /-- Standard quantum mechanical time-evolution operator U(t) = exp(-i H t) -/
1726 @[litlib_track "Unitary Time Evolution Operator"]
1727 noncomputable def CGD.Quantum.Holonamy.unitaryTimeEvolution
1728   (matrixExp : Matrix (Fin 2) (Fin 2) ℂ → Matrix (Fin 2) (Fin 2) ℂ)
1729   (Hamiltonian : Matrix (Fin 2) (Fin 2) ℂ) (t : ℝ) : Matrix (Fin 2) (Fin 2) ℂ :=
1730   matrixExp ((-Complex.I * (t : ℂ)) • Hamiltonian)
1731
1732
1733 -- MODULE: CGD.Quantum.Measurement.BornRule
1734 -----
1735 /--

```

```

1736 The exact analytical primitive (volume function) of the Bengtsson invariant Hopf metric.
1737 According to Litlib.Y2017.bengtsson2017geometry.Chapter03.Sec05_HopfFibration.Eq3_98,
1738 the invariant metric volume element scales natively as  $\sin(\theta)$ .
1739 The strict analytical primitive (antiderivative) of  $\sin(\theta)$  evaluates to  $-\cos(\theta)$ .
1740 -/
1741 noncomputable def CGD.Quantum.Measurement.hopfVolumePrimitive (theta : ℝ) : ℝ :=
1742   - Real.cos theta
1743
1744 --
1745 The Kinematic Born Rule Equivalence.
1746 Demonstrates that in Chiral Gauge Dynamics, the phase-space volume fraction
1747 (calculated strictly via the invariant Hopf metric primitive) bounded by the
1748 geometric correlation between the Physical Universe's gauge state and a
1749 macroscopic detector frame mathematically evaluates exactly to the quantum
1750 mechanical Born rule projection.
1751
1752 This derives the fundamental mechanism of quantum probability directly from
1753 the macroscopic continuous geometry of the gauge field, without requiring
1754 wave-function collapse or arbitrary dynamic thresholds.
1755 -/
1756 @[litlib_track "Kinematic Born Rule Equivalence"]
1757 theorem CGD.Quantum.Measurement.kinematicBornRuleEquivalence
1758   (pu : PhysicalUniverse)
1759   (evaluateBoundary : Sl2cGaugeField → SU2Group)
1760   (detector_frame : SU2Group)
1761   (theta : ℝ)
1762   -- The physical angle is strictly defined by the SU(2) trace metric between the universe and the detector
1763   (h_angle : Real.cos theta = (geometricBellCorrelation (evaluateBoundary pu.toUniverse.sd_sector)
1764     ↪ detector_frame).re) :
1764   -- The phase-space volume fraction of the universe...
1765   (hopfVolumePrimitive Real.pi - hopfVolumePrimitive theta) /
1766   (hopfVolumePrimitive Real.pi - hopfVolumePrimitive 0) =
1767   -- ...is mathematically identical to the linear projection...
1768   (1 + (geometricBellCorrelation (evaluateBoundary pu.toUniverse.sd_sector) detector_frame).re) / 2 ∧
1769   -- ...which identically equals the quantum mechanical Born rule probability projection
1770   (1 + (geometricBellCorrelation (evaluateBoundary pu.toUniverse.sd_sector) detector_frame).re) / 2 =
1771   ↪ (Real.cos (theta / 2))^2
1772
1773 -- MODULE: CGD.Quantum.Schroedinger

```

```

1774 -----
1775 noncomputable def CGD.Quantum.P_minus : Matrix (Fin 4) (Fin 4) Complex :=
1776   (1 / 2 : Complex) • (1 - gamma0)
1777
1778 noncomputable def CGD.Quantum.P_plus : Matrix (Fin 4) (Fin 4) Complex :=
1779   (1 / 2 : Complex) • (1 + gamma0)
1780
1781 /--
1782   The exact algebraic chiral split of the emergent Dirac equation.
1783   By applying the projection operators, the relativistic Dirac equation natively factors
1784   into a coupled system for the large and small components.
1785   -/
1786 @[litlib_track "Algebraic Dirac Chiral Split"]
1787 theorem CGD.Quantum.algebraicDiracChiralSplit (dPsi : Fin 4 → SpacetimePoint → Matrix (Fin 4) (Fin 4)
1788   ↪ Complex)
1789   (Psi : SpacetimePoint → Matrix (Fin 4) (Fin 4) Complex) (m : Complex) (x : SpacetimePoint) :
1790   localDiracOp (fun a => dPsi a x) = m • Psi x →
1791   let D0_mod := modulatedTemporalDeriv (dPsi 0 x) (Psi x) m
1792   let D_space := spatialDiracOp (fun mu => dPsi mu x)
1793   (P_plus * D0_mod + P_plus * gamma0 * D_space = 2 • m • (P_plus * Psi x)) ∧
1794   (P_minus * D0_mod + P_minus * gamma0 * D_space = 0)
1795 /--
1796   An algebraic decomposition of the emergent Dirac equation.
1797
1798   By applying the standard chiral projection operators (P_plus, P_minus), the
1799   relativistic equation natively factors into coupled relations for the large and
1800   small components. This verifies that the standard algebraic structures required
1801   for non-relativistic limits (which produce the 1/2m Schrödinger/Pauli Hamiltonian)
1802   are natively supported by the macroscopic gauge-covariant geometry.
1803   -/
1804 @[litlib_track "Exact Schroedinger Reduction"]
1805 theorem CGD.Quantum.exactSchroedingerReduction (dPsi : Fin 4 → SpacetimePoint → Matrix (Fin 4) (Fin 4)
1806   ↪ Complex)
1807   (Psi : SpacetimePoint → Matrix (Fin 4) (Fin 4) Complex) (m : Complex) (x : SpacetimePoint) :
1808   localDiracOp (fun a => dPsi a x) = m • Psi x →
1809   let D0_mod := modulatedTemporalDeriv (dPsi 0 x) (Psi x) m
1810   let Psi_small := P_plus * Psi x
1811   (2 • m • Psi_small = P_plus * D0_mod + gammaVec 1 * (P_minus * dPsi 1 x) + gammaVec 2 * (P_minus * dPsi 2
1812     ↪ x) + gammaVec 3 * (P_minus * dPsi 3 x)) ∧

```

```

1811 (P_minus * D0_mod = gammaVec 1 * (P_plus * dPsi 1 x) + gammaVec 2 * (P_plus * dPsi 2 x) + gammaVec 3 *
    ↪ (P_plus * dPsi 3 x))
1812
1813 /--
1814 The core Dirac operator evaluated in the local tangent space frame (using tangent space index 'a'
1815 instead of coordinate index 'mu'). In curved space, dP represents the tetrad-contracted derivative.
1816 -/
1817 noncomputable def CGD.Quantum.localDiracOp (dP : Fin 4 → Matrix (Fin 4) (Fin 4) Complex) : Matrix (Fin 4)
    ↪ (Fin 4) Complex :=
1818   ∑ a, gammaVec a * dP a
1819
1820 noncomputable def CGD.Quantum.modulatedTemporalDeriv (dPsi0 Psi : Matrix (Fin 4) (Fin 4) Complex) (m :
    ↪ Complex) : Matrix (Fin 4) (Fin 4) Complex :=
1821   dPsi0 + m • Psi
1822
1823 noncomputable def CGD.Quantum.spatialDiracOp (dPsi : Fin 4 → Matrix (Fin 4) (Fin 4) Complex) : Matrix (Fin
    ↪ 4) (Fin 4) Complex :=
1824   gammaVec 1 * dPsi 1 + gammaVec 2 * dPsi 2 + gammaVec 3 * dPsi 3
1825
1826
1827 -- MODULE: CGD.Quantum.Vacuum
1828 -----
1829 /--
1830 In pure connection emergent gravity, spacetime volume is not a fixed background; it is dynamically generated
    ↪ by the gauge field. Therefore, the existence of macroscopic space (det g ≠ 0) mathematically forbids the
    ↪ trivial vacuum (A = 0). Spacetime itself requires a non-zero gauge condensate.
1831 -/
1832 @[litlib_track "Spacetime Vacuum Condensate"]
1833 theorem CGD.Quantum.kinematicVacuumCondensate (pu : PhysicalUniverse) (x : SpacetimePoint) (hx : x ∈
    ↪ pu.bulk) :
1834   pu.toUniverse.sd_sector.val ≠ (fun _ _ => 0)
1835
1836
1837 -- MODULE: CGD.Quantum.YangMills
1838 -----
1839 /--
1840 Evaluates the chaotic non-linear self-interaction of the homogenous gauge field ansatz.
1841 -/
1842 @[litlib_track "Yang-Mills Chaos Bound"]
1843 theorem CGD.Quantum.kinematicYangMillsChaos :

```

```

1844   ∀ (x : SpacetimePoint),
1845     Matrix.trace ([homogeneousChaosAnsatz 1 x, homogeneousChaosAnsatz 2 x].val *
1846       [homogeneousChaosAnsatz 1 x, homogeneousChaosAnsatz 2 x].val) =
1847     -8 * (x 1 : ℂ)^2 * (x 2 : ℂ)^2
1848
1849
1850 -- MODULE: Litlib.Math.Dirac
1851 -----
1852 /--
1853 Physical Interpretation:
1854 The temporal Dirac Gamma matrix  $\gamma^0$  in the Weyl/Chiral basis. Flips parity.
1855 -/
1856 noncomputable def Litlib.Math.Dirac.gamma0 : Matrix (Fin 4) (Fin 4) Complex :=
1857   let m : Matrix (Fin 2 ⊕ Fin 2) (Fin 2 ⊕ Fin 2) Complex :=
1858     Matrix.fromBlocks 0 1 1 0
1859     Matrix.reindex chiralIso chiralIso m
1860
1861 /--
1862 Physical Interpretation:
1863 The covariant 4-vector of Gamma matrices  $\gamma^\mu$ .
1864 -/
1865 noncomputable def Litlib.Math.Dirac.gammaVec (mu : Fin 4) : Matrix (Fin 4) (Fin 4) Complex :=
1866   match mu.val with
1867   | 0 => gamma0
1868   | 1 => gammaSpatial 0
1869   | 2 => gammaSpatial 1
1870   | 3 => gammaSpatial 2
1871   | _ => 0
1872
1873
1874 -- MODULE: Litlib.Y1951.papapetrou1951spinning.Signature
1875 -----
1876 /--
1877 Geometric Non-Degeneracy Constraint: The macroscopic metric density determinant must be strictly non-zero.
1878 Velocity Non-Degeneracy Constraint: The 4-velocity must not be identically zero.
1879 Algebraic Domain Constraint: Parameterized over a generic Field `F` with Characteristic Zero (`CharZero`).
1880 Physical Domain Binding: Papapetrou (1951) Eq (2.12): If a covariant tensor distribution T represents a
1881 ↔ single-pole
1882 test particle, and it is covariantly conserved with respect to the background connection

```

```

1882  (expressed equivalently via the inverse metric contraction), then its worldline must exactly satisfy the
      ↪ geodesic equation.
1883  -/
1884  class Litlib.Y1951.papapetrou1951spinning.Eq2_12[13]
1885    (M : Type*) [TopologicalSpace M]
1886    (F : Type*) [Field F] [CharZero F]
1887    (Metric : M → Matrix (Fin 4) (Fin 4) F)
1888    (invMetric : M → Matrix (Fin 4) (Fin 4) F)
1889    (Christoffel : M → (Fin 4 → Fin 4 → Fin 4 → F))
1890    (T : Fin 4 → Fin 4 → M → F)
1891    (partialDeriv : Fin 4 → (M → F) → M → F)
1892    (worldline : F → M)
1893    (u_up : F → (Fin 4 → F))
1894    (du_up_ds : F → (Fin 4 → F))
1895    (isSinglePole : (Fin 4 → Fin 4 → M → F) → (F → M) → Prop)
1896    where
1897    metric_nondegenerate : ∀ p, (Metric p).det ≠ 0
1898    invMetric_prop : ∀ p, Metric p * invMetric p = 1
1899    u_norm_nonzero : ∀ s, ∑ μ, ∑ ν, Metric (worldline s) μ ν * u_up s μ * u_up s ν ≠ 0
1900    single_pole_eom :
1901      (∀ x b, ∑ a : Fin 4, ∑ c : Fin 4, invMetric x a c * (
1902        partialDeriv c (fun p => T a b p) x -
1903        ∑ d : Fin 4, (Christoffel x d c a * T d b x + Christoffel x d c b * T a d x)
1904      ) = 0) →
1905      isSinglePole T worldline →
1906      ∀ s α, du_up_ds s α + ∑ μ, ∑ ν, Christoffel (worldline s) α μ ν * u_up s μ * u_up s ν = 0
1907
1908
1909  -- MODULE: Litlib.Y1956.utiyama1956invariant.Signature
1910  -----
1911  class Litlib.Y1956.utiyama1956invariant.AppendixI_BilinearForm[12] where
1912    /--
1913    Invariant Bilinear Form (Appendix I):
1914    Constructs the uniquely non-degenerate invariant metric (the Killing form)
1915    for the Lie algebra generators. The `isLieAlgebra` constraint physically restricts
1916    the domain to semi-simple representations, mathematically preventing pathological
1917    central extensions and non-compact ghosts from trivializing the gauge structure.
1918    -/
1919    spans
1920    (M : Type*) [Ring M] [Algebra ℂ M]

```



```

1921 (Trace : M → ℂ)
1922 (isLieAlgebra : M → Prop)
1923 (hNonDegenerate : ∃ x, isLieAlgebra x ∧ x ≠ 0) :
1924 ∀ (B : M → M → ℂ),
1925 (∀ c x y, isLieAlgebra x → isLieAlgebra y → B (c • x) y = c * B x y) →
1926 (∀ x1 x2 y, isLieAlgebra x1 → isLieAlgebra x2 → isLieAlgebra y → B (x1 + x2) y = B x1 y + B x2 y) →
1927 (∀ x y1 y2, isLieAlgebra x → isLieAlgebra y1 → isLieAlgebra y2 → B x (y1 + y2) = B x y1 + B x y2) →
1928 (∀ x y (U : Mx), isLieAlgebra x → isLieAlgebra y →
1929   isLieAlgebra ((U : M) * x * (↑U-1 : M)) →
1930   isLieAlgebra ((U : M) * y * (↑U-1 : M)) →
1931   B ((U : M) * x * (↑U-1 : M)) ((U : M) * y * (↑U-1 : M)) = B x y) →
1932 ∃ (k : ℂ), ∀ x y, isLieAlgebra x → isLieAlgebra y → B x y = k * Trace (x * y)
1933
1934 class Litlib.Y1956.utiyama1956invariant.AppendixI_Expansion[12] where
1935 /--
1936   Utiyama Expansion Theorem (Appendix I):
1937   A gauge-invariant, quadratic (renormalizable) Lagrangian constructed from the
1938   field strength tensor uniquely decomposes into a linear combination of the
1939   traces of the field strength squared. This strictly binds the dynamical terms
1940   to the invariant metric of the gauge group, establishing the algebraic necessity
1941   of the Yang-Mills action.
1942 -/
1943 yieldsTraceExpansion
1944 (M : Type*) [Ring M] [Algebra ℂ M]
1945 (Trace : M → ℂ)
1946 (isLieAlgebra : M → Prop)
1947 (hNonDegenerate : ∃ x, isLieAlgebra x ∧ x ≠ 0)
1948 (L : (Fin 4 → Fin 4 → M) → ℂ)
1949 (hTraceSpans : ∀ (B : M → M → ℂ),
1950   (∀ c x y, isLieAlgebra x → isLieAlgebra y → B (c • x) y = c * B x y) →
1951   (∀ x1 x2 y, isLieAlgebra x1 → isLieAlgebra x2 → isLieAlgebra y → B (x1 + x2) y = B x1 y + B x2 y) →
1952   (∀ x y1 y2, isLieAlgebra x → isLieAlgebra y1 → isLieAlgebra y2 → B x (y1 + y2) = B x y1 + B x y2) →
1953   (∀ x y (U : Mx), isLieAlgebra x → isLieAlgebra y →
1954     isLieAlgebra ((U : M) * x * (↑U-1 : M)) →
1955     isLieAlgebra ((U : M) * y * (↑U-1 : M)) →
1956     B ((U : M) * x * (↑U-1 : M)) ((U : M) * y * (↑U-1 : M)) = B x y) →
1957   ∃ (k : ℂ), ∀ x y, isLieAlgebra x → isLieAlgebra y → B x y = k * Trace (x * y))
1958 (hLQuadScale : ∀ (c : ℂ) (F : Fin 4 → Fin 4 → M),
1959   (∀ μ ν, isLieAlgebra (F μ ν)) → L (fun μ ν => c • F μ ν) = c2 * L F)
1960 (hLQuadAdd : ∀ (F G : Fin 4 → Fin 4 → M),

```

```

1961     (∀ μ v, isLieAlgebra (F μ v)) → (∀ μ v, isLieAlgebra (G μ v)) →
1962     L (fun μ v => F μ v + G μ v) + L (fun μ v => F μ v - G μ v) = 2 * L F + 2 * L G
1963 (hLGauge : ∀ (F : Fin 4 → Fin 4 → M) (U : Mx),
1964     (∀ μ v, isLieAlgebra (F μ v)) →
1965     (∀ μ v, isLieAlgebra ((U : M) * F μ v * (†U-1 : M))) →
1966     L (fun μ v => (U : M) * F μ v * (†U-1 : M)) = L F) :
1967   ∃ (T : Fin 4 → Fin 4 → Fin 4 → Fin 4 → ℂ),
1968     ∀ F, (∀ μ v, isLieAlgebra (F μ v)) →
1969     L F = ∑ μ : Fin 4, ∑ v : Fin 4, ∑ ρ : Fin 4, ∑ σ : Fin 4, T μ v ρ σ * Trace (F μ v * F ρ σ)
1970
1971
1972 -- MODULE: Litlib.Y1965.spivak1965calculus.Chapter05.IntegrationOnChains
1973 -----
1974 class Litlib.Y1965.spivak1965calculus.DivergenceTheoremR4Compact[50]
1975   (f : (Fin 4 → ℝ) → (Fin 4 → ℂ))
1976   where
1977   /--
1978   Compactly Supported Divergence Free Constraint:
1979   By Stokes' Theorem, the integral of an exact differential form over a manifold
1980   without boundary evaluates to zero if the form has compact support.
1981   Here, we enforce continuous differentiability and explicitly require the closure
1982   of the support to be compact. This rigorously ensures the Lebesgue integral of
1983   the divergence over Euclidean space is mathematically well-posed and evaluates
1984   to zero, precluding boundary term singularities.
1985   -/
1986   hf_smooth : ContDiff ℝ 1 f
1987   hf_compact : IsCompact (closure {x | f x ≠ 0})
1988
1989   integral_div_zero :
1990     ∫ x : Fin 4 → ℝ, (∑ i : Fin 4, deriv (fun t => f (Function.update x i t) i) (x i)) @MeasureTheory.volume
1991     ↪ = 0
1992
1993 -- MODULE: Litlib.Y1975.belavin1975pseudoparticle.Signature
1994 -----
1995 class Litlib.Y1975.belavin1975pseudoparticle.Eq8[25]
1996   (BoundaryManifold Group : Type*) [TopologicalSpace BoundaryManifold] [TopologicalSpace Group]
1997   [Nonempty BoundaryManifold] [Nonempty Group]
1998   (isSmooth : (BoundaryManifold → Group) → Prop)
1999   (windingNumber : (BoundaryManifold → Group) → ℤ)

```

```

2000 (cartanMaurerIntegral : (BoundaryManifold → Group) → ℝ)
2001 [Litlib.Y2003.nakahara2003geometry.CartanMaurerTopology (BoundaryManifold → Group) isSmooth windingNumber
2002 ↦ cartanMaurerIntegral] where
2003
2004 /--
2005 Topological Non-Triviality Constraint: Ensure the existence of at least one
2006 map with a unit winding number to verify the principal bundle possesses
2007 a non-trivial topological structure.
2008 -/
2009 h_exists_degree_one : ∃ (g : BoundaryManifold → Group), windingNumber g = 1
2010
2011 /--
2012 Physical Interpretation: Belavin 1975, Page 86: "Hence q is the number of
2013 times the SU(2) is covered under this mapping." Any homeomorphism between
2014 topologically equivalent mapping spaces preserves the absolute degree of the map.
2015 -/
2016 degree_of_homeomorph :
2017   ∀ (f : BoundaryManifold → Group),
2018   IsHomeomorphism f →
2019   windingNumber f = 1 ∨ windingNumber f = -1
2020
2021 /--
2022 Topological Well-Posedness: A mathematically strict definition of a
2023 topological homeomorphism—a bijection between two topological spaces
2024 that is continuous and has a continuous inverse.
2025 -/
2026 structure Litlib.Y1975.belavin1975pseudoparticle.IsHomeomorphism[25] {X Y : Type*} [TopologicalSpace X]
2027 ↦ [TopologicalSpace Y] (f : X → Y) : Prop where
2028   bij : Function.Bijective f
2029   cont : Continuous f
2030   inv_cont : ∃ (g : Y → X), Function.LeftInverse g f ∧ Function.RightInverse g f ∧ Continuous g
2031
2032 -- MODULE: Litlib.Y1976.rudin1976principles.Chapter09.Sec08_DerivativesOfHigherOrder
2033 -----
2034 class Litlib.Y1976.rudin1976principles.ClairautTheoremNDimensional[14] where
2035   /--
2036   Exercise 29 (page 243): The n-dimensional, k-th order generalization of Clairaut's theorem.
2037
2038   Rigorous Formalization Constraint: We employ Mathlib's `ContDiff0n` to assert that

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```

2038  the function is  $k$ -times continuously differentiable. Instead of defining ad-hoc
2039  index-based partial derivatives (which can obscure topological boundaries), we utilize
2040  Mathlib's iteratedFDeriv, evaluating it on a tuple of vectors v. The theorem
2041  establishes that permuting the order of these vectors (via fun i => v (σ i))
2042  leaves the result invariant, serving as the rigorous, coordinate-free equivalent
2043  of permuting partial derivative indices.
2044  -/
2045  symmetry_of_higher_partials
2046    (n m : ℕ)
2047    (E : Set (Fin n → ℝ))
2048    (hOpen : IsOpen E)
2049    (f : (Fin n → ℝ) → ℝ)
2050    (hf : ContDiffOn ℝ m f E)
2051    (x : Fin n → ℝ) (hx : x ∈ E)
2052    (v : Fin m → (Fin n → ℝ))
2053    (σ : Equiv.Perm (Fin m)) :
2054    iteratedFDeriv ℝ m f x (fun i => v (σ i)) = iteratedFDeriv ℝ m f x v
2055
2056
2057  -- MODULE: Litlib.Y1976.rudin1976principles.Chapter11.LebesgueIntegral
2058  -----
2059  class Litlib.Y1976.rudin1976principles.LeibnizIntegralRule[14]
2060    (f : ℝ → (Fin 4 → ℝ) → ℝ)
2061    where
2062  /--
2063  Leibniz Integral Rule via Dominated Convergence (Theorem 11.28):
2064
2065  Topological Gatekeeping (Override Applied): A naive formalization requires
2066  only pointwise differentiability, which allows pathological functions to generate
2067  locally unbounded derivatives that Mathlib's default integral evaluates to zero
2068  (Garbage-In Exploit). To prevent this, we mathematically enforce ContDiff ℝ 1
2069  (joint continuous differentiability) on f over the product space, which
2070  guarantees the local boundedness of ∂f/∂t necessary for the rigorous application
2071  of the Dominated Convergence Theorem. The variation is compactly supported
2072  to prevent escape to spatial infinity.
2073  -/
2074  hf_smooth : ContDiff ℝ 1 (fun p : ℝ × (Fin 4 → ℝ) => f p.1 p.2)
2075  hf_integrable : ∀ t, MeasureTheory.Integrable (f t) MeasureTheory.volume
2076  hf_variation_compact : ∃ K : Set (Fin 4 → ℝ), IsCompact K ∧ ∀ t x, x ∈ K → f t x = f 0 x
2077

```

```

2078   leibniz_commute :
2079     ∀ t, deriv (fun s => ∫ x, f s x ∂MeasureTheory.volume) t =
2080       ∫ x, deriv (fun s => f s x) t ∂MeasureTheory.volume
2081
2082
2083   -- MODULE: Litlib.Y1983.witten1983current.Signature
2084   -----
2085   def AsymptoticVacuumSU2 (U : (Fin 3 → ℝ) → Matrix (Fin 2) (Fin 2) ℂ) : Prop :=
2086     ∀ ε : ℝ, ε > 0 → ∃ R : ℝ, R > 0 ∧ ∀ x : Fin 3 → ℝ,
2087       (x 0)^2 + (x 1)^2 + (x 2)^2 > R^2 →
2088       ∀ i j : Fin 2, Complex.normSq ((U x) i j - (1 : Matrix (Fin 2) (Fin 2) ℂ) i j) < ε^2
2089
2090   noncomputable def BaryonNumberIntegrandSU2 (U : (Fin 3 → ℝ) → Matrix (Fin 2) (Fin 2) ℂ) (x : Fin 3 → ℝ) : ℂ
2091     ↦ :=
2092     ∑ i : Fin 3, ∑ j : Fin 3, ∑ k : Fin 3,
2093       (eps3 i j k) *
2094       Matrix.trace (
2095         (U x)^-1 * (partialDerivSU2 i U x) *
2096         ((U x)^-1 * (partialDerivSU2 j U x)) *
2097         ((U x)^-1 * (partialDerivSU2 k U x))
2098       )
2099   /--
2100   Finkelstein-Rubinstein Spin-Statistics for SU(2) (Page 436):
2101   While the Wess-Zumino term vanishes for SU(2), the 4th homotopy group is non-trivial:  $\pi_4(SU(2)) = \mathbb{Z}_2$ .
2102   An adiabatic  $2\pi$  spatial rotation of a degree-1 SU(2) soliton traces the non-trivial loop in configuration
2103     ↦ space.
2104   Weighting this non-trivial topological history with a factor of -1 mathematically quantizes the soliton as a
2105     ↦ fermion.
2106   -/
2107
2108   class FinkelsteinRubinsteinQuantization
2109     [MeasureSpace (Fin 3 → ℝ)]
2110     (path_parity : (ℝ → (Fin 3 → ℝ) → Matrix (Fin 2) (Fin 2) ℂ) → ℕ)
2111     (U_0 : (Fin 3 → ℝ) → Matrix (Fin 2) (Fin 2) ℂ)
2112     (_h_SU2 : ∀ x, IsSU2 (U_0 x))
2113     (_h_smooth : ∀ i j : Fin 2, Differentiable ℝ (fun x => U_0 x i j))
2114     (_h_vacuum : AsymptoticVacuumSU2 U_0)
2115     (_h_integrable : Integrable (BaryonNumberIntegrandSU2 U_0))
2116     (_h_degree_one : baryon_number_SU2 U_0 _h_SU2 _h_smooth _h_vacuum _h_integrable = 1)
2117     (U_rot : ℝ → (Fin 3 → ℝ) → Matrix (Fin 2) (Fin 2) ℂ)

```

```

2115   (_h_rot_SU2 : ∀ t x, IsSU2 (U_rot t x))
2116   (_h_is_2pi_rot : ∀ t x, U_rot t x = U_0 (rotZ t x))
2117   (_h_fr_parity : path_parity U_rot = 1) : Prop where
2118   is_fermion : quantum_weight path_parity U_rot = -1
2119
2120   def IsSU2 (A : Matrix (Fin 2) (Fin 2) ℂ) : Prop :=
2121     Matrix.det A = 1 ∧ star A * A = 1
2122
2123   noncomputable def baryon_number_SU2
2124     [MeasureSpace (Fin 3 → ℝ)]
2125     (U : (Fin 3 → ℝ) → Matrix (Fin 2) (Fin 2) ℂ)
2126     (_h_SU2 : ∀ x, IsSU2 (U x))
2127     (_h_smooth : ∀ i j : Fin 2, Differentiable ℝ (fun x => U x i j))
2128     (_h_vacuum : AsymptoticVacuumSU2 U)
2129     (_h_integrable : Integrable (BaryonNumberIntegrandSU2 U)) : ℂ :=
2130     (1 / (24 * Real.pi ^ 2)) *
2131     ∫ (x : Fin 3 → ℝ), BaryonNumberIntegrandSU2 U x
2132
2133   /--
2134   Evaluates the quantum phase amplitude derived from the topological parity
2135   (the mapping class in  $\pi_4$ ) of a configuration path.
2136   -/
2137   noncomputable def quantum_weight (path_parity : (ℝ → (Fin 3 → ℝ) → Matrix (Fin 2) (Fin 2) ℂ) → ℕ)
2138     (U_path : ℝ → (Fin 3 → ℝ) → Matrix (Fin 2) (Fin 2) ℂ) : ℂ :=
2139     (-1 : ℂ) ^ (path_parity U_path)
2140
2141   /--
2142   Explicit SO(3) spatial rotation matrix around the Z-axis, parameterized by an angle t.
2143   This continuous coordinate transformation is strictly required to evaluate the
2144   spatial rotation of a configuration space field.
2145   -/
2146   noncomputable def rotZ (t : ℝ) (x : Fin 3 → ℝ) : Fin 3 → ℝ :=
2147     fun i =>
2148       if i.val = 0 then (Real.cos t) * x 0 - (Real.sin t) * x 1
2149       else if i.val = 1 then (Real.sin t) * x 0 + (Real.cos t) * x 1
2150       else x 2
2151
2152
2153   -- MODULE: Litlib.Y1984.urbantke1984integrability.Signature
2154   -----

```

```

2155 class Litlib.Y1984.urbantke1984integrability.Eq10_Symmetry[7] where
2156   /--
2157   Urbantke Metric Symmetry Identity:
2158   Demonstrates that the Urbantke quasimetric  $g_{\mu\nu}$ , when constructed from an
2159   antisymmetric curvature tensor  $F$  and its dual, is inherently symmetric ( $g_{\mu\nu} = g_{\nu\mu}$ ).
2160   This algebraic property is a foundational requirement for  $g_{\mu\nu}$  to act as a
2161   valid pseudo-Riemannian conformal metric.
2162   -/
2163   symmetricQuasimetric
2164     (F : Fin 3 → Fin 4 → Fin 4 → ℂ)
2165     (F_dual : Fin 3 → Fin 4 → Fin 4 → ℂ)
2166     (epsilon3 : Fin 3 → Fin 3 → Fin 3 → ℂ)
2167     (g : Fin 4 → Fin 4 → ℂ)
2168     (hF_anti : ∀ a μ ν, F a μ ν = - F a ν μ)
2169     (hF_dual_anti : ∀ a μ ν, F_dual a μ ν = - F_dual a ν μ)
2170     (hepsilon3_anti : ∀ a b c, epsilon3 a b c = - epsilon3 b a c ∧ epsilon3 a b c = - epsilon3 a c b)
2171     (hg_def : ∀ μ ν, g μ ν = (-1 / 6 : ℂ) * Finset.sum Finset.univ (fun a => Finset.sum Finset.univ (fun b
2172   ↪ => Finset.sum Finset.univ (fun c =>
2173   Finset.sum Finset.univ (fun α => Finset.sum Finset.univ (fun β =>
2174   epsilon3 a c b * F a μ α * F_dual c α β * F b β ν)))))) :
2175     ∀ μ ν, g μ ν = g ν μ
2176
2177 -- MODULE: Litlib.Y1991.capovilla1991pure.Signature
2178 -----
2179 class Litlib.Y1991.capovilla1991pure.Theorem_Volume_Element_Identity[9]
2180   (Spacetime : Type*) [TopologicalSpace Spacetime]
2181   (sqrt_g : Spacetime → ℂ)
2182   (mu : Spacetime → ℂ)
2183   (eta : Spacetime → ℂ)
2184   (Psi_3x3 : Spacetime → Matrix (Fin 3) (Fin 3) ℂ)
2185   (M_3x3 : Spacetime → Matrix (Fin 3) (Fin 3) ℂ) where
2186   /-- Field Non-Degeneracy Constraint: Precludes division by zero and ensures metric non-degeneracy. -/
2187   h_sqrt_g_nondeg : ∀ x, sqrt_g x ≠ 0
2188   h_mu_nondeg : ∀ x, mu x ≠ 0
2189   h_M_nondeg : ∀ x, Matrix.det (M_3x3 x) ≠ 0
2190   h_Psi_nondeg : ∀ x, Matrix.det (Psi_3x3 x) ≠ 0
2191
2192 -- Eq 2.16
2193 eq2_16_holds : ∀ x, (eta x)^2 = mu x / Matrix.det (M_3x3 x)

```

```

2194 -- Eq 2.21
2195 eq2_21_holds : ∀ x, eta x = (sqrt_g x * Matrix.det (Psi_3x3 x))-1
2196 -- Eq 2.11 taking determinant: det(Psi) = mu(-3/2) * det(M(1/2)) => det(Psi)2 = mu(-3) * det(M)
2197 eq2_11_det_holds : ∀ x, (Matrix.det (Psi_3x3 x))2 = (mu x)-1 * (mu x)-1 * (mu x)-1 * Matrix.det (M_3x3 x)
2198
2199 /--
2200 Physical Volume Density Evaluation:
2201 The text on page 67 implicitly defines and utilizes this identity, showing that the
2202 Lagrange multiplier  $\mu$  enforcing tracelessness in the matrix action natively encodes
2203 the emergent square root metric determinant  $\sqrt{g}$ . Computed algebraically
2204 from comparing the evaluation of  $\eta$  across equations 2.16 and 2.21.
2205 -/
2206 volume_element_identity : ∀ x, (sqrt_g x)2 = (mu x)2
2207
2208
2209 -- MODULE: Litlib.Y2000.hall2000elementary.Signature
2210 -----
2211 class Litlib.Y2000.hall2000elementary.DerivativeExponential[51]
2212   (n : Type*) [Fintype n] [DecidableEq n]
2213   [NormedAddCommGroup (Matrix n n ℂ)] [NormedSpace ℝ (Matrix n n ℂ)]
2214   (exp : Matrix n n ℂ → Matrix n n ℂ) where
2215   hIsExp : ∀ X, Tendsto (fun m : ℕ => ∑ k ∈ Finset.range m, (1 / (Nat.factorial k : ℂ)) • Xk) atTop (nhds
2216     ↪ (exp X))
2217   smooth_exp :
2218     ∀ X : Matrix n n ℂ, ContDiff ℝ T (fun (t : ℝ) => exp ((t : ℂ) • X))
2219 /--
2220 # Proposition: Derivative of the Exponential Mapping
2221 **Literature:** Hall 2000, Proposition 3.4, Page 29.
2222 **Physical Interpretation:** The matrix exponential maps a tangent vector (Lie algebra generator) to a
2223 ↪ continuous path in the Lie group. The temporal derivative of this path evaluated at the identity
2224 ↪ uniquely recovers the generator  $X$ . This ensures the rigorous binding between macroscopic group
2225 ↪ transformations and infinitesimal generators.
2226
2227 **Mathematical Boundaries:** Enforces that the mapping from  $\mathbb{R}$  to the space of matrices is smooth
2228 ↪ (ContDiff). The derivative applies strictly under the scalar action of the real line.
2229 -/
2230
2231 deriv_exp :
2232   ∀ (X : Matrix n n ℂ) (t : ℝ),
2233     deriv (fun (s : ℝ) => exp ((s : ℂ) • X)) t = X * exp ((t : ℂ) • X) ∧
2234     deriv (fun (s : ℝ) => exp ((s : ℂ) • X)) t = exp ((t : ℂ) • X) * X

```



```

2229
2230 deriv_exp_zero :
2231    $\forall X : \text{Matrix } n \ n \ \mathbb{C}, \text{ deriv } (\text{fun } (s : \mathbb{R}) \Rightarrow \exp ((s : \mathbb{C}) \bullet X)) \ 0 = X$ 
2232
2233
2234 -- MODULE: Litlib.Y2003.nakahara2003geometry.Chapter07.Sec04_LeviCivita
2235 -----
2236 class Litlib.Y2003.nakahara2003geometry.Theorem_ContractedBianchi[11]
2237   (Point Index : Type _) [Fintype Index] [DecidableEq Index] [Nonempty Index] [Nonempty Point]
2238   (isSmooth : (Point →  $\mathbb{C}$ ) → Prop)
2239   (partialDeriv : Index → (Point →  $\mathbb{C}$ ) → Point →  $\mathbb{C}$ ) where
2240
2241   -- The fundamental rules of calculus required for the geometry
2242   derivCommute :  $\forall \mu \nu f x, \text{ isSmooth } f \rightarrow \text{ isSmooth } (\text{partialDeriv } \mu f) \rightarrow \text{ isSmooth } (\text{partialDeriv } \nu f) \rightarrow$ 
2243     partialDeriv  $\mu$  (fun p => partialDeriv  $\nu$  f p) x = partialDeriv  $\nu$  (fun p => partialDeriv  $\mu$  f p) x
2244   derivLeibniz :  $\forall \mu f_1 f_2 x, \text{ isSmooth } f_1 \rightarrow \text{ isSmooth } f_2 \rightarrow$ 
2245     partialDeriv  $\mu$  (fun p => f1 p * f2 p) x = partialDeriv  $\mu$  f1 x * f2 x + f1 x * partialDeriv  $\mu$  f2 x
2246
2247   -- The General Theorem: If a metric satisfies the Levi-Civita premises, it satisfies Bianchi.
2248   all_metrics_satisfy_bianchi :
2249      $\forall (g\_inv : \text{Index} \rightarrow \text{Index} \rightarrow \text{Point} \rightarrow \mathbb{C})$ 
2250     (christoffel : Index → Index → Index → Point →  $\mathbb{C}$ )
2251     (ricci : Index → Index → Point →  $\mathbb{C}$ )
2252     (scalarCurv : Point →  $\mathbb{C}$ )
2253     (G : Index → Index → Point →  $\mathbb{C}$ ),
2254     -- PREMISES (Algebraic):
2255     ( $\forall x \ i \ j, g \ i \ j \ x = g \ j \ i \ x$ ) →
2256     ( $\forall x \ i \ j, g\_inv \ i \ j \ x = g\_inv \ j \ i \ x$ ) →
2257     ( $\forall x \ i \ j, (\sum k, g \ i \ k \ x * g\_inv \ k \ j \ x) = \text{if } i = j \text{ then } 1 \text{ else } 0$ ) →
2258     ( $\forall x \ \rho \mu \nu, \text{ christoffel } \rho \mu \nu \ x = (1/2 : \mathbb{C}) * \sum \sigma, g\_inv \ \rho \sigma \ x * ($ 
2259       partialDeriv  $\mu$  (fun p => g  $\sigma$   $\nu$  p) x + partialDeriv  $\nu$  (fun p => g  $\mu$   $\sigma$  p) x - partialDeriv
2260        $\hookrightarrow \sigma$  (fun p => g  $\mu \nu$  p) x)) →
2261     ( $\forall x \ \mu \nu, \text{ ricci } \mu \nu \ x = \sum \rho, ($ 
2262       partialDeriv  $\rho$  (fun p => christoffel  $\rho \mu \nu$  p) x - partialDeriv  $\nu$  (fun p => christoffel  $\rho \mu$ 
2263        $\hookrightarrow \rho$  p) x +
2264        $\sum \lambda, (\text{christoffel } \rho \lambda \rho \ x * \text{christoffel } \lambda \mu \nu \ x - \text{christoffel } \rho \lambda \nu \ x * \text{christoffel } \lambda \mu \rho \ x))) \rightarrow$ 
2265     ( $\forall x, \text{ scalarCurv } x = \sum \alpha, \sum \beta, g\_inv \ \alpha \beta \ x * \text{ ricci } \alpha \beta \ x$ ) →
2266     ( $\forall x \ \mu \nu, G \ \mu \nu \ x = \text{ ricci } \mu \nu \ x - (1/2 : \mathbb{C}) * g \ \mu \nu \ x * \text{ scalarCurv } x$ ) →
2267     -- PREMISES (Differential Smoothness):

```

```

2266     (∀ i j, isSmooth (fun p => g i j p)) →
2267     (∀ i j, isSmooth (fun p => g_inv i j p)) →
2268     (∀ rho mu nu, isSmooth (fun p => christoffel rho mu nu p)) →
2269     -- CONCLUSION:
2270     ∀ (nu : Index) (x : Point),
2271     ∑ mu : Index, ∑ alpha : Index, g_inv mu alpha x * (
2272     partialDeriv alpha (fun p => G mu nu p) x -
2273     ∑ lambda : Index, (christoffel lambda alpha mu x * G lambda nu x + christoffel lambda alpha nu x *
2274     ↪ G mu lambda x)
2275     ) = 0
2276
2277 -- MODULE: Litlib.Y2003.nakahara2003geometry.Chapter10.Sec05_GaugeTheories
2278 -----
2279 class Litlib.Y2003.nakahara2003geometry.CartanMaurerTopology[11]
2280   (GroupMap : Type _) [TopologicalSpace GroupMap] [Nonempty GroupMap]
2281   (isSmooth : GroupMap → Prop)
2282   (windingNumber : GroupMap → ℤ)
2283   (cartanMaurerIntegral : GroupMap → ℝ) where
2284   degreeTheorem :
2285     ∀ g : GroupMap, isSmooth g → cartanMaurerIntegral g = (windingNumber g : ℝ)
2286   homotopyInvariance
2287     (H : ℝ → GroupMap)
2288     (hCont : Continuous H) :
2289     ∀ t1 t2 : ℝ, windingNumber (H t1) = windingNumber (H t2)
2290
2291
2292 -- MODULE: Litlib.Y2010.wald2010general.AppendixD.Conformal
2293 -----
2294 /-- Conformal Ricci Transformation (Wald Eq D.8 for n=4).
2295 This explicitly maps the Ricci tensor of a conformally transformed metric to the
2296 original Ricci tensor using the fully expanded covariant derivatives of the conformal
2297 factor. Explicit index expansion (Fin 4) is utilized to prevent opaque function
2298 exploits and ensure dimensional rigor. -/
2299 class Litlib.Y2010.wald2010general.ConformalRicciTransformation[18]
2300   (M : Type _)
2301   (g g_tilde g_inv : M → Fin 4 → Fin 4 → ℝ)
2302   (Ricci Ricci_tilde : M → Fin 4 → Fin 4 → ℝ)
2303   (Omega : M → ℝ)
2304   (nabla : (M → ℝ) → M → Fin 4 → ℝ)

```

```

2305     (nabla_nabla : (M → ℝ) → M → Fin 4 → Fin 4 → ℝ)
2306     (isLeviCivitaRicci : (M → (Fin 4 → Fin 4 → ℝ)) → (M → (Fin 4 → Fin 4 → ℝ)) → Prop)
2307     where
2308     /-- Topological Gatekeeping (Conformal Factor):
2309     The conformal factor `Omega` must be strictly positive everywhere on the manifold.
2310     This mathematically prevents evaluation of undefined limits for the natural
2311     logarithm `ln(Omega)` present in the connection difference tensor. -/
2312     omega_pos : ∀ p : M, 0 < Omega p
2313
2314     /-- Inverse Metric Constraint: `g_inv` is explicitly verified as the strictly
2315     valid matrix inverse of the original metric `g` via Kronecker delta contraction.
2316     This provides the explicit algebraic structure required for downstream theorem proving. -/
2317     valid_inverse : ∀ (p : M) (a c : Fin 4),
2318       (∑ b : Fin 4, g p a b * g_inv p b c) = if a = c then 1 else 0
2319
2320     /-- Geometric Connection Constraint:
2321     Both Ricci tensors must be explicitly bound to the torsion-free Levi-Civita
2322     connections of their respective metrics. -/
2323     levi_civita_bound : isLeviCivitaRicci g Ricci
2324     levi_civita_bound_tilde : isLeviCivitaRicci g_tilde Ricci_tilde
2325
2326     /-- Conformal Ricci Transformation (Explicit 4D Expansion):
2327     The exact algebraic relationship between the transformed Ricci tensor and the
2328     original Ricci tensor, expanded for n=4 dimensions. The derivative operators
2329     are explicitly applied to the natural logarithm of the conformal factor to
2330     prevent decoupled type signature exploits. -/
2331     eq_D8 : ∀ (p : M) (a c : Fin 4),
2332       Ricci_tilde p a c = Ricci p a c
2333         - 2 * nabla_nabla (fun x => Real.log (Omega x)) p a c
2334         - g p a c * (∑ d : Fin 4, ∑ e : Fin 4, g_inv p d e * nabla_nabla (fun x => Real.log (Omega x)) p d e)
2335         + 2 * nabla (fun x => Real.log (Omega x)) p a * nabla (fun x => Real.log (Omega x)) p c
2336         - 2 * g p a c * (∑ d : Fin 4, ∑ e : Fin 4, g_inv p d e * nabla (fun x => Real.log (Omega x)) p d *
2337           ↪ nabla (fun x => Real.log (Omega x)) p e)
2338
2339 -- MODULE: Litlib.Y2011.krasnov2011plebanski.Signature
2340 -----
2341 /--

```

```

2342 Physical Interpretation: Formulates the vacuum Einstein equations in the Plebański formalism. The trace of
      ↪ the self-dual curvature matrix is strictly proportional to the cosmological constant, and its
      ↪ anti-self-dual part strictly vanishes.
2343 Mathematical Boundaries: This replaces the traditional differential, tensorial Ricci-flatness conditions with
      ↪ a purely algebraic constraint on the curvature components in the self-dual basis.
2344 -/
2345 class Litlib.Y2011.krasnov2011plebanski.Eq15[4]
2346   (plebanski_vacuum :  $\mathbb{C} \rightarrow (\text{Fin } 3 \rightarrow \text{Fin } 3 \rightarrow \mathbb{C}) \rightarrow (\text{Fin } 3 \rightarrow \text{Fin } 3 \rightarrow \mathbb{C}) \rightarrow \mathbf{Prop}$ )
2347   where
2348   plebanski_vacuum_iff :  $\forall (\text{Lambda} : \mathbb{C}) (\text{F}_{ij} \text{ F\_bar}_{ij} : \text{Fin } 3 \rightarrow \text{Fin } 3 \rightarrow \mathbb{C}),$ 
2349     plebanski_vacuum Lambda F_ij F_bar_ij  $\leftrightarrow$ 
2350      $((\sum i : \text{Fin } 3, \text{F}_{ij} i i) = -\text{Lambda} \wedge (\forall i j, \text{F\_bar}_{ij} i j = 0))$ 
2351
2352 /--
2353 Physical Interpretation: Defines the projection of the trace-free macroscopic stress-energy tensor onto the
      ↪ mixed self-dual/anti-self-dual basis. This term serves as the source coupling matter to the gravitational
      ↪ field in the Plebański formalism.
2354 Mathematical Boundaries: The construction explicitly requires raising the indices of the anti-self-dual basis
      ↪ forms ('Sigma_bar') using the inverse background metric ('g_inv'), strictly binding the definition to
      ↪ domains where the metric is non-degenerate.
2355 -/
2356 class Litlib.Y2011.krasnov2011plebanski.Eq16[4]
2357   (Sigma :  $\text{Fin } 3 \rightarrow \text{Fin } 4 \rightarrow \text{Fin } 4 \rightarrow \mathbb{C}$ )
2358   (Sigma_bar :  $\text{Fin } 3 \rightarrow \text{Fin } 4 \rightarrow \text{Fin } 4 \rightarrow \mathbb{C}$ )
2359   (g_inv :  $\text{Fin } 4 \rightarrow \text{Fin } 4 \rightarrow \mathbb{C}$ )
2360   (T_tilde :  $\text{Fin } 4 \rightarrow \text{Fin } 4 \rightarrow \mathbb{C}$ )
2361   (T_ij :  $\text{Fin } 3 \rightarrow \text{Fin } 3 \rightarrow \mathbb{C}$ )
2362   where
2363   T_ij_def :  $\forall (i j : \text{Fin } 3),$ 
2364     T_ij i j =
2365        $\sum \mu : \text{Fin } 4, \sum \nu : \text{Fin } 4, \sum \rho : \text{Fin } 4, \sum \alpha : \text{Fin } 4, \sum \beta : \text{Fin } 4,$ 
2366         T_tilde rho mu *
2367         Sigma i nu rho *
2368         g_inv mu alpha *
2369         g_inv nu beta *
2370         Sigma_bar j alpha beta
2371
2372 /--

```

```

2373 Physical Interpretation: The full non-vacuum Einstein equations coupled to macroscopic matter in the
    ↪ Plebański formulation. The trace of the self-dual curvature is determined by the cosmological constant
    ↪ and the trace of the stress-energy tensor. The anti-self-dual part is proportional to the trace-free
    ↪ stress-energy tensor.

2374 Mathematical Boundaries: By coupling the curvature to the stress-energy components `T` and `T_ij`, these
    ↪ equations are strictly bound to domains where macroscopic matter fields are well-defined.

2375 -/

2376 class Litlib.Y2011.krasnov2011plebanski.Eq17[4]
2377   (Lambda : ℂ)
2378   (G : ℂ)
2379   (F_ij : Fin 3 → Fin 3 → ℂ)
2380   (F_bar_ij : Fin 3 → Fin 3 → ℂ)
2381   (T : ℂ)
2382   (T_ij : Fin 3 → Fin 3 → ℂ)
2383   (plebanski_matter_eqs : Prop)
2384   where
2385   einstein_eqs_iff : plebanski_matter_eqs ↔
2386     ((∑ i : Fin 3, F_ij i i) = -Lambda - 2 * (Real.pi : ℂ) * G * T ∧
2387      (∀ i j, F_bar_ij i j = -2 * (Real.pi : ℂ) * G * T_ij i j))
2388
2389 /--
2390 Physical Interpretation: Establishes the bridge between the Plebański formulation and the standard metric
    ↪ formulation of general relativity (Page 6). It mathematically proves that for a Levi-Civita
    ↪ (torsion-free) connection, the algebraic self-dual curvature conditions (vanishing anti-self-dual part
    ↪ and trace proportional to the cosmological constant) are rigorously equivalent to the tensorial
    ↪ trace-reversed vacuum Einstein field equations ( $R_{ab} = \Lambda g_{ab}$ ).

2391 Mathematical Boundaries: The equivalence strictly requires a non-degenerate spacetime metric `g` with a
    ↪ well-defined inverse (`g_inv`) to prevent dimensional collapse. This constraint, alongside the explicit
    ↪ requirement that the connection is Levi-Civita (`isLeviCivitaRicci`), securely binds the theorem to
    ↪ standard Riemannian geometries where the isomorphism between the self-dual curvature and the Riemann
    ↪ curvature physically holds.

2392 -/

2393 class Litlib.Y2011.krasnov2011plebanski.PlebanskiToEinsteinEquivalence[4]
2394   (g : Fin 4 → Fin 4 → ℝ)
2395   (g_inv : Fin 4 → Fin 4 → ℝ)
2396   (Ricci : Fin 4 → Fin 4 → ℝ)
2397   (Lambda : ℝ)
2398   (F_ij F_bar_ij : Fin 3 → Fin 3 → ℂ)
2399   (plebanski_vacuum : ℂ → (Fin 3 → Fin 3 → ℂ) → (Fin 3 → Fin 3 → ℂ) → Prop)
2400   (isLeviCivitaRicci : (Fin 4 → Fin 4 → ℝ) → (Fin 4 → Fin 4 → ℝ) → Prop)

```

```

2401     where
2402     equivalence_iff :
2403       (∀ mu nu, (∑ alpha : Fin 4, g mu alpha * g_inv alpha nu) = if mu = nu then 1 else 0) →
2404       isLeviCivitaRicci g Ricci →
2405       (plebanski_vacuum (Lambda : ℂ) F_ij F_bar_ij ↔ (∀ a b, Ricci a b = Lambda * g a b))

```
